Cooperative Control of Heterogeneous DC/DC Boost Converters for Ship Degaussing Systems

Detailed Technical Report

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Topic: Cooperative Current Control

Application: Ship Degaussing System

System: Heterogeneous DC/DC Boost-Converter Network

Key Focus

Cooperative current synchronization using
Mixed NI+PR output-feedback control

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1 Introduction

1.1 Background

The rapid development of power electronics has resulted in the widespread use of DC/DC power converters in many modern electrical and industrial applications. DC/DC converters are essential for regulating voltage and current levels and for supplying electrical loads with the required power.

They are used in applications such as electric vehicles, DC microgrids, renewable-energy systems, and shipbuilding. Among the different converter topologies, the **DC/DC boost converter** is particularly important when a controlled lower DC voltage or current is required from a higher DC source.

The work considered here focuses specifically on the application of multiple DC/DC boost converters in a **ship degaussing system**.

1.2 Ship Degaussing Application

A ship, particularly one with a steel hull, naturally produces a magnetic field. This magnetic signature can be detected by magnetic sensors and may increase the vulnerability of the vessel to magnetically sensitive mines or other detection systems.

Degaussing is a technique used to reduce or compensate for this magnetic signature by generating controlled magnetic fields around the ship. This is achieved by passing controlled currents through specially installed **degaussing coils**.

In the considered system, multiple DC/DC boost converters are used to supply current to independent degaussing coils. The current produced by each converter directly influences the magnetic field generated by its corresponding coil.

Therefore, maintaining the required current accurately and consistently is an important requirement of the degaussing system.

In particular, the output currents of the different converters need to be coordinated so that they can follow a common reference current, even when the reference changes with time.

Thus, the control problem is not simply to regulate each boost converter independently. Instead, the converters must operate as a **cooperative multi-converter system**, where information about their output currents is exchanged through a communication network.
The control system must ensure that the individual converters work together and achieve the desired current synchronization.

1.3 Need for Multiple-Converter Systems

Using several converters instead of a single converter provides flexibility for supplying multiple independent loads.

However, coordinating several converters introduces additional control challenges. Each converter may have different electrical parameters, such as load resistance, inductance, capacitance, or source characteristics. Consequently, the dynamic behaviour of one converter may differ from that of another.

Such a system is therefore referred to as a **heterogeneous multi-agent system (MAS)** because the individual agents do not necessarily have identical dynamics. In the considered application, each boost converter can be treated as an individual agent, while the complete collection of converters forms the multi-agent system.

The converters communicate with one another through a communication network and exchange information related to their output currents.

This distributed exchange of information allows each converter to adjust its control input based not only on its own output but also on the outputs of neighbouring converters.

The objective is to make the converters cooperate and converge toward the same desired current.

1.4 Introduction to Cooperative Control

Cooperative control is a control methodology in which multiple agents interact with one another to accomplish a common objective.

It has become an important research area in multi-agent systems, with applications involving consensus, formation tracking, flocking, and leader-following systems.

For a multi-converter system, cooperative control provides a framework through which individual converters can coordinate their behaviour.

Instead of designing completely independent controllers for every converter, information exchanged through the communication network can be incorporated into the control law.

In this work, the output current of each converter is considered as the important output variable.

The converters exchange their output-current information, and the controller uses the differences between neighbouring converter outputs to reduce the mismatch between them.

At the same time, the converters are connected to a reference signal so that the common output current converges to the desired reference.

Individual Converters $\longrightarrow$ **Information Exchange** $\longrightarrow$ **Cooperative Control** $\longrightarrow$ **Current Consensus**

where **current consensus** means that the output currents of the individual converters converge to the same desired reference.

1.5 Motivation for Current Synchronization

Current synchronization is particularly important in the ship degaussing application because each converter supplies a degaussing coil.

If the converters do not produce their required currents accurately, the resulting magnetic field may not have the desired characteristics.

The control problem becomes more challenging when the load demand is changing. A controller must therefore be capable of responding to a **time-varying reference signal** rather than only a constant reference.
In addition, practical converter systems can experience external disturbances, which may temporarily alter the output current.

Consequently, an effective cooperative control strategy should satisfy several requirements:

1. The output currents of all converters should converge toward a common value.
2. The common current should track the desired reference.
3. The system should remain stable during the tracking process.
4. The controller should operate despite differences between the converter dynamics.
5. The system should maintain acceptable performance when external disturbances occur.
6. The control scheme should be suitable for time-varying reference signals.

The paper specifically identifies **unequal current sharing, stability problems, and limited robustness** as challenges associated with conventional approaches to interconnected converter systems.

1.6 Limitations of Conventional Control Approaches

When each converter is controlled independently, the controller primarily considers the behaviour of its own converter.

Such an approach may not adequately address the interaction between several interconnected converters.

For a heterogeneous converter network, the situation becomes more complicated because the converters do not have identical dynamics.

A controller that produces satisfactory performance for one converter may not automatically provide the same response for another converter having different electrical parameters.

Furthermore, the presence of varying load conditions and external disturbances can affect the output-current response.

Therefore, simply regulating individual converter outputs is not sufficient when the overall objective is **coordinated current tracking and consensus**.

The paper also points out that much of the existing cooperative-control research has concentrated on consensus analysis, whereas greater attention is required to simultaneously address **consensus and stability**, particularly for nonidentical multi-converter degaussing applications.

This motivates the development of a cooperative controller specifically capable of handling heterogeneous converter dynamics while maintaining stability.

1.7 Motivation for Negative Imaginary (NI) Theory

The **Negative Imaginary (NI) systems theory** provides the theoretical foundation for the proposed control strategy.

The motivation for applying NI theory comes from an important property of DC/DC converters: their average or large-signal models can exhibit **Strictly Negative Imaginary (SNI)** behaviour for combinations of circuit parameters such as resistance R, inductance L, and capacitance C.

This property makes NI-based control particularly suitable for the considered application.
Rather than treating the boost converters as completely arbitrary nonlinear systems, their frequency-domain properties can be exploited when designing the cooperative controller.

Another important advantage is that NI-based cooperative control can be applied to both **homogeneous and heterogeneous multi-agent systems** and can accommodate time-varying reference signals.

Therefore, an NI-based distributed output-feedback controller can be developed for the heterogeneous boost-converter network.

The proposed work further extends this idea by employing a **mixed NI+Positive Real (PR) controller**.

The controller is designed so that it possesses the required frequency-domain properties for establishing the stability of the interconnected heterogeneous converter system.

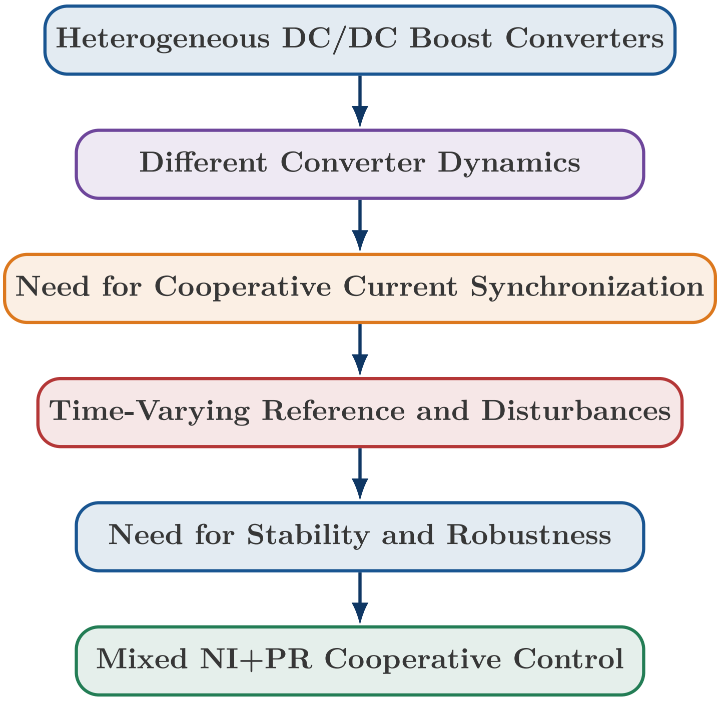
1.8 Research Gap and Motivation

Previous studies have investigated cooperative control and NI-based approaches for multi-agent systems.

However, much of the earlier NI-based work considered networked agents with **homogeneous dynamics**, such as single- or double-integrator systems.

The present problem is different because the individual DC/DC boost converters are **heterogeneous**, meaning that their dynamics are not identical.

Therefore, the main research motivation can be summarized as follows:

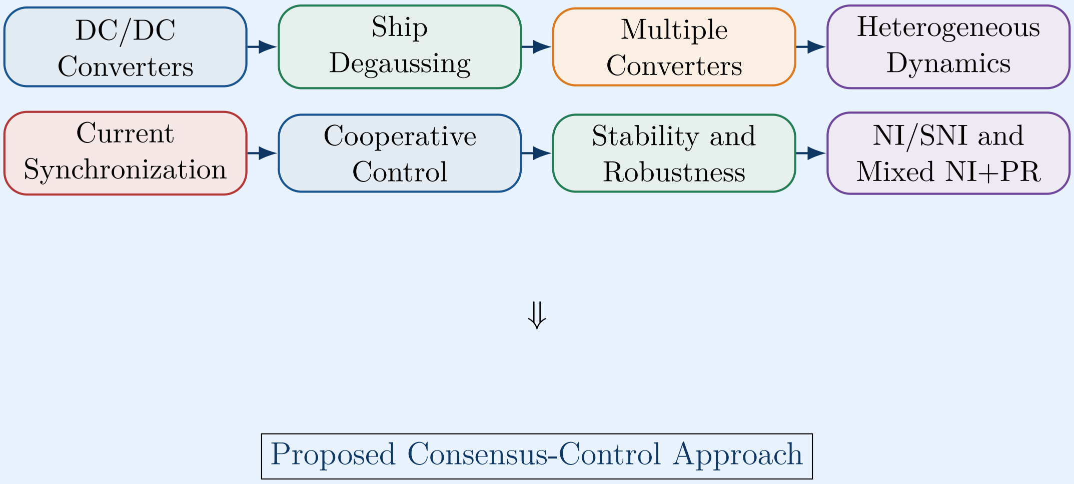


The proposed approach combines the SNI characteristics of the boost-converter plants with a homogeneous **mixed NI+PR output-feedback controller**.

The stability of the resulting interconnected system is then investigated using **characteristic loci theory**, rather than relying on a conventional Lyapunov-based proof.

Summary of the Introduction

Logical Progression of the Work



2 Notations and Symbols

This section defines the **mathematical notation and symbols** used throughout the report. These definitions are important because the later sections on NI/SNI systems, graph theory, transfer-function matrices, characteristic loci, and stability analysis use these symbols repeatedly.

2.1 Complex and Real Numbers

2.1.1 Complex Numbers

The set of all complex numbers is represented by

$$\boxed{\mathbb{C}}$$

A complex number can generally be written as

$$p = a + jb$$

where:

- $a =$ real part,
- $b =$ imaginary part,
- $j = \sqrt{-1}$.

The real and imaginary parts are denoted by

$$\boxed{\Re(p)}$$

and

$$\boxed{\Im(p)}$$

respectively.

2.1.2 Real Numbers

The set of all real numbers is represented by

$$\boxed{\mathbb{R}}$$

Examples of real numbers are

$$-2, \quad 0, \quad 1.5, \quad 10, \quad \pi.$$

2.2 Closed Right Half-Plane

The closed right half of the complex plane is represented by

$$\boxed{\overline{\mathbb{C}}_+}$$

It contains all complex numbers satisfying

$$\Re(s) \geq 0.$$

Similarly, the closed right half of the real line is represented by

$$\boxed{\overline{\mathbb{R}}_+}.$$

This region becomes important later when discussing **stability**, because the location of poles relative to the right-half plane determines the stability properties of a system.

2.3 Positive Real Numbers

The set of positive real numbers excluding zero is represented by

$$\boxed{\mathbb{R}_+ = \mathbb{R} \setminus \{0\}}$$

Thus,

$$k \in \mathbb{R}_+$$

means that

$$k > 0.$$

For example,

$$k = 1,\ 1.5,\ 10$$

are all members of $\mathbb{R}_+$.

This notation is particularly important when defining the controller gain k.

2.4 Real and Complex Matrices

The notation

$$\boxed{\mathbb{R}^{m\times n}}$$

represents the set of all $m \times n$ matrices containing real-valued elements.

For example,

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

belongs to

$$\mathbb{R}^{2\times 2}.$$

Similarly,

$$\boxed{\mathbb{C}^{m\times n}}$$

represents the set of $m \times n$ matrices whose elements can be complex numbers.

2.5 Rational Functions

The symbol

$$\boxed{\mathbb{R}(s)}$$

represents the set of **rational functions of the variable s with real coefficients**.

A typical transfer function is

$$G(s) = \frac{N(s)}{D(s)}$$

where $N(s)$ and $D(s)$ are polynomials in s.

For example,

$$G(s) = \frac{s+2}{s^2+3s+5}$$

is an element of $\mathbb{R}(s)$.

2.6 Rational Transfer-Function Matrices

The notation

$$\boxed{\mathbb{R}(s)^{n\times m}}$$

represents the set of $n \times m$ matrices whose elements are rational functions of s.

For example,

$$G(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix}$$

is a rational transfer-function matrix.

This notation is useful because the complete network of boost converters is represented using a **transfer-function matrix** rather than a single transfer function.

2.7 RH_∞

One of the most important notations is

$$\boxed{RH_\infty}$$

which represents the set of transfer-function matrices that are:

- real,
- rational,
- proper, and
- asymptotically stable.

In other words,

$$RH_\infty = \{\text{real, rational, proper, asymptotically stable TFMs}\}.$$

This notation is used to describe stable converter models and controller components.

For example, the problem formulation assumes that each converter transfer function satisfies

$$G_i(s) \in RH_\infty.$$

This means that each $G_i(s)$ is a stable, proper, real-rational transfer function.

2.8 Block Diagonal Matrix

The notation

$$\boxed{\text{diag}_{i=1}^{N}\{G_i\}}$$

represents a **block diagonal matrix** whose diagonal blocks are

$$G_1, G_2, \ldots, G_N.$$

For example, for four boost converters,

$$\mathcal{G}(s) = \text{diag}\left\{G_1(s), G_2(s), G_3(s), G_4(s)\right\}$$

can be written as

$$\mathcal{G}(s) = \begin{bmatrix} G_1(s) & 0 & 0 & 0 \\ 0 & G_2(s) & 0 & 0 \\ 0 & 0 & G_3(s) & 0 \\ 0 & 0 & 0 & G_4(s) \end{bmatrix}.$$

This representation is especially useful for the **heterogeneous boost-converter system**.

If

$$G_1(s) \neq G_2(s) \neq G_3(s) \neq G_4(s),$$

then the system represents heterogeneous converter dynamics.

2.9 Matrix Element Notation

A matrix is represented as

$$\boxed{A = [a_{ij}]}$$

where a_{ij} represents the element located in:

- the i^{th} row,
- the j^{th} column.

For example,

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

Here,

$$a_{12}$$

is the element in row 1, column 2.

This notation is particularly useful for the **adjacency matrix**

$$A = [a_{ij}]$$

of the converter communication network.

2.10 Kronecker Product

The symbol

$$\boxed{\otimes}$$

represents the **Kronecker product**.

If

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

then

$$A \otimes B$$

means that every element of A is multiplied by the complete matrix B.

The Kronecker product is important in multi-agent systems because it allows the **communication-network matrices** to be combined with the **individual converter dynamics**.

For example, expressions of the form

$$(L + P) \otimes G(s)$$

can represent the interaction between the converter dynamics and the communication network.

2.11 Eigenvalue

The notation

$$\boxed{\lambda_i[A]}$$

represents the i^{th} eigenvalue of the matrix A.

If

$$Av = \lambda v,$$

then λ is an eigenvalue of A.

For an $N \times N$ matrix, there are N eigenvalues, counting multiplicities:

$$\lambda_1[A], \lambda_2[A], \ldots, \lambda_N[A].$$

Eigenvalues are particularly important in the characteristic-loci analysis of the multi-agent boost-converter system.

2.12 Determinant

The notation

$$\boxed{\det[G]}$$

represents the **determinant** of the square matrix G.

For a 2×2 matrix,

$$G = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det[G] = ad - bc.$$

The determinant plays an important role in characteristic-loci theory because the characteristic loci are associated with the eigenvalues of the loop transfer-function matrix.

2.13 Magnitude of a Transfer Function

The notation

$$\boxed{|G(s)|}$$

represents the **magnitude** of the transfer function $G(s)$.

When

$$s = j\omega,$$

the magnitude becomes

$$|G(j\omega)|.$$

It indicates how strongly the system amplifies or attenuates a signal at a particular frequency.

For example,

$$|G(j\omega)| > 1$$

means amplification at that frequency, whereas

$$|G(j\omega)| < 1$$

means attenuation.

2.14 Phase Angle

The notation

$$\boxed{\angle G(s)}$$

represents the **phase angle** of the transfer function.

For sinusoidal analysis,

$$\angle G(j\omega)$$

gives the phase shift introduced by the system at frequency ω.

The magnitude and phase together describe the frequency-domain behaviour of the converter and controller.

For the NI/SNI analysis, phase angle is particularly important because NI systems have characteristic phase behaviour in the frequency domain.

2.15 Laplace Variable s

Although not separately explained in the notation paragraph, the variable

$$\boxed{s}$$

is the complex Laplace-domain variable:

$$s = \sigma + j\omega.$$

Here:

- σ = real component,
- ω = angular frequency,
- $j = \sqrt{-1}$.

When analysing frequency response,

$$s = j\omega.$$

Therefore, expressions such as

$$G(j\omega)$$

represent the transfer function evaluated along the imaginary axis of the s-plane.

2.16 Why These Notations Are Important for This Report

These symbols will appear repeatedly in the upcoming sections.

For example, the heterogeneous boost-converter system can be represented as

$$\boxed{G_r(s) = \mathrm{diag}_{i=1}^{N}\{G_i(s)\}}$$

where each

$$G_i(s) \in RH_\infty.$$

The communication network is represented using a matrix

$$A = [a_{ij}],$$

from which the Laplacian matrix is obtained as

$$L = D - A.$$

The interaction between the network and the converter dynamics can then involve a Kronecker product such as

$$(L + P) \otimes G_r(s).$$

The characteristic-loci analysis subsequently examines the eigenvalues

$$\lambda_i[\mathcal{L}(s)]$$

of the resulting loop transfer-function matrix.

The magnitude and phase,

$$|\lambda_i(j\omega)|$$

and

$$\angle\lambda_i(j\omega),$$

are then used to study the location of the characteristic loci in the complex plane.

Quick Reference Table

Symbol	Meaning
$\mathbb{C}$	Set of complex numbers
$\mathbb{R}$	Set of real numbers
$\overline{\mathbb{C}}_+$	Closed right half of complex plane
$\overline{\mathbb{R}}_+$	Closed right half of real line
$\mathbb{R}_+$	Positive real numbers excluding zero
$\mathbb{R}^{m\times n}$	$m \times n$ real matrices
$\mathbb{C}^{m\times n}$	$m \times n$ complex matrices
$\mathbb{R}(s)$	Real rational functions of s
$\mathbb{R}(s)^{n\times m}$	Rational transfer-function matrices
RH_∞	Real, rational, proper, asymptotically stable TFMs
$\text{diag}\{G_i\}$	Block diagonal matrix
$A = [a_{ij}]$	Matrix with element a_{ij}
$\otimes$	Kronecker product
$\lambda_i[A]$	i^{th} eigenvalue of A
$\det[G]$	Determinant of G
$\|G(s)\|$	Magnitude of $G(s)$
$\angle G(s)$	Phase angle of $G(s)$
s	Laplace-domain variable
j	Imaginary unit, $j^2 = -1$

These definitions establish the mathematical language needed for the next sections, particularly **Mixed NI+PR Theory, Characteristic Loci Theory, Graph Theory**, and the mathematical modelling of the heterogeneous DC/DC boost-converter system.

3 Ship Degaussing System

3.1 Introduction to Ship Degaussing

A ship, particularly one constructed with a steel hull, naturally develops a magnetic field due to the presence of ferromagnetic materials and the electrical systems operating onboard. This magnetic field forms a **magnetic signature** around the vessel. Such a signature can be detected by magnetic sensors and may make the ship vulnerable to detection by magnetically sensitive devices or mines.

> **Degaussing** is a technique used to reduce or compensate for the magnetic signature of a ship. The basic principle is to generate a controlled magnetic field that opposes or counteracts the magnetic field produced by the ship itself. This is achieved by installing electrical coils at appropriate locations around the ship and controlling the current flowing through these coils.

The fundamental concept can be represented as

Ship Magnetic Field + **Compensating Magnetic Field** → **Reduced Magnetic Signature**

The compensating magnetic field is controlled by regulating the current supplied to the degaussing coils.

3.2 Magnetic Signature of a Ship

The **magnetic signature** represents the magnetic field produced by the ship and its associated components. A steel-hulled ship can acquire a significant magnetic signature because the Earth's magnetic field and the ship's operating conditions can magnetize the steel structure.

The resulting magnetic field can vary depending on several operating and environmental conditions. Consequently, the magnetic signature cannot always be considered constant.

This creates a requirement for a controllable degaussing system capable of adjusting the magnetic compensation according to the required operating condition.

The purpose of the degaussing system is therefore not simply to generate a fixed magnetic field. Instead, it must generate an appropriate compensating field whose magnitude can change according to the required degaussing condition.

3.3 Degaussing Coils

Degaussing coils are electrical windings installed on or within different regions of the ship. When electric current flows through a coil, a magnetic field is produced around it.

The magnitude of the generated magnetic field depends on the current flowing through the coil and the physical characteristics and arrangement of the winding.

In simplified form,

$$\boxed{I_{\text{coil}} \longrightarrow B_{\text{coil}}}$$

where

- I_{coil} = current through the degaussing coil,
- B_{coil} = magnetic field generated by the coil.

The coil current is therefore one of the most important controlled variables in the degaussing system.

By appropriately controlling the coil currents, a compensating magnetic field can be generated to oppose the ship's unwanted magnetic field.

3.4 Principle of Magnetic Compensation

The principle of degaussing can be understood by considering two magnetic fields:

1. The magnetic field generated by the ship.
2. The compensating magnetic field generated by the degaussing coils.

If the magnetic field produced by the ship is represented by

$$B_{\text{ship}},$$

and the magnetic field generated by the degaussing system is represented by

$$B_{\text{coil}},$$

then the resulting magnetic field can be conceptually represented as

$$\boxed{B_{\text{net}} = B_{\text{ship}} + B_{\text{coil}}}$$

The degaussing system attempts to make

$$\boxed{B_{\text{net}} \to 0}$$

or, more practically, to reduce the externally observable magnetic signature to an acceptably low level.

Therefore, controlling the current through the degaussing coils is directly related to controlling the magnetic compensation produced by the system.

3.5 Role of DC/DC Boost Converters

In the proposed **boost-converter-based system**, DC/DC boost converters are used as the power-conditioning and current-control stages associated with the degaussing coils.

A boost converter is a DC/DC power electronic converter capable of producing an output voltage higher than its input voltage under ideal continuous-conduction operation. Its basic voltage-conversion relationship is

$$\boxed{V_o = \frac{V_{in}}{1 - D}}$$

where

- V_{in} = input DC voltage,
- V_o = output voltage,
- D = duty ratio of the switching signal.

The duty ratio is therefore an important control variable:

$$\boxed{D \uparrow \quad \Rightarrow \quad V_o \uparrow}$$

within the assumptions and operating range of the boost converter.

For the degaussing application considered in this report, the boost converter provides the controlled electrical power required by the associated degaussing load. The converter controller regulates the switching action so that the required output current can be obtained.

Thus, the power-flow path can be represented as

DC Source → Boost Converter → Degaussing Coil → Controlled Magnetic Field

The original paper develops this cooperative-control concept for DC/DC converters in a degaussing application and notes that boost converters, like buck converters, can possess stable NI/SNI characteristics in their average models.

3.6 Multiple Boost Converters in the Degaussing System

A practical degaussing system can contain multiple independently controlled coil sections. Therefore, instead of using a single converter, the system can be represented as a collection of converter-load units.

For the proposed boost-converter configuration,

$$\begin{array}{ccc} \text{Boost Converter 1} & \rightarrow & \text{Coil 1} \\ \text{Boost Converter 2} & \rightarrow & \text{Coil 2} \\ \vdots & & \vdots \\ \text{Boost Converter N} & \rightarrow & \text{Coil N} \end{array}$$

Each converter can have its own load and electrical parameters. Hence, the individual converter dynamics may not be identical.

This creates a **heterogeneous multi-agent system**, in which each boost converter can be treated as an individual agent.

The general structure is therefore

$$\boxed{G(s) = \text{diag}\left\{G_1(s), G_2(s), \ldots, G_N(s)\right\}}$$

where each

$$G_i(s)$$

represents the dynamics of the i^{th} boost converter and its associated load.

The source paper similarly models the degaussing system as a multi-agent network in which individual converters act as agents and exchange their output-current information through a communication network.

3.7 Independent Loads Associated with the Converters

Each boost converter is associated with an individual degaussing-coil load. Therefore, the loads do not necessarily have identical electrical characteristics.

For example,

$$\boxed{R_{L1} \neq R_{L2} \neq R_{L3} \neq R_{L4}}$$

As a result, even if all converters receive the same reference command, their uncontrolled responses may differ.

This is the reason the system is considered **heterogeneous**.

The general arrangement can be represented as

$$\begin{array}{ccc} G_1(s) & \rightarrow & R_{L1} \\ G_2(s) & \rightarrow & R_{L2} \\ G_3(s) & \rightarrow & R_{L3} \\ \vdots & & \vdots \\ G_N(s) & \rightarrow & R_{LN} \end{array}$$

where R_{Li} represents the load associated with the i^{th} converter.

Different loads result in different converter responses. Therefore, a cooperative controller is required to coordinate the converters rather than relying solely on independent local control.

3.8 Importance of Accurate Current Control

The current flowing through each degaussing coil determines the magnetic field generated by that coil. Therefore, accurate current regulation is essential for obtaining the desired magnetic compensation.

If the output currents are different from their required values, the generated magnetic fields may also deviate from their desired values.

For example,

$$I_1 \neq I_2 \neq I_3 \neq I_4$$

can result in different magnetic-field contributions from the corresponding coils.

The objective is therefore to achieve

$$\boxed{I_1(t) \rightarrow I_{\text{ref}}(t)}$$

$$\boxed{I_2(t) \rightarrow I_{\text{ref}}(t)}$$

$$\boxed{\vdots}$$

$$\boxed{I_N(t) \rightarrow I_{\text{ref}}(t)}$$

so that

$$\boxed{I_1(t) \approx I_2(t) \approx \cdots \approx I_N(t) \approx I_{\text{ref}}(t)}$$

when the desired operating condition requires equal current references.

The source paper emphasizes that maintaining the dynamic and steady-state integrity of the converter output current/load is necessary for consistent magnetic-field generation.

3.9 Need for Current Synchronization

The major control challenge arises because the boost converters may have different dynamics and independent loads.

Suppose four converters have outputs

$$y_1(t), \quad y_2(t), \quad y_3(t), \quad y_4(t).$$

The desired condition is

$$\boxed{y_1(t) = y_2(t) = y_3(t) = y_4(t) = r_0(t)}$$

in the steady-state tracking sense.

Therefore,

$$\boxed{\lim_{t\to\infty} [y_i(t) - r_0(t)] = 0, \qquad i = 1, 2, \ldots, N}$$

This represents both **consensus** and **reference tracking**.

The converters must exchange information so that each agent can determine how its current differs from the currents of its neighbouring agents.

A simplified representation is

Current Measurement → **Communication** → **Consensus Controller** → **Duty-Ratio Control** → **Boost Converter**

The controller therefore does not treat every converter as an isolated unit. Instead, it coordinates the complete converter network.

3.10 Communication Between Boost Converter Agents

The converters can be represented as nodes in a communication graph.

For example,

$$\boxed{\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)}$$

where:

- $\mathcal{V}$ represents the converter agents,

- $\mathcal{E}$ represents communication links,
- $A = [a_{ij}]$ is the adjacency matrix.

If converter j can transmit information to converter i, then

$$\boxed{a_{ij} > 0}$$

The information exchanged between converters primarily concerns their output currents.

For example, converter i can compare its own current with the current of converter j:

$$\boxed{y_j - y_i}$$

The resulting difference becomes part of the cooperative control action.

Thus, if one converter deviates from the desired collective behaviour, information from neighbouring converters helps the controller correct that deviation.

The source paper uses an undirected connected leader-following graph and defines the Laplacian and pinning matrices for this purpose.

3.11 Time-Varying Reference Current

The required degaussing current does not necessarily need to remain constant. The control system can therefore be designed to follow a **bounded time-varying reference signal**.

Let the reference be

$$\boxed{r_0(t)}$$

Then the individual converter currents should satisfy

$$\boxed{y_i(t) \to r_0(t)}$$

For example, if the required reference changes from one level to another,

$$\boxed{r_0(t) : I_1 \to I_2 \to I_3}$$

the cooperative controller must coordinate the boost converters so that their output currents follow these changes.

The reference-tracking objective can therefore be written as

$$\boxed{\lim_{t\to\infty} [y_i(t) - r_0(t)] = 0}$$

for all converter agents.

The source paper specifically formulates its objective around a bounded reference signal with a finite steady-state value.

3.12 Disturbances in the Degaussing System

A practical power-electronic system may experience external disturbances. Such disturbances can cause temporary deviations in the output current.

For the cooperative boost-converter system, a disturbance can be represented as

$$\boxed{d_i(t)}$$

acting on the output of the i^{th} converter.

The resulting output can temporarily deviate from the reference:

$$\boxed{y_i(t) \neq r_0(t)}$$

The controller should then restore the converter output toward the desired reference.

The disturbance-rejection objective can therefore be represented as

$d_i(t) \rightarrow$ **Temporary Current Deviation** $\rightarrow$ **Controller Action** $\rightarrow y_i(t) \rightarrow r_0(t)$

The source study evaluates robustness by applying single-pulse disturbances at the converter outputs at different times and observing whether the converters continue to track the bounded time-varying reference.

3.13 Complete Degaussing System Structure

The complete boost-converter-based system can be conceptually represented as follows:

Time-Varying Reference $r_0(t)$

$\downarrow$

Communication / Consensus Network

$\downarrow$

Distributed Cooperative Controller

$\downarrow$

Boost Converter 1	$\cdots$	Boost Converter N
$\downarrow$		$\downarrow$
Coil 1	$\cdots$	Coil N

$\downarrow$

Individual Magnetic Fields

$\downarrow$

Compensation of Ship Magnetic Signature

The essential relationship is

Reference Current $\rightarrow$ **Cooperative Control** $\rightarrow$ **Boost Converters** $\rightarrow$ **Degaussing-Coil Currents** $\rightarrow$ **Magnetic-Field Compensation**

3.14 Role of the Degaussing System in the Overall Control Problem

The ship degaussing application provides the practical motivation for the entire cooperative-control problem.

The control requirements can be summarized as:

Multiple Boost Converters

$\downarrow$

Different Converter/Load Dynamics

$\downarrow$

Independent Degaussing Coils

$\downarrow$

Need for Coordinated Current

$\downarrow$

Time-Varying Reference

$\downarrow$

External Disturbances

$\downarrow$

Distributed Cooperative Control

$\downarrow$

Current Consensus + Stability

Therefore, the ship degaussing system is an appropriate application for investigating cooperative control of heterogeneous DC/DC converters. Each boost converter acts as an individual agent, the degaussing coil represents its associated load, and the communication network allows the converter agents to coordinate their output currents.

4 Objectives and Contributions

This section explains **what the research intends to achieve and what is new or significant about the proposed approach**. The work is motivated by the challenges associated with coordinating multiple heterogeneous DC/DC boost converters while maintaining stability, tracking a time-varying reference, and remaining robust to disturbances. The proposed approach employs a **mixed Negative Imaginary (NI) + Positive Real (PR) cooperative control scheme** for this purpose.

4.1 Main Objective of the Work

The primary objective is to design a **distributed dynamic consensus controller** for multiple DC/DC boost converters used in a ship degaussing system.

The controller must make the output currents of all boost converters converge to the same desired reference current. Importantly, the reference may vary with time.

Mathematically, the desired objective is

$$\boxed{\lim_{t\to\infty}[y_i(t) - r_0(t)] = 0, \qquad i = 1, 2, \ldots, N}$$

where:

- $y_i(t)$ = output current of the i^{th} boost converter,
- $r_0(t)$ = desired reference current,
- N = number of boost converter agents.

Therefore, the controller must achieve two objectives simultaneously:

$$\boxed{\text{Current Consensus} + \text{Reference Tracking}}$$

Current consensus means that the output currents of all boost converters converge toward a common value, while reference tracking ensures that this common value follows the desired reference signal.

Thus,

$$y_1(t) \to r_0(t), \tag{1}$$
$$y_2(t) \to r_0(t), \tag{2}$$
$$\vdots \tag{3}$$
$$y_N(t) \to r_0(t). \tag{4}$$

Consequently,

$$y_1(t) \approx y_2(t) \approx \cdots \approx y_N(t)$$

while simultaneously,

$$y_i(t) \approx r_0(t).$$

4.2 Mixed NI+PR Cooperative Control Scheme

The first major contribution is the development of a **mixed NI+PR cooperative control scheme** for heterogeneous DC/DC boost converters.

The average or large-signal model of several DC/DC converter systems, including boost converters, can exhibit stable Negative Imaginary or Strictly Negative Imaginary properties for suitable circuit parameters. This provides motivation for exploiting NI-system characteristics in the design of the cooperative controller.

Instead of relying only on a conventional controller, the proposed approach uses a controller possessing both **Negative Imaginary and Positive Real characteristics** over appropriate frequency regions. This type of controller is referred to as a **"mixed" NI+PR controller**.

An important feature of the proposed approach is that the **plant dynamics are heterogeneous**, meaning that the individual boost converters may have different dynamic characteristics, while the controller configuration remains **homogeneous**.

In other words,

$$\boxed{\underbrace{G_1(s), G_2(s), \ldots, G_N(s)}_{\text{Different boost-converter dynamics}} + \underbrace{C(s), C(s), \ldots, C(s)}_{\text{Same controller}}}$$

The use of a common controller for different converter dynamics is particularly significant for multi-converter systems in which individual converters may operate under different load conditions or possess different electrical parameters.

4.3 Consensus Under Varying Load Demand

A second important objective is to maintain current consensus even when the load demand changes.

In a practical ship degaussing system, the required current may not remain constant. Therefore, the reference signal can be considered a **time-varying bounded signal**.

The proposed controller is designed to ensure that all boost-converter output currents follow the changing reference:

$$y_1(t) \to r_0(t), \tag{5}$$
$$y_2(t) \to r_0(t), \tag{6}$$
$$\vdots \tag{7}$$
$$y_N(t) \to r_0(t). \tag{8}$$

Consequently,

$$y_1(t) \approx y_2(t) \approx \cdots \approx y_N(t)$$

while simultaneously,

$$y_i(t) \approx r_0(t).$$

Therefore, even when the desired current changes with time, the boost converters are required to coordinate their responses and maintain synchronization.

The achievement of output-current consensus under varying load demand is one of the central objectives of the proposed cooperative control strategy.

4.4 Output-Feedback Control

Another major contribution is that the proposed controller uses **output feedback** rather than requiring complete state information.

For a boost converter, complete information about all internal states may not always be directly available in a practical implementation. Measuring every internal state can increase the number of sensors and the complexity of the control hardware.

The proposed approach instead relies on measurable output information, particularly the output current, together with information exchanged between neighbouring converter agents.

The basic control-information flow can therefore be represented as

Individual Converters $\longrightarrow$ **Information Exchange** $\longrightarrow$ **Cooperative Control** $\longrightarrow$ **Current Consensus**

Each converter receives information about its own output and, depending on the network topology, information from neighbouring converters. The controller processes this information and generates the corresponding control action.

This output-feedback structure makes the approach suitable for applications where **full-state feedback is unavailable**. Output feedback is explicitly identified as one of the important contributions of the proposed scheme.

4.5 Control of Heterogeneous Boost Converters

A particularly important contribution is the application of the proposed cooperative control strategy to **four nonidentical boost converter agents**.

The converters are considered heterogeneous because their dynamic characteristics are different. In the considered simulation framework, the four converters have different load resistances:

$$R_{L1} = 0.25\,\Omega, \tag{9}$$
$$R_{L2} = 0.50\,\Omega, \tag{10}$$
$$R_{L3} = 0.75\,\Omega, \tag{11}$$
$$R_{L4} = 1.00\,\Omega. \tag{12}$$

The proposed controller, however, maintains the same configuration for all converter agents.

Therefore, the control architecture can be represented as

$$\boxed{\begin{array}{l} \text{Boost Converter 1} \\ \text{Boost Converter 2} \\ \text{Boost Converter 3} \\ \text{Boost Converter 4} \end{array} + \text{Common Cooperative Controller}}$$

where the converter plants can have different dynamics while the controller remains homogeneous.

This approach avoids the need to independently design a completely different controller for every converter. Instead, one common controller structure is used to coordinate the heterogeneous boost converters.

This is important because practical multi-converter systems may contain converters operating under different electrical conditions.

4.6 Stability Analysis Using Characteristic Loci Theory

Another major contribution is the theoretical verification of the **closed-loop stability** of the proposed cooperative control system.

The stability analysis does not rely only on simulation results. Instead, the proposed methodology uses **characteristic loci theory** to investigate the stability of the interconnected converter system.

For the complete network, the loop transfer function is considered and its characteristic loci are analyzed in the complex plane. The critical point associated with the negative-feedback stability condition is then examined.

The fundamental stability requirement is that the characteristic loci should not encircle the critical point under the specified controller-gain conditions.

The theoretical result establishes that there exists a finite positive gain k^* such that

$$\boxed{0 < k \leq k^*}$$

ensures asymptotic stability of the proposed consensus-control system under the stated assumptions.

Therefore, the contribution is not simply that the controller performs well in simulation. Rather, the proposed approach provides a **theoretical stability foundation** for the cooperative control of heterogeneous boost converters.

4.7 Robustness Against Output Disturbances

The proposed control scheme is also evaluated in the presence of external disturbances.

To examine its robustness, **single-pulse disturbances** are introduced at the output of the individual boost converters at different time instants.

The disturbance magnitude is selected as

$$\boxed{10\%}$$

of the maximum value of the reference signal.

The purpose of this test is to determine whether the boost-converter network can recover its coordinated output-current response after a disturbance occurs.

The desired behaviour is that, following the disturbance, the converter outputs should return toward the reference:

$$y_i(t) \to r_0(t).$$

At the same time, the converters should continue to maintain consensus with one another.

Thus, the disturbance test evaluates both:

- reference-tracking capability, and
- cooperative synchronization capability.

The study demonstrates that the proposed cooperative control strategy continues to track the bounded time-varying reference in the presence of the applied output disturbances.

Important limitation: The maximum disturbance magnitude that the system can tolerate is not determined in the study. A detailed investigation of disturbance tolerance is therefore an area for future research.

4.8 Investigation of Different Switching Frequencies

The proposed control scheme is also evaluated at different switching frequencies to investigate its practical performance.

The switching frequencies considered are

$$30 \text{ kHz}, \quad 20 \text{ kHz}, \quad 10 \text{ kHz}, \quad 7.5 \text{ kHz}, \quad 5 \text{ kHz}.$$

The switching frequency has a significant effect on the output-current waveform of a power converter.

At higher switching frequencies, the output current generally exhibits reduced switching-related variations. In the considered study, the effect of chattering is negligible at the higher frequencies of **20 kHz and 30 kHz**.

However, when the switching frequency is reduced, the output current exhibits increased chattering. Excessive chattering can degrade the quality of the output response and may affect system performance and reliability.

At the same time, very high switching frequencies can increase the practical difficulty of implementing the converter because of switching losses and hardware limitations.

Therefore, a compromise must be established between:

Output-Current Quality ↔ Practical Switching Frequency

Based on the study, **10 kHz** is selected for the main simulation cases.

4.9 Comprehensive Simulation Validation

The effectiveness of the proposed mixed NI+PR cooperative control scheme is evaluated under multiple operating conditions rather than under a single ideal simulation.

Three major simulation cases are considered.

4.9.1 Case 1 — Reference Tracking Without Output Disturbance

In the first case, the boost converters operate without externally introduced output disturbances.

The objective is to verify whether all converters can:

- track the desired reference,
- achieve current consensus,
- maintain stability, and
- provide an acceptable transient response.

The individual output currents are expected to converge toward the common reference signal.

4.9.2 Case 2 — Reference Tracking With Output Disturbance

In the second case, external single-pulse disturbances are introduced at different time instants.

The purpose is to evaluate the robustness of the cooperative controller.

The important performance indicators include:

- disturbance rejection,
- recovery time,
- transient deviation,
- current synchronization after disturbance, and
- steady-state tracking.

The converters should recover from the disturbances and return toward the desired reference while maintaining cooperative behaviour.

4.9.3 Case 3 — Reference Tracking Under Different Switching Frequencies

In the third case, the boost converters are evaluated at different switching frequencies:

$$30 \text{ kHz}, \ 20 \text{ kHz}, \ 10 \text{ kHz}, \ 7.5 \text{ kHz}, \ 5 \text{ kHz}.$$

The purpose is to determine how the switching frequency influences the output-current response and the practical feasibility of the proposed control strategy.

This provides a more comprehensive assessment of the controller because its performance is examined under different operating conditions rather than only under an idealized fixed-frequency condition.

4.10 Overall Contribution

The complete contribution of the work can be summarized through the following sequence:

Heterogeneous DC/DC Boost Converters
↓
NI/SNI Plant Characteristics
↓
Homogeneous Mixed NI+PR Controller
↓
Distributed Output Feedback
↓
Cooperative Current Consensus
↓
Characteristic-Loci Stability Analysis
↓
Disturbance Robustness
↓
Switching-Frequency Evaluation

The main significance of the proposed approach is that a **single homogeneous mixed NI+PR cooperative controller** can be used to coordinate heterogeneous DC/DC boost-converter agents.

The objective is for the output currents of the individual boost converters to achieve consensus with respect to a time-varying reference while maintaining asymptotic stability under the conditions established by the theoretical analysis.

The overall control concept can therefore be expressed as

Heterogeneous Boost Converters → Communication → Cooperative Control → Current Consensus → Reference Tracking

4.11 Key Contributions of the Work

The major contributions can finally be summarized as follows:

4.11.1 Development of a Mixed NI+PR Cooperative Controller

A mixed Negative Imaginary and Positive Real controller is developed for the cooperative control of heterogeneous DC/DC boost converters.

4.11.2 Output-Feedback Implementation

The proposed control architecture uses output feedback, reducing the requirement for complete internal-state information.

4.11.3 Consensus Control of Nonidentical Boost Converters

Multiple heterogeneous boost-converter agents are coordinated using a common homogeneous controller so that their output currents converge toward a common reference.

4.11.4 Robustness Evaluation

The proposed controller is evaluated in the presence of output disturbances introduced at different times to examine its ability to maintain reference tracking and recover from disturbances.

4.11.5 Switching-Frequency Investigation

The performance of the proposed controller is evaluated at different switching frequencies to investigate output-current chattering and practical operating considerations.

4.11.6 Overall Objective

Thus, the overall objective can be stated as

$$\boxed{\text{Design a stable and distributed cooperative controller}}$$

such that

$$\boxed{y_1(t), y_2(t), \ldots, y_N(t) \to r_0(t)}$$

despite

heterogeneous converter dynamics + **varying reference** + **external disturbances**

This forms the foundation for the subsequent sections dealing with the **boost-converter mathematical model, NI/SNI analysis, graph-theoretic formulation, controller design, characteristic-loci stability analysis, and simulation results**.

5 Multi-Agent System and Cooperative Control

5.1 Concept of Multi-Agent Systems

A **Multi-Agent System (MAS)** is a system composed of multiple individual agents that interact with one another to accomplish a common objective. Each agent has its own dynamics, input, output, and control action, while communication between agents allows them to coordinate their behaviour.

In a conventional control system, a single controller generally regulates a single plant. In contrast, a multi-agent system contains several interconnected subsystems, and the overall system behaviour depends not only on the individual dynamics of each agent but also on the **interaction and communication among the agents**.

In the proposed application, the complete system consists of multiple DC/DC boost converters. Each boost converter is treated as an individual agent, and the collection of all converters forms the multi-agent system.

The general representation is

$$\boxed{\mathcal{G}(s) = \operatorname{diag}\left\{G_1(s), G_2(s), \ldots, G_N(s)\right\}}$$

where $G_i(s)$ represents the dynamic model of the i^{th} boost converter.

The objective is not to control the converters independently, but to make them **cooperate** so that their output currents achieve a common desired behaviour.

The concept of treating converters as agents and exchanging their load-current information through a communication network follows the multi-agent formulation used in the source work.

5.2 Boost Converter as an Individual Agent

In the proposed system, each DC/DC boost converter is considered an independent **agent**.

For the i^{th} converter, the plant can be represented by

$$\boxed{G_i(s)}$$

with:

- $u_i(t)$ — control input,
- $y_i(t)$ — output current,
- $r_0(t)$ — desired reference current.

The control input of a boost converter is generally associated with its **duty ratio**,

$$\boxed{0 < D_i < 1}$$

which determines the switching behaviour of the converter.

Thus, the basic operation of an individual agent can be represented as

$$\boxed{u_i(t) \rightarrow \text{Boost Converter} \rightarrow y_i(t)}$$

where the controller adjusts $u_i(t)$ according to the measured output and the information received from other converter agents.

Each converter may have different circuit or load parameters. Consequently,

$$\boxed{G_i(s) \neq G_j(s), \qquad i \neq j}$$

This difference in dynamics is what makes the complete converter network a **heterogeneous multi-agent system**.

5.3 Leader-Follower Architecture

The proposed cooperative-control structure follows a **leader-follower architecture**.

In this arrangement, one external system or reference generator acts as the **leader**, while the individual boost converters act as **followers**.

The leader provides the desired reference signal

$$\boxed{r_0(t)}$$

The follower converters then attempt to track this reference while simultaneously coordinating with neighbouring converters.

The architecture can be represented as

$$\boxed{\text{Leader} \quad r_0(t) \quad \longrightarrow \quad \begin{array}{c} \text{Boost Converter 1} \\ \text{Boost Converter 2} \\ \vdots \\ \text{Boost Converter N} \end{array}}$$

However, the followers do not necessarily receive the reference independently in the same manner. Instead, the information is distributed through the communication topology and **pinning connections**.

For a follower converter i, the controller considers two important types of information:

1. Difference between its output and neighbouring converter outputs.
2. Difference between its output and the reference signal.

These two components are responsible for achieving both **consensus** and **reference tracking**.

The source work explicitly formulates the problem using a leader-following MAS with a connected undirected communication graph.

5.4 Communication Between Boost Converter Agents

Communication between the converter agents is an essential part of cooperative control.

Each boost converter measures its own output current and exchanges relevant information with its neighbouring converters.

For example, consider two converters i and j. The difference between their output currents is

$$\boxed{y_j(t) - y_i(t)}$$

If

$$y_j(t) > y_i(t),$$

then converter i can use this information to modify its control action and reduce the difference.

Similarly, if

$$y_j(t) < y_i(t),$$

the control action can move in the opposite direction.

Therefore, the interaction between neighbouring converters can be represented as

$$\boxed{y_j - y_i \rightarrow \text{Cooperative Controller} \rightarrow u_i}$$

This information exchange enables the converters to gradually reduce their output-current differences.

The source paper describes the converter agents as exchanging output, specifically load-current, information through a communication network.

5.5 Communication Graph

The communication network can be mathematically represented using graph theory.

Let the communication graph be

$$\boxed{\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)}$$

where:

- $\mathcal{V}$ = set of converter agents,
- $\mathcal{E}$ = set of communication links,
- $A = [a_{ij}]$ = adjacency matrix.

The adjacency coefficient a_{ij} indicates whether converter j can communicate information to converter i.

For an undirected communication network,

$$\boxed{a_{ij} = a_{ji}}$$

If two converters can exchange information, then

$$\boxed{a_{ij} > 0}$$

If there is no communication link,

$$\boxed{a_{ij} = 0}$$

The communication topology therefore determines **which converter knows about which neighbouring converter**.

5.6 Neighbouring Agents

For a particular converter i, its neighbouring-agent set can be represented as

$$\boxed{\mathcal{N}_i = \{j \mid (v_j, v_i) \in \mathcal{E}\}}$$

The controller of converter i uses the output information from these neighbouring agents.

For example, if converter 2 communicates with converters 1 and 3, then

$$\boxed{\mathcal{N}_2 = \{1, 3\}}$$

Its controller can therefore use

$$\boxed{y_1 - y_2}$$

and

$$\boxed{y_3 - y_2}$$

to determine the required cooperative correction.

This means that each converter does not need information from every other converter. It only needs information from its designated neighbours according to the communication topology.

5.7 Adjacency, Degree, and Laplacian Matrices

The communication network can be represented mathematically using three important matrices.

5.7.1 Adjacency Matrix

The adjacency matrix is

$$\boxed{A = [a_{ij}]}$$

It describes the communication links between the converters.

5.7.2 Degree Matrix

The degree matrix is

$$\boxed{D = \mathrm{diag}\{d_1, d_2, \ldots, d_N\}}$$

where

$$\boxed{d_i = \sum_{j=1}^{N} a_{ij}}$$

It represents the total communication weight associated with each agent.

5.7.3 Laplacian Matrix

The Laplacian matrix is defined as

$$\boxed{L = D - A}$$

and plays an important role in consensus control.

The Laplacian matrix captures the relative differences between neighbouring agents and is therefore fundamental to the mathematical formulation of cooperative synchronization.

5.8 Pinning Matrix

In a leader-follower architecture, some follower agents are directly connected to the leader.

This connection is represented using the **pinning gain** p_i.

The pinning matrix is

$$\boxed{P = \mathrm{diag}\{p_1, p_2, \ldots, p_N\}}$$

where p_i represents the strength of the connection between the leader and follower i.

If a converter is directly connected to the reference source,

$$\boxed{p_i > 0}$$

If it is not directly connected,

$$\boxed{p_i = 0}$$

The combined matrix

$$\boxed{L + P}$$

therefore contains information about both:

- communication among the follower converters,
- connection of followers to the reference/leader.

For the assumed connected undirected graph, the source formulation requires

$$\boxed{L + P > 0}$$

5.9 Output-Current Information Exchange

The most important information exchanged between the boost converters is their **output current**.

Let the output currents be

$$\boxed{y_1(t), y_2(t), \ldots, y_N(t)}$$

Each converter compares its output with the outputs of its neighbours.

For converter i, the cooperative information can be represented as

$$\boxed{\sum_{j=1}^{N} a_{ij}\left[y_j(t) - y_i(t)\right]}$$

This term represents the total influence of the neighbouring converters on converter i.

If the currents are already equal,

$$y_j(t) - y_i(t) = 0,$$

and therefore the consensus component becomes zero.

Hence,

$$\boxed{y_1 = y_2 = \cdots = y_N \quad \Rightarrow \quad \text{Consensus error} = 0}$$

This is the basic mechanism through which the converter agents synchronize their output currents.

5.10 Consensus Concept

The fundamental concept of **consensus** is that all agents eventually agree on a common value.

For the boost-converter system, consensus means

$$\boxed{y_1(t) = y_2(t) = \cdots = y_N(t)}$$

in the steady-state sense.

However, in this application, simple agreement between converters is not sufficient. The common value must also correspond to the desired reference.

Therefore, the actual objective is

$$\boxed{\begin{aligned} y_1(t) &\to r_0(t), \\ y_2(t) &\to r_0(t), \\ &\vdots \\ y_N(t) &\to r_0(t) \end{aligned}}$$

and consequently,

$$\boxed{y_1(t) \approx y_2(t) \approx \cdots \approx y_N(t) \approx r_0(t)}$$

This is known as **consensus with respect to the reference**.

5.11 Consensus Error

The tracking or consensus error of the i^{th} boost converter can be defined as

$$\boxed{\xi_i(t) = y_i(t) - r_0(t)}$$

or, collectively,

$$\boxed{\xi(t) = \begin{bmatrix} \xi_1(t) \\ \xi_2(t) \\ \vdots \\ \xi_N(t) \end{bmatrix}}$$

The desired condition is

$$\boxed{\lim_{t \to \infty} \xi_i(t) = 0}$$

for every converter.

Therefore,

$$\boxed{\lim_{t \to \infty} [y_i(t) - r_0(t)] = 0}$$

This means that every converter eventually tracks the reference and the output currents become synchronized.

The source work uses the same reference-based consensus objective in its formal problem statement.

5.12 Cooperative Control Mechanism

The cooperative controller combines the information obtained from neighbouring converters with the reference information.

For converter i, the general cooperative signal can be expressed as

$$\boxed{e_i(t) = \sum_{j=1}^{N} a_{ij}\left[y_j(t) - y_i(t)\right] + p_i\left[r_0(t) - y_i(t)\right]}$$

The first term,

$$\sum_{j=1}^{N} a_{ij}\left[y_j - y_i\right],$$

is the **consensus term**.

It forces the converter output toward the outputs of its neighbours.

The second term,

$$p_i(r_0 - y_i),$$

is the **reference-tracking or pinning term**.

It forces the converter output toward the desired reference.

Therefore,

$$\boxed{\text{Cooperative Error} = \text{Consensus Error} + \text{Reference Error}}$$

This structure corresponds to the distributed control law used in the source work.

5.13 Current Synchronization Among Boost Converters

Current synchronization is particularly important because the boost converters are associated with independent degaussing loads.

Consider four boost converters with output currents

$$\boxed{i_{o1}, \quad i_{o2}, \quad i_{o3}, \quad i_{o4}}$$

Initially, because the converters may have different parameters and loads,

$$\boxed{i_{o1} \neq i_{o2} \neq i_{o3} \neq i_{o4}}$$

After the cooperative controller becomes active, the converters exchange current information and modify their control inputs.

The desired result is

$$\boxed{i_{o1} \to r_0}$$

$$\boxed{i_{o2} \to r_0}$$

$$\boxed{i_{o3} \to r_0}$$

$$\boxed{i_{o4} \to r_0}$$

Therefore,

$$\boxed{i_{o1} \approx i_{o2} \approx i_{o3} \approx i_{o4}}$$

while following the desired reference.

This synchronized behaviour ensures that the multiple converter agents operate collectively rather than as isolated systems.

5.14 Homogeneous and Heterogeneous Multi-Agent Systems

An important distinction in multi-agent control is between **homogeneous** and **heterogeneous** systems.

5.14.1 Homogeneous MAS

A multi-agent system is homogeneous when all agents possess identical dynamics:

$$\boxed{G_1(s) = G_2(s) = \cdots = G_N(s)}$$

In such a system, every converter would ideally have identical dynamic characteristics.

5.14.2 Heterogeneous MAS

A multi-agent system is heterogeneous when the individual agents have different dynamics:

$$\boxed{G_i(s) \neq G_j(s)}$$

for at least some $i \neq j$.

In the proposed boost-converter system, differences in parameters such as load resistance, inductance, capacitance, or operating conditions can lead to different converter dynamics.

Thus,

$$\boxed{\mathcal{G}(s) = \mathrm{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}}$$

represents a heterogeneous converter network.

The source paper explicitly distinguishes homogeneous and heterogeneous MASs and identifies nonidentical converter dynamics as the motivation for the considered heterogeneous setup.

5.15 Heterogeneous Plants With a Homogeneous Controller

One of the important ideas in this work is that the **plant can be heterogeneous while the controller remains homogeneous**.

That is,

$$\boxed{G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s)}$$

while

$$\boxed{C_1(s) = C_2(s) = \cdots = C_N(s) = C(s)}$$

This is highly desirable because it allows the same controller structure to be implemented across multiple converter agents even though their electrical characteristics are different.

The overall architecture can therefore be represented as

$$\begin{array}{ccccc} G_1(s) & G_2(s) & G_3(s) & \cdots & G_N(s) \\ \uparrow & \uparrow & \uparrow & & \uparrow \\ C(s) & C(s) & C(s) & \cdots & C(s) \end{array}$$

The source paper's formal problem statement similarly considers a heterogeneous plant with a homogeneous controller configuration.

5.16 Overall Cooperative-Control Structure

The complete operation of the heterogeneous boost-converter MAS can be summarized as

Reference $r_0(t)$

$\downarrow$

Leader / Reference Generator

$\downarrow$

Communication Network

$\downarrow$

Output-Current Information Exchange

$\downarrow$

Mixed NI+PR Cooperative Controller

$\downarrow$

$$\begin{array}{ccc} \text{Boost Converter 1} & \cdots & \text{Boost Converter N} \\ \downarrow & & \downarrow \\ y_1(t) & \cdots & y_N(t) \end{array}$$

$\downarrow$

Current Consensus

The controller continuously uses the differences between converter currents and the reference to generate appropriate control actions.

5.17 Key Difference Between Independent and Cooperative Control

The difference can be illustrated as follows.

5.17.1 Independent Control

$$\boxed{r_i \rightarrow C_i(s) \rightarrow G_i(s) \rightarrow y_i}$$

Each converter operates independently.

5.17.2 Cooperative Control

$$\boxed{\begin{array}{c} r_0 \\ \downarrow \\ \text{Cooperative Controller} \\ \uparrow \\ \text{Neighbouring Converter Information} \\ \downarrow \\ G_i(s) \rightarrow y_i \end{array}}$$

Here, each converter considers both its own output and information from neighbouring agents.

Therefore, cooperative control provides an additional layer of coordination that is particularly important for heterogeneous converter systems.

5.18 Summary

The heterogeneous boost-converter system can therefore be understood as a **network of interacting agents**. Each boost converter represents an individual agent with its own dynamics and output current. The agents exchange output-current information through a communication network represented by graph-theoretic concepts such as the adjacency matrix, Laplacian matrix, and pinning matrix.

The cooperative controller uses two fundamental pieces of information:

$$\boxed{\text{Neighbour-to-Neighbour Current Difference}}$$

and

$$\boxed{\text{Reference-to-Converter Current Difference}}$$

These produce the overall cooperative control action:

$$\boxed{\text{Consensus + Reference Tracking}}$$

leading to the desired condition

$$\boxed{\lim_{t\to\infty}[y_i(t) - r_0(t)] = 0, \qquad i = 1, 2, \dots, N}$$

Thus, the main objective of the multi-agent cooperative-control framework is to ensure that **heterogeneous DC/DC boost converters with different dynamics can work together as a coordinated system and achieve synchronized output-current tracking of a common time-varying reference**.

Preliminaries

6 Negative Imaginary (NI) Systems Theory

Negative Imaginary (NI) systems theory provides the theoretical foundation for the cooperative controller developed for the heterogeneous DC/DC converter system. In this report, the theory is particularly relevant because the average models of several DC/DC converters, including **boost converters**, can exhibit stable NI or Strictly Negative Imaginary (SNI) properties for suitable combinations of circuit parameters R, L, and C.

The main advantage of using NI theory is that it provides a frequency-domain framework for studying the stability of interconnected systems. This is useful for the present application because several heterogeneous boost converters are connected through a cooperative communication network and controlled using a common controller.

6.1 Introduction to Negative Imaginary Systems

A **Negative Imaginary (NI) system** is a class of dynamical systems whose frequency-domain response satisfies a particular phase relationship.

Consider a transfer function

$$G(s).$$

For frequency-domain analysis, substitute

$$s = j\omega,$$

giving

$$G(j\omega).$$

The NI property is related to the imaginary part of this frequency response. For a scalar system, the essential condition can be expressed as

$$\boxed{\mathrm{Im}[G(j\omega)] \leq 0}$$

over the relevant positive-frequency range, subject to the pole conditions of the NI definition.

For a matrix-valued transfer function, the corresponding condition is expressed using

$$j[G(j\omega) - G(j\omega)^*] \geq 0,$$

where $G(j\omega)^*$ denotes the conjugate transpose.

Equivalently, the imaginary part of the system satisfies the appropriate negative-imaginary condition.

The source paper uses the formal NI/SNI framework as the foundation of its cooperative-control analysis and refers to established NI definitions rather than reproducing them in the paper because of space limitations.

6.2 Strictly Negative Imaginary (SNI) Systems

A **Strictly Negative Imaginary (SNI) system** satisfies a stricter version of the NI condition.

For a scalar transfer function, the frequency-domain behaviour can be expressed conceptually as

$$\boxed{\mathrm{Im}[G(j\omega)] < 0, \qquad \omega > 0}$$

provided that the system satisfies the required pole and stability conditions.

The distinction can therefore be summarized as

$$\boxed{\begin{aligned} \text{NI:} \quad & \mathrm{Im}[G(j\omega)] \leq 0 \\ \text{SNI:} \quad & \mathrm{Im}[G(j\omega)] < 0 \end{aligned}}$$

for the applicable positive frequencies.

The SNI property is particularly important in this work because the individual converter plants are assumed to satisfy the SNI property.

For the heterogeneous converter network,

$$G_r(s) = \mathrm{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\},$$

each individual converter satisfies

$$\boxed{G_i(s) \in SNI}$$

The source paper states that the individual agents $G_i(s)$ are SNI and satisfy

$$G_i(0) > 0.$$

6.3 Frequency-Domain Interpretation

The NI property is most conveniently understood in the frequency domain.

Let

$$s = j\omega.$$

The transfer function becomes

$$G(j\omega) = \Re[G(j\omega)] + j\Im[G(j\omega)].$$

For an NI system, the imaginary component satisfies the negative-imaginary condition.

For an SNI system,

$$\mathrm{Im}[G(j\omega)] < 0$$

over the relevant frequency range.

Another way of interpreting the property is through the **phase angle**.

For an NI system, the phase contribution associated with its characteristic response lies within the appropriate lower-half-plane region. For SNI systems, the phase is strictly within that region for positive frequencies.

The source paper states that the characteristic loci of NI systems have phase angles in

$$[-\pi, 0]$$

while SNI systems have phase angles in

$$(-\pi, 0)$$

for $\omega > 0$.

Thus,

$$\boxed{\text{SNI system} \Rightarrow \text{Characteristic response lies in the lower-half-plane region}}$$

This frequency-domain property becomes extremely useful when analysing the stability of interconnected converter systems.

6.4 NI and SNI Properties in DC/DC Boost Converters

The application of NI theory to the present report is motivated by the behaviour of the average models of DC/DC converters.

The source paper notes that the average, or large-signal, models of several DC/DC converter topologies—including

$$\boxed{\text{buck}, \quad \text{boost}, \quad \text{buck-boost}}$$

can inherently satisfy stable NI or SNI properties for combinations of circuit parameters R, L, and C.

For this report, the converter topology is changed from buck to **boost**. Therefore, the relevant plant is a heterogeneous collection of boost converters whose average models are considered within the NI/SNI framework.

The general representation is

$$\boxed{G_r(s) = \operatorname{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}}$$

where

$$G_i(s)$$

represents the average model of the i^{th} boost converter.

The individual converters may have different parameters, resulting in

$$G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s).$$

Nevertheless, each individual plant is assumed to possess the required SNI characteristics.

6.5 Why the Average Model Is Used

A practical boost converter contains high-frequency switching behaviour. Direct analysis of the switching waveform can make the mathematical model complicated.

For control-system analysis, an **average model** is commonly used to represent the low-frequency behaviour of the converter.

The average model captures the dominant dynamics relevant to the control problem while avoiding the need to explicitly analyse every switching transition.

Thus, the control design can focus on

$$\boxed{\text{Average Boost-Converter Dynamics}}$$

rather than directly analysing the high-frequency switching waveform.

The source paper explicitly refers to the average model as the **large-signal model** when discussing the NI/SNI characteristics of DC/DC converters.

For the boost-converter version of this report, the actual boost average model and its transfer function will be derived separately in the mathematical-modelling section.

6.6 Positive DC Gain Requirement

An important condition in the proposed control framework is the value of the plant transfer function at zero frequency.

For each converter,

$$\boxed{G_i(0) > 0}$$

Here,

$$G_i(0)$$

represents the DC or zero-frequency gain of the i^{th} converter.

For the complete heterogeneous plant,

$$G_r(0) > 0$$

is required in the sense specified by the theoretical formulation.

This condition is important because it contributes to the location of the characteristic loci near zero frequency and is subsequently used in the stability proof.

The source paper explicitly assumes

$$G_i(0) > 0$$

for every converter agent.

6.7 NI Versus SNI

The distinction between NI and SNI can be summarized as follows:

Property	**NI System**	**SNI System**
Imaginary response	Non-positive	Strictly negative
Phase behaviour	Can reach boundary	Strictly inside NI region
Frequency response	NI condition	Stronger NI condition
Stability analysis	Useful	Particularly useful for feedback interconnection
Present application	Controller/plant theory	Converter plant

Conceptually,

$$\boxed{SNI \subset NI}$$

because an SNI system satisfies a stricter form of the NI condition.

The source work specifically considers SNI converter agents and develops a controller with a mixed NI+PR property.

6.8 Why NI Theory Is Suitable for the Proposed System

NI theory is suitable for the heterogeneous boost-converter system for several reasons.

1. Converter Models Can Possess SNI Properties

The average models of DC/DC converters, including boost converters, can possess stable NI/SNI characteristics.

Therefore, the converter dynamics naturally fit within the NI control framework.

2. It Can Handle Heterogeneous Agents

The converter agents do not have to possess identical dynamics.

Thus,

$$G_i(s) \neq G_j(s)$$

can be allowed while the cooperative control framework is maintained.

The source paper specifically develops its approach for a **heterogeneous SNI multi-agent system**.

3. It Supports Distributed Output Feedback

The proposed approach uses output information rather than requiring complete state information.

This is advantageous when only converter output currents are readily measurable.

4. It Supports Time-Varying Reference Tracking

The control framework is designed to achieve

$$\boxed{\lim_{t\to\infty}[y_i(t) - r_0(t)] = 0}$$

for the specified bounded reference signal.

5. It Provides a Frequency-Domain Stability Framework

The NI property can be combined with characteristic-loci analysis to investigate the stability of the interconnected system.

6.9 NI-Based Cooperative Control

The key idea is not to use NI theory only for analysing one converter. Instead, the NI properties of the individual converters are incorporated into the **cooperative multi-agent control problem**.

Consider N heterogeneous boost converters:

$$G_1(s), G_2(s), \dots, G_N(s).$$

Each converter has its own output

$$y_i(s).$$

The converters exchange information through a communication network.

The resulting control architecture can be represented as

Reference $r_0(t)$

$\downarrow$

Communication Network

$\downarrow$

NI-Based Cooperative Controller

$\downarrow$

Heterogeneous Boost Converters

$\downarrow$

$y_1, y_2, \ldots, y_N$

The controller uses both

$$y_j - y_i$$

and

$$r_0 - y_i$$

to coordinate the converter agents.

6.10 Relationship Between NI Theory and Stability

One of the most important reasons for using NI theory is its relationship with feedback stability.

Suppose an SNI plant is interconnected with an appropriate NI controller.

The NI framework provides conditions under which the feedback interconnection can remain stable.

The source paper explicitly uses the fact that the interconnection of an SNI plant and an NI controller can be stable under suitable conditions. It then extends this idea by searching for controllers that have both NI and PR characteristics.

For the proposed system,

$$\boxed{\text{SNI Boost-Converter Plant} + \text{Mixed NI+PR Controller}}$$

forms the fundamental feedback relationship.

The controller is subsequently selected in the form

$$\boxed{C(s) = \frac{k}{s} C_p(s)}$$

where $k > 0$ and $C_p(s)$ satisfies the required properties.

6.11 Characteristic-Loci Interpretation

The NI properties also help constrain the location of the characteristic loci in the complex plane.

For an SNI system, the phase satisfies

$$\boxed{\angle G_i(j\omega) \in (-\pi, 0), \qquad \omega > 0}$$

Therefore, its frequency response is restricted to the lower-half-plane region.

For a network of converters, the eigenvalues of the resulting loop transfer-function matrix form the **characteristic loci**.

The source paper notes that the characteristic loci of NI/SNI multi-agent systems occupy specific regions of the complex plane and uses this property in the subsequent stability proof.

This gives the following conceptual chain:

SNI Converter $\longrightarrow$ Restricted Phase Behaviour $\longrightarrow$ Restricted Characteristic Loci $\longrightarrow$ Stability Analysis

6.12 NI Theory and Heterogeneous Multi-Agent Systems

For a homogeneous multi-agent system,

$$G_1(s) = G_2(s) = \cdots = G_N(s).$$

However, the present application considers

$$\boxed{G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s)}$$

because the boost converters can have different electrical and load parameters.

Therefore,

$$G_r(s) = \text{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}$$

represents a heterogeneous SNI multi-agent plant.

A significant feature of the proposed method is that the **plant is heterogeneous while the controller is homogeneous**:

$$\boxed{G_i(s) \neq G_j(s)}$$

but

$$\boxed{C_i(s) = C_j(s) = C(s)}$$

The source paper identifies this heterogeneous-plant/homogeneous- controller combination as the central setting of its theoretical result.

6.13 Advantages of NI-Based Control

The NI-based approach offers several advantages for the proposed boost-converter system.

6.13.1 Natural Compatibility With Converter Dynamics

The average models of several DC/DC converters can possess NI/SNI properties, making NI theory naturally applicable.

6.13.2 Frequency-Domain Stability Analysis

The approach allows stability to be investigated through frequency-domain properties and characteristic loci.

6.13.3 Compatibility With Heterogeneous Systems

Different converter dynamics can be handled within the same cooperative framework.

6.13.4 Output-Feedback Implementation

The controller can operate using output information, avoiding the requirement for complete state feedback. The source paper explicitly highlights this as an advantage of the proposed scheme.

6.13.5 Cooperative Current Synchronization

The NI-based controller can be incorporated into a distributed consensus structure so that multiple converter currents converge toward the same reference.

6.13.6 Time-Varying Reference Capability

The framework accommodates a bounded time-varying reference with a finite steady-state value.

6.14 Relationship Between Phase Characteristics and Stability

The phase characteristics of NI systems are central to the stability analysis.

For the SNI converter plants,

$$\boxed{\angle G_i(j\omega) \in (-\pi, 0)}$$

for positive frequencies.

The mixed NI+PR controller has a phase behaviour that allows the combined loop transfer function to remain within a specified region of the characteristic-loci plane.

The overall loop transfer function takes the form

$$\boxed{L(s) = \frac{kC_p(s)}{s}(L+P)G_r(s)}$$

in the formulation used by the source paper.

The characteristic loci are then obtained from the eigenvalues of this loop transfer-function matrix:

$$\boxed{\gamma_i(s) = \lambda_i[L(s)]}$$

The objective of the stability analysis is to show that these characteristic loci do not encircle the critical point

$$\boxed{-\frac{1}{k} + j0}$$

under the prescribed gain condition.

The source paper establishes that there exists a finite $k^* > 0$ such that

$$\boxed{0 < k \leq k^*}$$

ensures asymptotic stability of the proposed closed-loop system.

6.15 Overall NI-Based Control Concept

The role of NI theory in the complete boost-converter system can be summarized as

Average Model of Boost Converter

$\downarrow$

SNI Property

$\downarrow$

Heterogeneous SNI Multi-Agent Plant

$\downarrow$

Mixed NI+PR Controller

$\downarrow$

Characteristic Loci Analysis

$\downarrow$

Asymptotic Stability

$\downarrow$

Consensus and Reference Tracking

Therefore, NI theory serves as the **theoretical bridge between the physical dynamics of the boost converters and the stability of the cooperative control system**.

6.16 Summary

The Negative Imaginary systems framework provides an appropriate theoretical basis for the cooperative control of heterogeneous DC/DC boost converters. The average models of DC/DC converters, including boost converters, can exhibit stable NI/SNI properties for appropriate R, L, and C parameter combinations.

The essential concepts can be summarized as

$$\boxed{\text{NI} \Rightarrow \text{Non-positive imaginary response}}$$

and

$$\boxed{\text{SNI} \Rightarrow \text{Strictly negative imaginary response}}$$

with the SNI boost-converter plants satisfying

$$\boxed{G_i(0) > 0}$$

These frequency-domain properties restrict the location of the characteristic loci and can consequently be used to establish closed-loop stability.

For the proposed heterogeneous boost-converter system, the overall strategy is therefore

$$\boxed{\underbrace{\text{Heterogeneous SNI Boost Converters}}_{\text{Plant}} + \underbrace{\text{Mixed NI+PR Controller}}_{\text{Controller}} + \underbrace{\text{Communication Network}}_{\text{Cooperation}} \rightarrow \text{Stable Current Consensus}}$$

This provides the theoretical foundation for the **mixed NI+PR cooperative controller** introduced in the next section.

7 Positive Real (PR) Systems Theory

Positive Real (PR) systems theory provides an important frequency-domain framework for analysing dynamical systems and their interconnections. PR systems are widely used in control theory and stability analysis because their frequency-domain properties provide useful information about the behaviour of interconnected systems.

In this report, PR theory is primarily introduced as a **frequency-domain concept** that will later be used in the analysis of the proposed controller.

7.1 Introduction to Positive Real Systems

A **Positive Real (PR) system** is a class of dynamical systems whose frequency-domain response satisfies a non-negative real-part condition.

Consider a transfer function

$$G(s).$$

For frequency-domain analysis, the Laplace variable is replaced by

$$\boxed{s = j\omega}$$

giving

$$G(j\omega).$$

A scalar transfer function is Positive Real when its frequency response satisfies the fundamental condition

$$\boxed{\mathrm{Re}\left[G(j\omega)\right] \geq 0}$$

for the relevant frequencies, together with the required pole conditions.

Therefore, the real part of the frequency response of a PR system is non-negative.

7.2 Frequency-Domain Interpretation

A complex transfer function can be written as

$$\boxed{G(j\omega) = \mathrm{Re}[G(j\omega)] + j\,\mathrm{Im}[G(j\omega)]}$$

For a PR system,

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

Therefore, the frequency response lies in the **right half of the complex plane**, subject to the formal PR-system conditions.

This can be represented as

$$\text{Positive Real System} \quad \Longrightarrow \quad \mathrm{Re}[G(j\omega)] \geq 0$$

Thus, the PR property primarily places a restriction on the **real part** of the system's frequency response.

7.3 Mathematical Definition

For a real-rational transfer function $G(s)$, the Positive Real property requires appropriate analyticity and frequency-domain conditions.

In the scalar case, the important condition can be summarized as

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

for frequencies for which the frequency response is defined.

The transfer function must also satisfy the appropriate pole conditions associated with Positive Realness.

For a matrix-valued transfer function $G(s)$, the corresponding condition is expressed in terms of the Hermitian part:

$$\boxed{G(j\omega) + G(j\omega)^* \geq 0}$$

where $G(j\omega)^*$ denotes the conjugate transpose.

Equivalently,

$$\mathrm{Re}[G(j\omega)] \geq 0$$

in the appropriate matrix sense.

7.4 Strictly Positive Real Systems

A stronger form of the Positive Real property is known as **Strictly Positive Real (SPR)**.

For the scalar case, the distinction can be expressed as

$$\boxed{\text{PR:} \qquad \mathrm{Re}[G(j\omega)] \geq 0}$$

whereas the strict condition is conceptually

$$\boxed{\text{SPR:} \qquad \mathrm{Re}[G(j\omega)] > 0}$$

Thus, an SPR system satisfies a stronger positivity condition than a PR system.

The relationship can be represented as

$$\boxed{\text{SPR} \subset \text{PR}}$$

The strictness of the inequality provides additional margin from the boundary of the right-half-plane region.

7.5 Geometrical Interpretation

The Positive Real property can be understood using the complex plane.

Let

$$\boxed{G(j\omega) = a + jb}$$

Here,

$$a = \mathrm{Re}[G(j\omega)]$$

and

$$b = \mathrm{Im}[G(j\omega)].$$

For a PR system,

$$\boxed{a \geq 0}$$

Therefore, the frequency response is restricted to the right-half-plane region.

This can be represented as

$$\mathrm{Re}[G(j\omega)] \geq 0 \quad \Longrightarrow \quad G(j\omega) \text{ lies in the right-half plane}$$

This provides a convenient graphical interpretation of Positive Realness.

7.6 Magnitude and Phase of a PR System

The frequency response can also be represented in polar form as

$$\boxed{G(j\omega) = |G(j\omega)|e^{j\angle G(j\omega)}}$$

where

$$|G(j\omega)|$$

is the magnitude and

$$\angle G(j\omega)$$

is the phase angle.

Since a PR system has a non-negative real part,

$$\mathrm{Re}[G(j\omega)] \geq 0,$$

its frequency response is constrained to the right-half-plane.

For a nonzero scalar response, this corresponds to the phase lying within the appropriate range around the positive real axis.

Thus, Positive Realness imposes both a **geometrical restriction** and a **frequency-domain constraint** on the system response.

7.7 PR Systems and Stability

Positive Real systems are important in control because their frequency-domain properties can be used to establish stability of interconnected systems.

Consider two interconnected systems. If the systems satisfy appropriate positive-realness and stability conditions, their interconnection can possess desirable stability properties.

Therefore, the general concept is

Positive Real Property $\longrightarrow$ Frequency-Domain Constraint $\longrightarrow$ Stability Analysis

This makes PR theory useful for analysing feedback systems and dynamical interconnections.

7.8 PR Systems in Control Applications

Positive Real theory is useful in several control-system contexts, particularly when analysing systems that exchange energy or interact through feedback.

The PR property provides information about whether the system's frequency response remains within the required region of the complex plane.

This is useful for:

- feedback stability analysis,
- interconnected dynamical systems,
- frequency-domain controller analysis,
- passive-system analysis, and
- stability verification.

In the present report, PR theory is primarily introduced as a **frequency-domain concept** that will later be used in the analysis of the proposed controller.

7.9 PR Property and Transfer Functions

Suppose the transfer function of a system is

$$\boxed{G(s) = \frac{N(s)}{D(s)}}$$

To examine its Positive Real property, the frequency response is obtained by substituting

$$s = j\omega.$$

Therefore,

$$\boxed{G(j\omega) = \frac{N(j\omega)}{D(j\omega)}}$$

The real part is then examined:

$$\boxed{\mathrm{Re}[G(j\omega)]}$$

If the required PR conditions are satisfied,

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

Hence, the frequency response remains within the required right-half-plane region.

7.10 Importance of PR Theory for the Present Report

For the development of the proposed control strategy, Positive Real theory provides a mathematical framework for describing a particular class of frequency responses.

The key property is

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

This condition restricts the frequency response of the system and provides useful information for subsequent stability analysis.

The PR concept will be particularly important when analysing the controller frequency response and its interaction with the converter plant.

7.11 Summary

A **Positive Real system** is a system whose frequency-domain response satisfies a non-negative real-part condition, together with the associated pole and analyticity requirements.

The fundamental scalar condition is

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

A stricter form is the Strictly Positive Real property:

$$\boxed{\mathrm{Re}[G(j\omega)] > 0}$$

Therefore,

PR $\longrightarrow$ Non-negative real part $\longrightarrow$ Right-half-plane frequency response

and

$$\boxed{\text{SPR} \quad \longrightarrow \quad \text{Strictly positive real part}}$$

The main points to remember are:

1. **PR systems are defined primarily through their frequency-domain real-part property.**
2. **For a scalar system, the key condition is**
$$\mathrm{Re}[G(j\omega)] \geq 0.$$
3. **The frequency response of a PR system lies in the right-half-plane region**, subject to the formal PR conditions.
4. **SPR is a stricter form of PR.**
5. **PR theory is useful for frequency-domain stability and interconnection analysis.**
6. **The PR property will be used later when discussing the frequency-domain characteristics of the controller.**

8 Mixed Negative Imaginary + Positive Real (NI+PR) Property

The **mixed Negative Imaginary and Positive Real (NI+PR)** property is introduced to describe a class of systems whose frequency-domain behaviour is not restricted to only one of the two properties. A system may exhibit **Positive Real (PR) behaviour over one frequency range** and **Negative Imaginary (NI) behaviour over another frequency range**.

This concept is important for the proposed control design because it provides greater flexibility in shaping the frequency response of the controller while maintaining the properties required for the stability analysis.

8.1 Motivation for the Mixed NI+PR Property

The conventional NI and PR properties describe different frequency-domain characteristics.

For a PR system, the real part of the frequency response satisfies

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

For an NI system, the imaginary part satisfies the corresponding NI condition.

Instead of requiring a system to satisfy one property over the entire frequency range, the **mixed NI+PR** concept allows the frequency range to be divided into different regions.

Thus, the basic idea is

$$\boxed{\text{Mixed NI+PR} = \text{PR behaviour in some frequency intervals} + \text{NI behaviour in other frequency intervals}}$$

The source introduces this as a new class of systems motivated by earlier work on mixed NI+PR systems.

8.2 Definition of a Mixed NI+PR System

Let

$$\boxed{G(s) \in \mathbb{R}(s)^{n \times n}}$$

be a causal and proper transfer-function matrix (TFM) of a finite-dimensional system.

The system $G(s)$ is called **mixed NI+PR** if it satisfies the conditions given below.

The definition consists of three main requirements:

1. Analyticity in the open right-half plane.
2. PR behaviour in one frequency region and NI behaviour in another.
3. Appropriate conditions on pure imaginary-axis poles.

8.3 Condition 1 – Analyticity

The first requirement is

$$\boxed{\text{All elements of } G(s) \text{ are analytic in } \Re[s] > 0}$$

The condition

$$\boxed{\Re[s] > 0}$$

represents the **open right-half plane** of the complex s-plane.

Being analytic in this region means that $G(s)$ does not have poles in the open right-half plane.

Therefore, the transfer-function matrix must not contain unstable poles in

$$\boxed{\Re[s] > 0}$$

This condition is important because right-half-plane poles are associated with instability.

8.4 Frequency-Domain Partition

The most important part of the mixed NI+PR definition is that the frequency range is divided into two regions.

Let

$$\boxed{\Omega \subseteq \mathbb{R}_+}$$

The set Ω represents the frequency interval or collection of frequency intervals where the system exhibits **Positive Real behaviour**.

The remaining frequency region is

$$\boxed{\mathbb{R}_+ \setminus \Omega}$$

This region is where the system exhibits **Negative Imaginary behaviour**.

Therefore,

$$\boxed{\mathbb{R}_+ = \Omega \cup (\mathbb{R}_+ \setminus \Omega)}$$

with

$$\boxed{\begin{array}{ccl} \omega \in \Omega & \longrightarrow & \text{PR property} \\ \omega \in \mathbb{R}_+ \setminus \Omega & \longrightarrow & \text{NI property} \end{array}}$$

This frequency-dependent behaviour is the defining feature of the mixed NI+PR system.

8.5 PR Region

For

$$\omega \in \Omega,$$

where $j\omega$ is not a pole of $G(s)$, the definition requires

$$\boxed{G(j\omega) + G(j\omega)^* \succeq 0}$$

Here,

$$G(j\omega)^*$$

denotes the **conjugate transpose** of $G(j\omega)$.

For a scalar transfer function, this condition becomes

$$\boxed{G(j\omega) + G(j\omega)^* = 2\,\mathrm{Re}[G(j\omega)]}$$

Therefore,

$$G(j\omega) + G(j\omega)^* \succeq 0$$

is equivalent to

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

Hence, in the frequency region Ω, the system behaves as a **Positive Real system**.

8.6 NI Region

For the remaining frequencies,

$$\omega \in \mathbb{R}_+ \setminus \Omega,$$

where $j\omega$ is not a pole of $G(s)$, the definition requires

$$\boxed{j\,[G(j\omega) - G(j\omega)^*] \succeq 0}$$

For a scalar transfer function,

$$\boxed{G(j\omega) - G(j\omega)^* = 2j\,\mathrm{Im}[G(j\omega)]}$$

Therefore,

$$\boxed{j[G(j\omega) - G(j\omega)^*] = -2\,\mathrm{Im}[G(j\omega)]}$$

Thus, the condition

$$j[G(j\omega) - G(j\omega)^*] \succeq 0$$

corresponds to

$$\boxed{\mathrm{Im}[G(j\omega)] \leq 0}$$

Therefore, outside the PR region Ω, the system satisfies the **Negative Imaginary property**.

8.7 Combined Mathematical Definition

The complete frequency-domain condition can therefore be summarized as

$$\boxed{\begin{cases} G(j\omega) + G(j\omega)^* \succeq 0, & \omega \in \Omega, \\ j[G(j\omega) - G(j\omega)^*] \succeq 0, & \omega \in \mathbb{R}_+ \setminus \Omega. \end{cases}}$$

For a scalar system, this becomes

$$\boxed{\begin{cases} \mathrm{Re}[G(j\omega)] \geq 0, & \omega \in \Omega, \\ \mathrm{Im}[G(j\omega)] \leq 0, & \omega \in \mathbb{R}_+ \setminus \Omega. \end{cases}}$$

This equation captures the fundamental meaning of **mixed NI+PR behaviour**.

8.8 Why Frequency Intervals Are Used

A conventional PR system must maintain the PR property over the required frequency range, while an NI system must maintain the NI property over its required frequency range.

The mixed NI+PR concept provides additional flexibility because the system does not have to exhibit the same property everywhere.

For example,

$$\boxed{\begin{aligned} 0 < \omega < \omega_1 &\quad\longrightarrow\quad \text{PR} \\ \omega_1 < \omega < \omega_2 &\quad\longrightarrow\quad \text{NI} \\ \omega > \omega_2 &\quad\longrightarrow\quad \text{PR/NI according to the specified design} \end{aligned}}$$

The exact frequency regions depend on the transfer function being considered.

This allows the controller designer to shape different portions of the frequency response according to the requirements of the closed-loop system.

8.9 Pure Imaginary-Axis Poles

The third part of the definition concerns **pure imaginary poles**.

A pure imaginary pole occurs at

$$\boxed{s = j\omega_0}$$

Such a pole lies directly on the imaginary axis of the s-plane.

The mixed NI+PR definition imposes special conditions on these poles because their presence can significantly influence frequency-domain stability analysis.

The source definition gives two possible cases for pure imaginary poles.

8.10 Case 1 – Pure Imaginary Pole of the Matrix

Suppose

$$\boxed{s = j\omega_0, \qquad \omega_0 \in (0, \infty)}$$

is a pole of $G(s)$.

Then it must be **at most a simple pole**, meaning that the pole cannot have multiplicity greater than one.

Furthermore, the residue matrix

$$\boxed{\lim_{s \to j\omega_0} (s - j\omega_0) jG(s)}$$

must be **Hermitian and positive semidefinite**.

Therefore,

$$\boxed{\lim_{s \to j\omega_0} (s - j\omega_0) j G(s) \succeq 0}$$

with the additional requirement that the resulting matrix be Hermitian.

Here,

$$\boxed{\succeq 0}$$

means **positive semidefinite**.

This condition ensures that any pure imaginary-axis poles have the appropriate residue behaviour required by the mixed NI+PR property.

8.11 Case 2 – Pure Imaginary Pole of an Individual Element

The definition also considers a pure imaginary pole of an individual element of $G(s)$.

If

$$j\omega$$

is a pure imaginary pole of an element of $G(s)$, then it must be a **simple pole**, and

$$\boxed{\lim_{s \to j\omega} (s - j\omega) j G(s) \geq 0}$$

This condition ensures that the residue associated with the pole satisfies the required non-negative condition.

Therefore, the mixed NI+PR property does not simply specify the frequency response away from poles; it also carefully defines the allowable behaviour at imaginary-axis poles.

8.12 Meaning of Hermitian Matrix

The definition uses the term **Hermitian matrix**.

A matrix H is Hermitian if

$$\boxed{H = H^*}$$

where H^* is the conjugate transpose.

For a real symmetric matrix, this reduces to

$$\boxed{H = H^T}$$

Hermitian matrices are important in frequency-domain system theory because their eigenvalues are real, allowing positive-semidefinite conditions to be meaningfully defined.

8.13 Meaning of Positive Semidefinite

A matrix H is positive semidefinite if

$$\boxed{x^* H x \geq 0}$$

for every appropriate vector x.

It is commonly written as

$$\boxed{H \succeq 0}$$

Therefore, when the definition states that the residue matrix is positive semidefinite, it means that the residue satisfies the required non-negative quadratic-form condition.

8.14 Graphical Interpretation

The mixed NI+PR property can be visualized using the complex plane.

For the PR region,

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

so the frequency response is restricted to the right-half-plane region.

For the NI region,

$$\boxed{\mathrm{Im}[G(j\omega)] \leq 0}$$

so the frequency response satisfies the lower-half-plane imaginary-part condition.

Therefore, different frequency ranges can be associated with different regions of the complex plane:

$$\boxed{\begin{array}{c}\text{Frequency Region } \Omega \\ \Downarrow \\ \mathrm{Re}[G(j\omega)] \geq 0 \\ \Downarrow \\ \text{PR condition}\end{array}}$$

and

$$\boxed{\begin{array}{c}\text{Frequency Region } \mathbb{R}_+ \setminus \Omega \\ \Downarrow \\ \mathrm{Im}[G(j\omega)] \leq 0 \\ \Downarrow \\ \text{NI condition}\end{array}}$$

8.15 Examples of Mixed NI+PR Systems

The source provides three representative examples of mixed NI+PR systems.

The first example is

$$\boxed{G_1(s) = \frac{s+1}{(s+6)(s^2+4s+8)}}$$

The second example is

$$\boxed{G_2(s) = \frac{s+1}{s^2+8s+32}}$$

The third example is

$$\boxed{G_3(s) = \frac{s+1}{(s+10)^3}}$$

These systems demonstrate that transfer functions can satisfy the mixed NI+PR property even though their frequency-domain behaviour changes according to the relevant frequency interval.

8.16 Importance of the Mixed NI+PR Property

The mixed property provides an additional degree of freedom in controller design.

Instead of requiring

$$\boxed{\text{NI over the entire frequency range}}$$

or

$$\boxed{\text{PR over the entire frequency range}}$$

the controller can be designed according to

$$\boxed{\text{PR in one frequency region} + \text{NI in another frequency region}}$$

This is useful when the plant and controller have different frequency-domain characteristics.

For the proposed heterogeneous converter system, this flexibility is valuable because the converter dynamics must be combined with the cooperative controller while preserving the conditions required for stability.

8.17 Mixed NI+PR and Characteristic Loci

The mixed NI+PR property is particularly relevant to **characteristic-loci analysis**.

Suppose the closed-loop system has a loop transfer matrix

$$L(s).$$

Its characteristic loci are obtained from the eigenvalues

$$\boxed{\lambda_i[L(j\omega)]}$$

As frequency changes, the eigenvalues trace paths in the complex plane.

Because the system possesses different NI and PR properties in different frequency regions, the characteristic loci can be analysed separately in those regions.

Conceptually,

Frequency Region → NI/PR Property → Characteristic-Locus Behaviour → Stability

This is why the mixed NI+PR property is important for the subsequent stability analysis.

8.18 Role in the Boost-Converter Control System

For the boost-converter version of this report, the individual boost-converter plants are considered within the NI/SNI framework, while the controller will be designed using the mixed NI+PR concept.

The overall structure is

$$\boxed{\text{Heterogeneous Boost Converters} + \text{Mixed NI+PR Controller}}$$

The controller is intended to interact with the converter plants in such a way that the resulting characteristic loci satisfy the required stability conditions.

The important point is that **the mixed NI+PR property describes the controller's frequency-domain behaviour; it is not a replacement for the boost-converter model**. The boost-converter transfer functions must still be derived separately.

8.19 Key Mathematical Conditions

For quick reference, the mixed NI+PR property can be summarized by the following conditions.

8.19.1 Condition 1 – Analyticity

$$\boxed{G(s) \text{ is analytic for } \Re[s] > 0}$$

8.19.2 Condition 2 – PR Region

For

$$\omega \in \Omega,$$

$$\boxed{G(j\omega) + G(j\omega)^* \succeq 0}$$

or, for a scalar system,

$$\boxed{\mathrm{Re}[G(j\omega)] \geq 0}$$

8.19.3 Condition 3 – NI Region

For

$$\omega \in \mathbb{R}_+ \setminus \Omega,$$

$$\boxed{j[G(j\omega) - G(j\omega)^*] \succeq 0}$$

or, for a scalar system,

$$\boxed{\mathrm{Im}[G(j\omega)] \leq 0}$$

8.19.4 Condition 4 – Imaginary-Axis Poles

Any pure imaginary pole must satisfy the required **simple-pole and non-negative residue conditions**.

8.20 Overall Concept

The complete concept can be represented as

$$\begin{array}{c}
G(s) \in \mathbb{R}(s)^{n\times n} \\
\downarrow \\
\text{Analytic in } \Re[s] > 0 \\
\downarrow \\
\text{Frequency Range Divided} \\
\downarrow \\
\begin{array}{cc}
\omega \in \Omega & \omega \in \mathbb{R}_+ \setminus \Omega \\
\downarrow & \downarrow \\
\text{PR} & \text{NI} \\
\downarrow & \downarrow \\
G + G^* \succeq 0 & j(G - G^*) \succeq 0
\end{array} \\
\downarrow \\
\text{Mixed NI+PR System}
\end{array}$$

Thus, the defining idea is

$$\boxed{\text{Different frequency regions} \implies \text{Different frequency-domain properties}}$$

8.21 Summary

A **mixed NI+PR system** is a causal and proper finite-dimensional system whose frequency-domain behaviour satisfies the PR property over one set of frequencies and the NI property over the remaining frequencies.

The central definition is

$$\boxed{\begin{cases} G(j\omega) + G(j\omega)^* \succeq 0, & \omega \in \Omega, \\ j[G(j\omega) - G(j\omega)^*] \succeq 0, & \omega \in \mathbb{R}_+ \setminus \Omega. \end{cases}}$$

For scalar systems, these conditions become

$$\boxed{\begin{cases} \mathrm{Re}[G(j\omega)] \geq 0, & \omega \in \Omega, \\ \mathrm{Im}[G(j\omega)] \leq 0, & \omega \in \mathbb{R}_+ \setminus \Omega. \end{cases}}$$

In addition, $G(s)$ must be analytic in

$$\boxed{\Re[s] > 0}$$

and any pure imaginary-axis poles must satisfy the specified simple-pole and residue conditions.

The key idea for the remainder of the report is therefore

$$\boxed{\text{Mixed NI+PR} = \text{PR behaviour in selected frequencies} + \text{NI behaviour in the remaining frequencies}}$$

This property provides the theoretical basis for designing the **mixed NI+PR controller** that will later be applied to the heterogeneous **DC/DC boost-converter** system.

9 Characteristic Loci Analysis

Characteristic loci theory is used to establish the **closed-loop stability** of the proposed heterogeneous boost-converter system. It is the MIMO extension of the classical Nyquist stability criterion. Instead of analysing a single scalar loop transfer function, characteristic loci analyse the **eigenvalues of a matrix-valued loop transfer function**.

For the proposed system, this analysis is particularly useful because multiple heterogeneous boost converters are interconnected through the cooperative controller.

9.1 Characteristic Loci of a MIMO System

Consider a square loop transfer-function matrix

$$\boxed{L(s) \in \mathbb{R}(s)^{n\times n}}$$

The characteristic loci are defined using the eigenvalues of $L(s)$:

$$\boxed{\gamma_i(s) = \lambda_i[L(s)], \qquad i = 1, 2, \ldots, n}$$

When

$$s = j\omega,$$

the eigenvalues become

$$\boxed{\gamma_i(j\omega) = \lambda_i[L(j\omega)]}$$

As the frequency ω varies from 0 to ∞, each eigenvalue traces a curve in the complex plane. These curves are called the **characteristic loci**.

Thus,

$$\boxed{\omega : 0 \to \infty \quad \Longrightarrow \quad \lambda_i[L(j\omega)] \text{ traces a characteristic locus}}$$

9.2 Critical Point

The characteristic-loci stability test is based on the critical point

$$\boxed{-1 + j0}$$

The closed-loop characteristic equation is

$$\boxed{\det[I + L(s)] = 0}$$

Using the eigenvalues of $L(s)$,

$$\boxed{\det[I + L(s)] = \prod_{i=1}^{n} [1 + \lambda_i(L(s))]}$$

Therefore, the closed-loop system becomes unstable if one of the characteristic loci passes through or appropriately encircles

$$\boxed{-1 + j0}$$

Consequently, the central question in the stability proof is

$$\boxed{\text{Do the characteristic loci encircle the critical point } -1 + j0?}$$

9.3 Nyquist Interpretation

The characteristic-loci approach is a generalization of the classical Nyquist criterion.

For a SISO system, we normally study

$$\boxed{L(j\omega)}$$

For a MIMO system, we instead study

$$\boxed{\lambda_i[L(j\omega)]}$$

Therefore,

$$\boxed{\text{SISO Nyquist} \quad \longrightarrow \quad L(j\omega)}$$

whereas

$$\boxed{\text{MIMO Characteristic Loci} \quad \longrightarrow \quad \lambda_i[L(j\omega)]}$$

The stability condition is based on the number of encirclements of the critical point.

The theorem used in the analysis states that the net number of counterclockwise encirclements of

$$-1 + j0$$

by the characteristic loci is related to the number of poles of $L(s)$ in the open right-half plane.

If $L(s)$ has no poles in the open RHP, the net number of encirclements must be zero for stability.

9.4 Characteristic-Loci Region for Mixed NI+PR Systems

For a mixed NI+PR system, the characteristic loci are restricted to particular regions of the complex plane.

The figure in the source divides the characteristic-loci behaviour into different regions associated with the frequency intervals.

The important regions used in the proof are

$$\boxed{\psi_0, \qquad \psi_{\pm I}, \qquad \psi_R}$$

These regions correspond to different frequency-domain behaviours of the mixed NI+PR system.

The stability proof therefore examines the characteristic loci separately in these three cases.

9.5 Case I – $s \in \psi_0$

The first case considers the behaviour of the characteristic loci near

$$\boxed{s = 0}$$

This is the **low-frequency or zero-frequency region**.

The behaviour near $s = 0$ is important because the proposed controller contains an integral term,

$$\boxed{\frac{k}{s}}$$

Therefore, as

$$s \to 0,$$

the magnitude of the controller can become very large.

For the controller

$$\boxed{C(s) = \frac{k}{s} C_p(s)}$$

we have

$$\boxed{\left|\frac{k}{s}\right| \to \infty \qquad \text{as } s \to 0}$$

Consequently, the characteristic loci can move toward very large magnitude values.

9.5.1 Infinite-Magnitude Behaviour

As

$$s \to 0,$$

the loop transfer function contains the factor

$$\frac{k}{s}.$$

Therefore,

$$\boxed{|L(s)| \to \infty}$$

under the relevant assumptions.

Hence,

$$\boxed{s \to 0 \quad \Longrightarrow \quad |\lambda_i[L(s)]| \to \infty}$$

This means that the characteristic loci begin at or approach infinity rather than beginning near the critical point

$$-1 + j0.$$

This is important because it prevents the low-frequency portion of the locus from directly passing through the critical point.

9.6 Positive Real-Axis Crossover

Although the characteristic loci can have very large magnitude near $s = 0$, their direction in the complex plane is also important.

The analysis considers the location of the characteristic loci on the **positive real axis**.

The relevant behaviour can be expressed conceptually as

$$\boxed{\lambda_i[L(j\omega)] \to \text{positive real-axis region}}$$

near the low-frequency boundary.

Therefore, the characteristic loci do not approach the critical point from the negative real direction.

Recall that the critical point is

$$-1 + j0.$$

The positive real axis lies on the opposite side of the origin:

$$+1, +2, +3, \ldots$$

rather than

$$-1, -2, -3, \ldots.$$

Consequently, the low-frequency characteristic loci do not produce an encirclement of

$$-1 + j0.$$

9.7 Interpretation of Case I

The first case can therefore be summarized as

$$\boxed{\begin{array}{c} s \in \psi_0 \\ \Downarrow \\ s \to 0 \\ \Downarrow \\ |\lambda_i[L(s)]| \to \infty \\ \Downarrow \\ \text{Positive-real-axis region} \end{array}}$$

Therefore, they remain away from the critical point

$$-1 + j0.$$

This establishes that the portion of the characteristic locus associated with the low-frequency region does not cause a critical-point encirclement.

9.8 Case II – $s \in \psi_{\pm I}$

The second case considers the main **frequency-domain behaviour** of the characteristic loci.

Here,

$$\boxed{s \in \psi_{\pm I}}$$

This region corresponds to the frequency intervals where the NI/PR properties impose restrictions on the characteristic-locus location.

The analysis examines three important quantities:

1. Magnitude,
2. Phase,
3. Location of the characteristic locus.

9.9 Magnitude in Case II

For

$$s = j\omega,$$

the characteristic eigenvalues are

$$\lambda_i[L(j\omega)].$$

Each eigenvalue can be represented in polar form:

$$\boxed{\lambda_i[L(j\omega)] = |\lambda_i[L(j\omega)]|e^{j\angle\lambda_i[L(j\omega)]}}$$

Therefore, the magnitude is

$$\boxed{|\lambda_i[L(j\omega)]|}$$

The magnitude determines the radial distance of the characteristic locus from the origin.

The phase,

$$\boxed{\angle\lambda_i[L(j\omega)]}$$

determines the angular position of the locus.

Together,

$$\boxed{\text{Magnitude} + \text{Phase} \quad \longrightarrow \quad \text{Location of Characteristic Locus}}$$

9.10 Phase Behaviour in Case II

The mixed NI+PR properties constrain the phase behaviour of the characteristic loci.

For an NI-type response, the relevant phase lies in the lower-half-plane region, while the PR property restricts the response according to its positive-real condition.

The figure provided in the source illustrates that the characteristic loci of mixed NI+PR systems are confined to the shaded regions.

In particular, the loci occupy the regions associated with the first, third, and fourth quadrants.

Thus,

$$\boxed{\text{Characteristic loci avoid the critical region around } -1 + j0}$$

This restriction is the main reason why the characteristic-loci analysis can be used to establish stability.

9.11 Green Stability Region

The green region shown in the characteristic-loci figure represents an important **stable region** of the complex plane.

The characteristic loci are constrained to remain within the regions allowed by the mixed NI+PR properties.

The critical point,

$$-1 + j0,$$

lies in the negative-real-axis region.

The important observation is that the characteristic loci do not freely move through this critical point.

Therefore, as frequency varies,

$$\boxed{\omega : 0 \to \infty}$$

the characteristic loci cannot form an encirclement of

$$-1 + j0$$

under the conditions established in the proof.

9.12 Case II – Main Stability Argument

The second case can be summarized as

$$\boxed{\begin{array}{c} s \in \psi_{\pm I} \\ \Downarrow \\ \text{Frequency-domain NI/PR restrictions} \\ \Downarrow \\ \text{Restricted magnitude and phase} \\ \Downarrow \\ \text{Characteristic loci remain in permitted regions} \\ \Downarrow \\ -1 + j0 \text{ is not crossed or encircled} \end{array}}$$

This is the central frequency-domain part of the characteristic-loci proof.

9.13 Case III – $s \in \psi_R$

The third case considers the behaviour as

$$\boxed{s \to \infty}$$

This corresponds to the **high-frequency region**.

At very high frequencies, the transfer functions of practical dynamical systems generally exhibit decreasing magnitude when they are proper.

For the loop transfer function $L(s)$, the characteristic eigenvalues therefore approach the origin under the conditions used in the analysis:

$$\boxed{\lambda_i[L(s)] \to 0 \qquad \text{as } s \to \infty}$$

9.14 Characteristic Loci Near the Origin

Since

$$\lambda_i[L(s)] \to 0,$$

the high-frequency characteristic loci approach

$$\boxed{0 + j0}$$

Therefore, the characteristic loci terminate or approach the origin as frequency tends to infinity.

This is important because the critical point is

$$-1 + j0,$$

which is one unit to the left of the origin.

Hence,

$$\boxed{0 + j0 \neq -1 + j0}$$

The high-frequency part of the characteristic locus therefore does not pass through the critical point.

9.15 Interpretation of Case III

The third case can be summarized as

$$\boxed{\begin{array}{c} s \in \psi_R \\ \Downarrow \\ s \to \infty \\ \Downarrow \\ |\lambda_i[L(s)]| \to 0 \\ \Downarrow \\ \lambda_i[L(s)] \to 0 + j0 \end{array}}$$

Thus, the characteristic loci return toward the origin at high frequency rather than moving toward

$$-1 + j0.$$

9.16 Combining the Three Cases

The complete characteristic-loci path can now be understood by combining all three cases.

9.16.1 Case I

$$s \in \psi_0$$

Near zero frequency,

$$\boxed{|\lambda_i[L(s)]| \to \infty}$$

The characteristic loci approach the appropriate positive-real-axis region.

9.16.2 Case II

$$s \in \psi_{\pm I}$$

At finite frequencies, the mixed NI+PR properties restrict the magnitude and phase of the characteristic loci.

The loci remain in the permitted regions of the complex plane.

9.16.3 Case III

$$s \in \psi_R$$

As

$$s \to \infty,$$

the characteristic loci approach the origin:

$$\boxed{\lambda_i[L(s)] \to 0}$$

9.17 Complete Characteristic-Loci Path

The three cases can therefore be represented as

$$
\boxed{\begin{array}{c}
s \to 0 \\
\Downarrow \\
|\lambda_i[L(s)]| \to \infty \\
\Downarrow \\
\text{Case I: } \psi_0 \\
\Downarrow \\
\text{Finite-frequency characteristic loci} \\
\Downarrow \\
\text{Case II: } \psi_{\pm I} \\
\Downarrow \\
\text{Restricted magnitude and phase} \\
\Downarrow \\
\text{Case III: } \psi_R \\
\Downarrow \\
s \to \infty \\
\Downarrow \\
\lambda_i[L(s)] \to 0
\end{array}}
$$

The characteristic loci therefore form a continuous trajectory from the appropriate low-frequency region toward the origin without producing the required encirclement of

$$-1 + j0.$$

9.18 Why the Critical Point Is Not Encircled

The final objective is to establish

$$\boxed{N = 0}$$

where N represents the net number of counterclockwise encirclements of the critical point

$$-1 + j0.$$

The three cases collectively establish this result.

9.18.1 From Case I

The low-frequency locus is located away from the critical point and approaches the appropriate positive-real region.

9.18.2 From Case II

The finite-frequency locus remains within the regions imposed by the mixed NI+PR characteristics.

9.18.3 From Case III

The high-frequency locus approaches the origin rather than the critical point.

Therefore, the complete characteristic locus cannot form a closed encirclement around

$$-1 + j0.$$

Hence,

$$\boxed{N = 0}$$

9.19 Connection With the Nyquist Stability Criterion

The characteristic-loci theorem states that

$$\boxed{N = P - Z}$$

under the corresponding Nyquist convention, where:

- N = net encirclements of the critical point,
- P = number of open-loop RHP poles,
- Z = number of closed-loop RHP poles.

In the formulation shown in the source, the net counterclockwise encirclement count is equated to the number of RHP poles of the loop transfer function, with the stated convention.

If

$$P = 0,$$

then the required net encirclement is

$$\boxed{N = 0}$$

Since the characteristic-loci analysis establishes

$$N = 0,$$

the closed-loop system has no RHP poles:

$$\boxed{Z = 0}$$

Therefore, the closed-loop system is **asymptotically stable**, provided the other assumptions of the theorem are satisfied.

9.20 Characteristic-Loci Analysis for the Boost-Converter System

For the present report, the plant consists of heterogeneous **boost converters**.

The overall plant can be represented as

$$\boxed{G_r(s) = \mathrm{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}}$$

The cooperative controller is incorporated into the loop transfer matrix.

In the framework used by the source,

$$\boxed{L(s) = \frac{kC_p(s)}{s}(L + P)G_r(s)}$$

For the boost-converter version,

$$G_i(s)$$

must represent the **boost-converter transfer functions** derived from the boost-converter model.

The characteristic loci are then

$$\boxed{\gamma_i(s) = \lambda_i[L(s)]}$$

The stability analysis examines

$$\gamma_i(j\omega)$$

for

$$\boxed{0 \leq \omega < \infty}$$

9.21 Role of the Gain k

The controller gain k affects the location of the characteristic loci.

The loop transfer function is

$$L(s) = \frac{kC_p(s)}{s}(L + P)G_r(s).$$

Therefore,

$$\boxed{k \uparrow \implies |L(j\omega)| \uparrow}$$

in the corresponding frequency range.

An excessively large gain can cause the characteristic loci to move toward the critical point.

Consequently, the stability analysis establishes the existence of an allowable gain range of the form

$$\boxed{0 < k \leq k^*}$$

The source theorem gives this finite upper bound on k as the condition ensuring asymptotic stability of the closed-loop system.

9.22 Physical Meaning for the Converter System

The characteristic-loci analysis is not merely a mathematical exercise. It determines whether the cooperative controller can safely coordinate the multiple boost converters.

The overall relationship is

$$\boxed{\text{Boost Converter Dynamics} + \text{Communication Network} + \text{Cooperative Controller}}$$

$$\Downarrow$$

$$\boxed{\text{Loop Transfer Matrix } L(s)}$$

$$\Downarrow$$

$$\boxed{\lambda_i[L(j\omega)]}$$

$$\Downarrow$$

$$\boxed{\text{Characteristic Loci}}$$

$$\Downarrow$$

$$\boxed{\text{No Encirclement of } -1+j0}$$

$$\Downarrow$$

$$\boxed{\text{Closed-Loop Stability}}$$

Thus, characteristic-loci theory provides the mathematical connection between the **frequency-domain properties of the controller and converters** and the **stability of the complete cooperative system**.

9.23 Summary of the Three Cases

Case	Frequency Region	Main Behaviour	Stability Significance
Case I	$s \in \psi_0$	$s \to 0$, characteristic loci have infinite magnitude	Loci remain away from $-1+j0$
Case II	$s \in \psi_{\pm I}$	Finite-frequency magnitude and phase are restricted by NI/PR properties	Loci remain in permitted regions
Case III	$s \in \psi_R$	$s \to \infty$, loci approach the origin	Loci do not approach $-1+j0$

The complete argument is therefore

Case I: $s \to 0$

$\Downarrow$

Loci start away from critical point

$\downarrow$

Case II: finite frequencies

$\Downarrow$

NI/PR properties restrict loci

$\downarrow$

Case III: $s \to \infty$

$\Downarrow$

Loci approach origin

$\downarrow$

$\boxed{\text{No encirclement of } -1 + j0}$

$\downarrow$

$\boxed{N = 0}$

$\downarrow$

$\boxed{\text{Closed-loop asymptotic stability}}$

Therefore, the **three-case characteristic-loci analysis** establishes that the characteristic loci of the proposed mixed NI+PR cooperative system do not encircle the critical point $-1 + j0$.

Under the theorem's assumptions, this gives the required stability result for the closed-loop heterogeneous converter system.

10 Graph Theory for Cooperative Control

Graph theory provides the mathematical framework for representing the **communication network** among the multiple boost-converter agents. Since the converters must exchange output-current information to achieve cooperative current synchronization, it is necessary to define which converter can communicate with which other converter.

The communication network is represented by a graph

$$\boxed{\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})}$$

where $\mathcal{V}$ represents the converter agents, $\mathcal{E}$ represents the communication links, and $\mathcal{A}$ represents the weighted adjacency matrix.

This formulation follows the graph-theoretic model used in the source paper.

10.1 Graph Representation of the Converter Network

Consider a system containing N boost converters.

Each converter is treated as one **node** of the communication graph:

$$\boxed{\mathcal{V} = \{v_1, v_2, \ldots, v_N\}}$$

Thus, for a four-converter system,

$$\boxed{\mathcal{V} = \{v_1, v_2, v_3, v_4\}}$$

Here,

$$v_i$$

represents the i^{th} boost-converter agent.

The complete communication network can therefore be represented as

$$\boxed{\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})}$$

The important point is that the graph represents **communication between converters**, not the physical power-flow connection between the converters.

10.2 Nodes

A **node** represents an individual agent in the multi-agent system.

For the proposed system,

$$\boxed{v_i \leftrightarrow i^{\text{th}} \text{ boost converter}}$$

Therefore,

$$v_1 \rightarrow \text{Boost Converter 1}$$

$$v_2 \rightarrow \text{Boost Converter 2}$$

and so on.

Each node has its own converter dynamics and measures its own output current.

For example,

$$y_i(t)$$

represents the output current associated with the i^{th} converter agent.

10.3 Edges

The set of communication links is represented by

$$\boxed{\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}}$$

An edge indicates that information can be exchanged between two agents.

For example,

$$(v_i, v_j) \in \mathcal{E}$$

means that a communication link exists between agents i and j.

Physically, this means that one converter has access to the required information from the other converter through the communication network.

For this application, the exchanged information is primarily the **output-current information**.

Therefore,

$$\boxed{\text{Communication Edge} \Rightarrow \text{Output-current Information Exchange}}$$

10.4 Physical Meaning of the Communication Topology

The communication topology answers an important practical question:

Which converter receives information from which other converter?

For example, consider

$$v_1 \longleftrightarrow v_2$$

and

$$v_2 \longleftrightarrow v_3.$$

This means that converter 1 and converter 2 can exchange information, and converter 2 and converter 3 can exchange information.

However, converter 1 does not directly communicate with converter 3.

Nevertheless, information can propagate through converter 2.

Thus,

$$\boxed{v_1 \to v_2 \to v_3}$$

allows converter 3 to indirectly become influenced by the information originating from converter 1.

This is why the connectivity of the communication network is important for achieving global consensus.

10.5 Adjacency Matrix

The communication links are mathematically represented using the **weighted adjacency matrix**

$$\boxed{\mathcal{A} = [a_{ij}]}$$

where

$$a_{ij} > 0$$

indicates that agents v_i and v_j can exchange information according to the graph convention.

If there is no communication link, then

$$\boxed{a_{ij} = 0}.$$

For an undirected network,

$$\boxed{a_{ij} = a_{ji}}$$

Therefore, the adjacency matrix is symmetric:

$$\boxed{\mathcal{A} = \mathcal{A}^T}$$

10.6 Example of an Adjacency Matrix

Consider four boost converters with a communication topology represented by

$$\boxed{\mathcal{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}}$$

This means:

- Converter 1 communicates with Converter 2.
- Converter 2 communicates with Converters 1 and 3.
- Converter 3 communicates with Converters 2 and 4.
- Converter 4 communicates with Converter 3.

The communication structure is therefore

$$\boxed{v_1 \longleftrightarrow v_2 \longleftrightarrow v_3 \longleftrightarrow v_4}$$

This is only an illustrative topology; the actual topology used in the simulation should be the one specified for the particular system being modelled.

10.7 Neighbouring Agents

The set of neighbours of agent v_i is represented by

$$\boxed{\mathcal{N}_{v_i} = \{v_j \in \mathcal{V} \mid (v_j, v_i) \in \mathcal{E}\}}$$

In simple terms,

$$\boxed{\mathcal{N}_{v_i} = \text{set of converters directly connected to converter } i}$$

For example, if

$$\mathcal{N}_{v_2} = \{v_1, v_3\},$$

then Converter 2 receives information from Converters 1 and 3.

Its cooperative controller can therefore compare

$$y_2$$

with

$$y_1$$

and

$$y_3.$$

10.8 Degree Matrix

The **degree matrix** represents the total communication weight associated with each node.

It is defined as

$$\boxed{\mathcal{D} = \mathrm{diag}\{d_1, d_2, \ldots, d_N\}}$$

where

$$\boxed{d_i = \sum_{j=1}^{N} a_{ij}}$$

Thus, d_i represents the total weight of the communication connections associated with converter i.

For an unweighted graph, it simply represents the number of neighbouring converters.

For example, if Converter 2 is connected to Converters 1 and 3, then

$$\boxed{d_2 = 2}.$$

For the illustrative adjacency matrix above,

$$\boxed{\mathcal{D} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

10.9 Laplacian Matrix

The **Laplacian matrix** is one of the most important matrices in cooperative control.

It is defined as

$$\boxed{\mathcal{L} = \mathcal{D} - \mathcal{A}}$$

The Laplacian matrix contains information about the differences between neighbouring agents.

For the illustrative four-converter topology,

$$\boxed{\mathcal{L} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}}$$

The Laplacian is particularly important because consensus control is fundamentally based on **relative differences between agents**.

10.10 Physical Meaning of the Laplacian

Consider the vector of converter output currents

$$\boxed{y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix}}$$

Then

$$\mathcal{L}y$$

contains information related to the differences between each converter and its neighbours.

For example, the i^{th} component can be written as

$$\boxed{(\mathcal{L}y)_i = \sum_{j=1}^{N} a_{ij}(y_i - y_j)}$$

Therefore,

$$\boxed{\mathcal{L}y = \text{network-wide current-difference information}}$$

If all converter currents become equal,

$$y_1 = y_2 = \cdots = y_N,$$

then

$$\boxed{\mathcal{L}y = 0}$$

This is one of the fundamental mathematical characteristics of consensus.

10.11 Pinning Gain

The multi-agent system considered here follows a **leader-following architecture**.

The desired reference signal is associated with the leader.

The connection between the leader and a follower converter is represented by a **pinning gain**

$$\boxed{p_i \in \mathbb{R}_+}$$

The value p_i represents the strength of the connection between the leader and the i^{th} follower.

If

$$p_i > 0,$$

then converter i is directly connected or **pinned** to the leader.

If

$$p_i = 0,$$

then converter i does not have a direct pinning connection.

Therefore,

$$\boxed{p_i \rightarrow \text{strength of leader-to-follower connection}}$$

10.12 Leader Node

The **leader** represents the source of the desired reference signal.

Let the reference be

$$\boxed{r_0(t)}$$

The leader therefore provides the desired behaviour that the follower converters must track.

Conceptually,

$$\boxed{\text{Leader} \rightarrow r_0(t)}$$

and

$$\boxed{r_0(t) \rightarrow \text{Follower Converters}}$$

The leader does not necessarily represent another physical boost converter. In the control formulation, it represents the reference or command that the follower agents are required to follow.

10.13 Follower Nodes

The boost converters are the **follower agents**.

For N converters,

$$\boxed{v_1, v_2, \ldots, v_N}$$

represent the follower nodes.

Each follower attempts to achieve

$$\boxed{y_i(t) \rightarrow r_0(t)}$$

At the same time, each follower uses information from neighbouring converters to maintain consensus.

Therefore, every follower has two important objectives:

$$\boxed{\text{Track the Leader}}$$

and

$$\boxed{\text{Reach Consensus with Neighbouring Followers}}$$

10.14 Pinning Gain Matrix

The individual pinning gains are collected into the **pinning gain matrix**

$$\boxed{\mathcal{P} = \mathrm{diag}\{p_1, p_2, \ldots, p_N\}}$$

For example, if only converters 1 and 4 are directly connected to the leader,

$$p_1 > 0, \qquad p_4 > 0,$$

while

$$p_2 = p_3 = 0.$$

Then

$$\boxed{\mathcal{P} = \begin{bmatrix} p_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & p_4 \end{bmatrix}}$$

This matrix identifies which converter agents have direct access to the leader reference.

10.15 Combined Matrix $\mathcal{L} + \mathcal{P}$

The Laplacian matrix describes communication **between follower converters**, while the pinning matrix describes the connection **between the leader and follower converters**.

Therefore, the combination

$$\boxed{\mathcal{L} + \mathcal{P}}$$

contains information about both:

$$\boxed{\text{Follower-to-Follower Communication} + \text{Leader-to-Follower Connection}}$$

This matrix appears directly in the cooperative-control and stability analysis.

The source paper assumes that

$$\boxed{\mathcal{L} + \mathcal{P} > 0}$$

under the stated connected leader-following graph assumption.

10.16 Connected Undirected Graph

The communication graph is assumed to be **undirected and connected**.

10.16.1 Undirected Graph

An undirected graph means that communication links are symmetric.

If converter i communicates with converter j, then

$$\boxed{a_{ij} = a_{ji} > 0}$$

Thus, information exchange is represented in both directions.

10.16.2 Connected Graph

A connected graph means that there exists a communication path between every pair of converter agents.

For example,

$$\boxed{v_1 \leftrightarrow v_2 \leftrightarrow v_3 \leftrightarrow v_4}$$

is connected even though v_1 and v_4 do not communicate directly.

Information can travel through intermediate nodes.

10.17 No Self-Loops

The graph is also assumed to have **no self-loops**.

A self-loop would mean that a converter is considered to have a communication link with itself.

Therefore,

$$\boxed{a_{ii} = 0}$$

For example,

$$\mathcal{A} = \begin{bmatrix} 0 & \cdots \\ \vdots & \ddots \end{bmatrix}$$

has zero diagonal elements.

This assumption is explicitly stated in the source paper's graph-theoretic formulation.

10.18 Why Network Topology Is Important

The communication topology determines how information propagates through the converter network.

Suppose Converter 1 experiences a current deviation:

$$\boxed{y_1(t) \neq r_0(t)}$$

If Converter 1 is connected to Converter 2, then Converter 2 can receive information about this deviation.

That information can then influence Converter 2's control action and propagate through the network.

Thus,

$$\boxed{\text{Current Deviation} \rightarrow \text{Neighbour Communication} \rightarrow \text{Cooperative Correction}}$$

A properly connected network ensures that information about the desired reference and converter outputs can propagate throughout the system.

10.19 Current Consensus Through the Graph

The graph structure directly enters the consensus-control law.

For converter i, the neighbour information can be represented as

$$\boxed{\sum_{j=1}^{N} a_{ij}(y_j - y_i)}$$

This term measures how different converter i's output is from its neighbours.

If

$$y_i = y_j$$

for all connected neighbours, then

$$y_j - y_i = 0.$$

Therefore,

$$\boxed{\sum_{j=1}^{N} a_{ij}(y_j - y_i) = 0}$$

Consequently, the cooperative correction associated with consensus becomes zero once the converters reach agreement.

10.20 Leader-Following Consensus

For the leader-following system, the cooperative error contains both neighbour information and leader information.

A general form is

$$\boxed{e_i = \sum_{j=1}^{N} a_{ij}(y_j - y_i) + p_i(r_0 - y_i)}$$

The first term

$$\sum_{j=1}^{N} a_{ij}(y_j - y_i)$$

promotes **consensus among the converters**.

The second term

$$p_i(r_0 - y_i)$$

promotes **tracking of the leader/reference**.

Therefore,

$$\boxed{\text{Cooperative Control} = \text{Consensus} + \text{Leader Tracking}}$$

10.21 Four-Boost-Converter Example

For the four-boost-converter system, suppose the agents are

$$\boxed{v_1, v_2, v_3, v_4}$$

A possible connected communication topology is

$$\boxed{v_1 \longleftrightarrow v_2 \longleftrightarrow v_3 \longleftrightarrow v_4}$$

Assume that v_1 and v_4 are pinned to the leader.

Then the information structure is conceptually

$$\boxed{\begin{array}{c} r_0(t) \\ \downarrow \qquad \downarrow \\ v_1 \leftrightarrow v_2 \leftrightarrow v_3 \leftrightarrow v_4 \\ \downarrow \qquad\qquad\qquad\qquad \downarrow \\ y_1 \qquad y_2 \qquad y_3 \qquad y_4 \end{array}}$$

Here:

- v_1 receives the reference directly.
- v_4 receives the reference directly.
- v_2 receives information through v_1 and v_3.
- v_3 receives information through v_2 and v_4.

This allows the reference information and output-current information to propagate through the entire network.

10.22 Graph Theory in the Overall Control Structure

The graph-theoretic structure can be summarized as

$$\boxed{\begin{array}{c} \text{Leader / Reference } r_0(t) \\ \downarrow \\ \text{Pinning Matrix } \mathcal{P} \\ \downarrow \\ \text{Follower Boost Converters} \\ \updownarrow \\ \text{Communication Graph } \mathcal{G} \\ \downarrow \\ \text{Adjacency Matrix } \mathcal{A} \\ \downarrow \\ \text{Laplacian Matrix } \mathcal{L} \\ \downarrow \\ \text{Cooperative Control} \end{array}}$$

The important mathematical relationships are

$$\boxed{\mathcal{L} = \mathcal{D} - \mathcal{A}}$$

and

$$\boxed{\mathcal{L} + \mathcal{P} > 0}$$

10.23 Connection With the Heterogeneous Boost-Converters

The graph describes **how the boost converters communicate**, while the transfer-function matrix describes **how the boost converters dynamically respond**.

The heterogeneous converter plant is

$$\boxed{G_r(s) = \mathrm{diag}\{G_1(s), G_2(s), \dots, G_N(s)\}}$$

The communication network is represented by

$$\mathcal{L}$$

and

$$\mathcal{P}.$$

Therefore, the overall cooperative system combines

$$\boxed{\text{Converter Dynamics} + \text{Communication Topology}}$$

This combination is what allows the system to achieve cooperative current tracking even when

$$\boxed{G_i(s) \neq G_j(s)}$$

10.24 Importance of $\mathcal{L} + \mathcal{P} > 0$

The condition

$$\boxed{\mathcal{L} + \mathcal{P} > 0}$$

is an important assumption in the stability analysis.

Physically, it indicates that the communication structure together with the leader connections provides sufficient connectivity for the cooperative system.

If the graph were disconnected in an inappropriate way, some converters could become isolated from the reference information.

For example,

$$\boxed{v_1 \leftrightarrow v_2 \qquad v_3 \leftrightarrow v_4}$$

with no connection between the two groups would divide the network into two disconnected components.

In such a situation, reference information from one group cannot propagate to the other group through the communication network.

Therefore, network connectivity is essential for achieving system-wide consensus.

10.25 Summary

Graph theory provides the mathematical description of the communication network used to coordinate the heterogeneous boost converters.

The main definitions are:

$$\boxed{\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})}$$

where:

- $\mathcal{V}$ = set of converter nodes,
- $\mathcal{E}$ = set of communication edges,
- $\mathcal{A}$ = weighted adjacency matrix.

The adjacency matrix is

$$\boxed{\mathcal{A} = [a_{ij}], \qquad a_{ij} = a_{ji} > 0}$$

when two agents are connected.

The degree matrix is

$$\boxed{\mathcal{D} = \mathrm{diag}\{d_1, \dots, d_N\}, \qquad d_i = \sum_{j=1}^{N} a_{ij}}$$

The Laplacian matrix is

$$\boxed{\mathcal{L} = \mathcal{D} - \mathcal{A}}$$

The pinning gains are

$$\boxed{p_i \in \mathbb{R}_+}$$

and the pinning matrix is

$$\boxed{\mathcal{P} = \mathrm{diag}\{p_1, \ldots, p_N\}}$$

The combined network matrix is

$$\boxed{\mathcal{L} + \mathcal{P} > 0}$$

For the proposed boost-converter system, the physical meaning is:

$$\boxed{\text{Graph} \rightarrow \text{Who communicates with whom}}$$

$$\boxed{\text{Adjacency Matrix} \rightarrow \text{Communication Connections}}$$

$$\boxed{\text{Laplacian} \rightarrow \text{Current Differences Between Neighbours}}$$

$$\boxed{\text{Pinning Matrix} \rightarrow \text{Connection to Reference/Leader}}$$

and finally,

$$\boxed{\text{Graph Connectivity} \rightarrow \text{Information Propagation} \rightarrow \text{Current Consensus} \rightarrow \text{Reference Tracking}}$$

This graph-theoretic framework is therefore the mathematical foundation for the **distributed cooperative control of the heterogeneous DC/DC boost-converter agents**.

11 Multiple-Agent NI and SNI Systems

The proposed control system consists of multiple boost-converter agents whose dynamics satisfy the **Negative Imaginary (NI)** or **Strictly Negative Imaginary (SNI)** property. This section extends the NI/SNI concepts from a single dynamical system to a **multi-agent system (MAS)** and establishes the mathematical properties required for the subsequent modelling and stability analysis.

In the proposed application, each **DC/DC boost converter is considered an individual agent**. Since the converters may have different electrical parameters and load conditions, the complete system is treated as a **heterogeneous multi-agent system**.

11.1 NI and SNI Multi-Agent Systems

Consider a multi-agent system whose individual agents have transfer functions

$$\boxed{G_1(s), G_2(s), \ldots, G_N(s)}$$

In a homogeneous MAS, all agents have identical dynamics:

$$\boxed{G_1(s) = G_2(s) = \cdots = G_N(s) = M(s)}$$

The corresponding individual-agent transfer function can therefore be represented by

$$\boxed{M(s) \in \mathbb{R}(s)^{n\times n}}$$

If $M(s)$ satisfies the NI or SNI property, the individual agents possess the corresponding NI/SNI frequency-domain characteristics.

For the boost-converter system considered in this report, the individual agents will later be represented by their respective boost-converter transfer functions.

11.2 Networked Multi-Agent Model

When the individual agents are interconnected according to the communication graph, the networked system can be represented using the Kronecker product.

The interconnected transfer-function model is given by

$$\boxed{\overline{M}(s) = (\mathcal{L} + \mathcal{P}) \otimes M(s)}$$

where:

- $\mathcal{L}$ = Laplacian matrix of the communication graph,

- $\mathcal{P}$ = pinning gain matrix,
- $M(s)$ = transfer function of an individual agent,
- $\otimes$ = Kronecker product.

This expression combines the **dynamics of the individual agents** with the **communication topology**.

Therefore,

$$\boxed{\text{Agent Dynamics} + \text{Communication Network} \rightarrow \text{Networked MAS Dynamics}}$$

11.3 Preservation of NI and SNI Properties

An important result for the multi-agent system is that the network interconnection can preserve the NI or SNI property.

If $M(s)$ satisfies the required NI or SNI property and the communication graph satisfies the assumed connectivity conditions, then

$$\boxed{\overline{M}(s) = (\mathcal{L} + \mathcal{P}) \otimes M(s)}$$

also preserves the corresponding NI or SNI property under the conditions established by the theory.

This is particularly useful because it means that the stability properties of the individual converter dynamics can be carried into the networked system.

Conceptually,

$$\boxed{M(s) \in NI \Rightarrow \overline{M}(s) \in NI}$$

and similarly,

$$\boxed{M(s) \in SNI \Rightarrow \overline{M}(s) \in SNI}$$

The source paper uses this result to establish the NI/SNI properties of the interconnected multi-agent system.

11.4 Role of the Communication Network

The communication network is represented by

$$\mathcal{L} + \mathcal{P}.$$

Here,

$$\mathcal{L}$$

describes communication between the follower agents, while

$$\mathcal{P}$$

describes the connections between the leader and selected follower agents.

Therefore,

$$\boxed{\mathcal{L} + \mathcal{P} = \text{Follower Communication} + \text{Leader Connection}}$$

The Kronecker product

$$(\mathcal{L} + \mathcal{P}) \otimes M(s)$$

then combines this network information with the agent dynamics.

11.5 Positive Definiteness of $\mathcal{L} + \mathcal{P}$

For the assumed connected leader-following graph, the matrix

$$\boxed{\mathcal{L} + \mathcal{P} > 0}$$

is positive definite.

This condition is important because the eigenvalues of

$$\mathcal{L} + \mathcal{P}$$

are positive:

$$\boxed{\lambda_i(\mathcal{L} + \mathcal{P}) > 0}$$

These eigenvalues play an important role in determining the characteristic loci of the networked system.

Thus, the communication topology directly affects the frequency-domain analysis through the eigenvalues of

$$\mathcal{L} + \mathcal{P}.$$

11.6 DC Gain of the Networked System

Another important property concerns the value of the transfer function at zero frequency.

If

$$M(0) > 0,$$

then the networked system satisfies

$$\boxed{\overline{M}(0) > 0}$$

Similarly, if the individual system satisfies

$$M(0) < 0,$$

the corresponding sign relationship is preserved under the network transformation according to the stated result.

For the present boost-converter application, the relevant requirement is that the individual converter models have the appropriate positive DC gain:

$$\boxed{G_i(0) > 0}$$

This condition becomes important in the subsequent characteristic-loci analysis.

11.7 Characteristic Loci of the Multi-Agent System

For a MIMO system, the frequency-domain behaviour is described using the eigenvalues of the transfer-function matrix.

For the networked system,

$$\overline{M}(s) = (\mathcal{L} + \mathcal{P}) \otimes M(s),$$

the characteristic loci are

$$\boxed{\lambda_i[\overline{M}(s)]}$$

When

$$s = j\omega,$$

the characteristic loci become

$$\boxed{\lambda_i[\overline{M}(j\omega)]}$$

As ω varies,

$$0 \leq \omega < \infty,$$

each eigenvalue traces a path in the complex plane.

These paths are used later to determine whether the feedback interconnection satisfies the required stability condition.

11.8 Phase Characteristics of NI Systems

For an NI system, the characteristic loci have a restricted phase behaviour.

The phase angle of the i^{th} characteristic locus can be written as

$$\phi_i(\lambda_i(j\omega)).$$

For an NI system, the source formulation gives

$$\boxed{\phi_i(\lambda_i(j\omega)) \in [-\pi, 0], \qquad \omega \geq 0}$$

Thus, the characteristic loci remain within the lower-half-plane angular region.

In terms of degrees,

$$\boxed{-180^\circ \leq \phi_i \leq 0^\circ}$$

11.9 Phase Characteristics of SNI Systems

For an SNI system, the phase condition is strict:

$$\boxed{\phi_i(\lambda_i(j\omega)) \in (-\pi, 0), \qquad \omega > 0}$$

Therefore,

$$\boxed{-180^\circ < \phi_i < 0^\circ}$$

The difference between NI and SNI can therefore be summarized as

$$\boxed{\begin{array}{lcl} \text{NI} & : & [-180^\circ, 0^\circ] \\ \text{SNI} & : & (-180^\circ, 0^\circ) \end{array}}$$

for the relevant positive-frequency range.

The SNI condition is stronger because the phase remains strictly inside the boundary.

11.10 Complex-Plane Interpretation

The phase restrictions have a direct geometrical interpretation.

For

$$\phi \in [-\pi, 0],$$

the corresponding characteristic locus lies in the lower-half-plane region, including the negative real axis.

For SNI,

$$\phi \in (-\pi, 0),$$

the locus lies strictly within the lower-half-plane region.

Therefore,

$$\boxed{\text{NI/SNI Property} \rightarrow \text{Restricted Phase} \rightarrow \text{Restricted Characteristic-Locus Region}}$$

This property is later used in the characteristic-loci stability analysis.

11.11 Homogeneous Multi-Agent Systems

The results described above initially consider **homogeneous multi-agent systems**.

In a homogeneous MAS,

$$\boxed{G_1(s) = G_2(s) = \cdots = G_N(s)}$$

Thus, every agent has identical dynamics.

For example,

$$G_1(s) = G_2(s) = G_3(s) = G_4(s) = M(s).$$

The complete plant can then be written as

$$\boxed{G_r(s) = \mathrm{diag}\{M(s), M(s), M(s), M(s)\}}$$

Homogeneous systems are mathematically simpler because the same transfer function describes every agent.

11.12 Heterogeneous Multi-Agent Systems

The actual system considered in this report is more general because the boost converters may not have identical dynamics.

A heterogeneous MAS satisfies

$$\boxed{G_i(s) \neq G_j(s)}$$

for at least some

$$i \neq j.$$

Thus,

$$\boxed{G_r(s) = \mathrm{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}}$$

Each

$$G_i(s)$$

represents the dynamics of an individual boost converter.

The differences can arise from different:

- inductances,
- capacitances,
- load resistances,
- operating points,
- converter parameters, or
- other electrical characteristics.

Therefore, the converters cannot simply be treated as identical copies of one another.

11.13 Heterogeneous Boost-Converters as Agents

For the proposed ship-degaussing application, the multiple boost converters are considered individual agents:

$$\boxed{\begin{array}{ccc} v_1 & \rightarrow & G_1(s) \\ v_2 & \rightarrow & G_2(s) \\ v_3 & \rightarrow & G_3(s) \\ \vdots & & \vdots \\ v_N & \rightarrow & G_N(s) \end{array}}$$

where $G_i(s)$ is the average transfer function of the i^{th} boost converter.

The complete heterogeneous plant is

$$\boxed{G_r(s) = \operatorname{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}}$$

The communication graph then determines how these converter agents cooperate.

11.14 Why Heterogeneity Is Important

If all boost converters had exactly the same dynamics, then

$$G_1(s) = G_2(s) = \cdots = G_N(s),$$

and achieving consensus would generally be simpler.

However, in the practical system,

$$G_i(s) \neq G_j(s)$$

may occur.

Suppose two converters have different load resistances:

$$R_1 \neq R_2.$$

Their dynamic responses will generally differ:

$$G_1(s) \neq G_2(s).$$

Even if both receive the same control command, their output currents may not respond identically.

This is why the cooperative controller must coordinate the converters rather than simply applying independent identical control actions.

11.15 NI/SNI Properties of Heterogeneous Agents

Although the individual boost converters can have different transfer functions,

$$G_1(s), G_2(s), \dots, G_N(s),$$

they can still belong to the same system class.

For example,

$$\boxed{G_i(s) \in SNI, \qquad i = 1, 2, \dots, N}$$

Therefore,

$$\boxed{\text{Different Dynamics} + \text{Common SNI Property}}$$

can be used as the basis for designing a common cooperative-control structure.

This is particularly useful because it allows the controller design to focus on the common NI/SNI characteristics rather than requiring the exact same converter model for every agent.

11.16 Homogeneous Controller for Heterogeneous Plants

A major feature of the proposed approach is that the plants can be heterogeneous while the controller structure can remain common.

Thus,

$$\boxed{G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s)}$$

while

$$\boxed{C_1(s) = C_2(s) = \cdots = C_N(s) = C(s)}$$

This produces the desired structure:

$$\boxed{\text{Heterogeneous Boost-Converter Plants} + \text{Common Cooperative Controller}}$$

The controller uses the communication network to compensate for the differences among the converter agents.

11.17 Relationship Between Network Eigenvalues and Converter Dynamics

The communication network and the converter dynamics interact through the Kronecker-product representation.

For a homogeneous system,

$$\overline{M}(s) = (\mathcal{L} + \mathcal{P}) \otimes M(s).$$

The eigenvalues of the network matrix,

$$\lambda_i(\mathcal{L} + \mathcal{P}),$$

scale the dynamics of the individual agents.

Therefore, the characteristic behaviour of the complete network depends on both

$$\boxed{\lambda_i(\mathcal{L} + \mathcal{P})}$$

and

$$\boxed{M(s)}$$

This provides the mathematical connection between **communication topology** and **converter dynamics**.

11.18 Connection With Characteristic-Loci Stability Analysis

The NI/SNI multi-agent properties provide the foundation for the next stability-analysis stage.

The overall sequence is

Individual Boost Converter

$$\Downarrow$$

$$\boxed{G_i(s) \in SNI}$$

$$\Downarrow$$

Heterogeneous SNI MAS

$$\Downarrow$$

Communication Graph

$$\Downarrow$$

$$\boxed{\mathcal{L} + \mathcal{P}}$$

$$\Downarrow$$

Networked Transfer Function

$$\Downarrow$$

$$\lambda_i[L(j\omega)]$$

$$\Downarrow$$

Characteristic Loci

$$\Downarrow$$

Stability Analysis

11.19 Application to the Ship-Degaussing System

The multi-agent NI/SNI framework is applied to the ship-degaussing system because each boost converter supplies an individual degaussing-coil load.

The system can therefore be represented as

Boost Converter 1 $\rightarrow$ Degaussing Coil 1
Boost Converter 2 $\rightarrow$ Degaussing Coil 2
$\vdots$
Boost Converter N $\rightarrow$ Degaussing Coil N

The output current of each converter is important because it determines the magnetic field contribution of the associated coil.

Therefore, the cooperative objective is

$$\boxed{y_1(t) \rightarrow r_0(t), \quad y_2(t) \rightarrow r_0(t), \quad \ldots, \quad y_N(t) \rightarrow r_0(t)}$$

The NI/SNI properties provide the frequency-domain framework needed to analyse the stability of this cooperative system.

11.20 Important Conditions

For the proposed system, the main theoretical conditions can be summarized as

$$\boxed{G_i(s) \in SNI, \qquad i = 1, \ldots, N}$$

and

$$\boxed{G_i(0) > 0}$$

The communication graph is assumed to satisfy

$$\boxed{\mathcal{L} + \mathcal{P} > 0}$$

The heterogeneous plant is represented by

$$\boxed{G_r(s) = \mathrm{diag}\{G_1(s), \ldots, G_N(s)\}}$$

These conditions form the basis for the subsequent controller design and characteristic-loci analysis.

11.21 Summary

The multiple-agent NI/SNI framework extends the properties of individual dynamical systems to a network of interacting converter agents.

For the proposed system:

$$\boxed{\text{Each Boost Converter} = \text{One Agent}}$$

and

$$\boxed{G_i(s) = \text{Dynamic Model of the } i^{\text{th}} \text{ Boost Converter}}$$

The communication network is described using

$$\mathcal{L} + \mathcal{P},$$

and for a homogeneous agent model $M(s)$, the networked system can be represented as

$$\boxed{\overline{M}(s) = (\mathcal{L} + \mathcal{P}) \otimes M(s)}$$

The NI/SNI property of the individual dynamics is preserved under the specified network structure.

The characteristic loci satisfy the phase restrictions

$$\boxed{\phi_i \in [-\pi, 0] \qquad \text{for NI}}$$

and

$$\boxed{\phi_i \in (-\pi, 0) \qquad \text{for SNI}}$$

For the heterogeneous boost-converter system,

$$\boxed{G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s)}$$

while the individual agents can still satisfy

$$\boxed{G_i(s) \in SNI}$$

Therefore, the overall concept is

$$\boxed{\text{Heterogeneous SNI Boost Converters} + \text{Connected Communication Network} \rightarrow \text{Networked SNI MAS} \rightarrow \text{Characteristic-Loci Stability Analysis}}$$

This establishes the theoretical connection between the **individual boost-converter dynamics**, the **communication topology**, and the **stability of the complete cooperative multi-agent system**.

12 Mathematical Modelling of the DC/DC Boost Converter

The mathematical model of the DC/DC boost converter is required to describe its dynamic behaviour and to obtain the transfer function used in the subsequent control-system and stability analysis.

A conventional boost converter consists of a DC input voltage, inductor, controlled switch, diode, output capacitor, and load resistance. The converter operates in two switching states, namely the **ON state** and the **OFF state**.

The two switching-state models are first derived independently. The resulting state-space equations are then averaged according to the duty ratio to obtain the large-signal averaged model. Finally, small-signal perturbation is applied around the steady-state operating point to obtain the duty-to-output transfer function.

12.1 Boost Converter Operating Principle

A conventional boost converter consists of:

- DC input voltage V_{in},
- inductor L,
- controlled switch,
- diode,
- output capacitor C, and
- load resistance R.

The converter has two switching states.

12.1.1 State 1: Switch ON

When the switch is ON, the diode is reverse biased. The inductor is directly connected to the input source and stores energy.

The inductor voltage is therefore

$$\boxed{V_L = V_{in}}$$

During this interval, the capacitor supplies the load.

12.1.2 State 2: Switch OFF

When the switch is OFF, the diode conducts. The inductor releases its stored energy through the diode to the output capacitor and load.

The inductor voltage becomes

$$\boxed{V_L = V_{in} - v_C}$$

Therefore, the boost converter has two state-space models, which are then averaged according to the duty ratio D.

12.2 Choice of State Variables

The inductor current and capacitor voltage are selected as the state variables.

Let

$$\boxed{x_1 = i_L}$$

represent the inductor current and

$$\boxed{x_2 = v_C}$$

represent the capacitor voltage.

Hence, the state vector is

$$\boxed{x = \begin{bmatrix} i_L \\ v_C \end{bmatrix}}$$

The output of the converter is selected as the capacitor voltage:

$$\boxed{y = v_C}$$

Therefore, the output matrices are

$$\boxed{C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \qquad D_s = \begin{bmatrix} 0 \end{bmatrix}}$$

Here, D_s denotes the state-space feedthrough matrix and is different from the converter duty ratio D.

12.3 State 1 — Switch ON

When the switch is ON, the diode is OFF.

The inductor is directly connected to the input source. Therefore,

$$\boxed{V_L = V_{in}}$$

Using the inductor voltage-current relationship,

$$V_L = L\frac{di_L}{dt},$$

we obtain

$$L\frac{di_L}{dt} = V_{in}.$$

Hence,

$$\boxed{\frac{di_L}{dt} = \frac{V_{in}}{L}}$$

During the ON interval, the capacitor supplies the load. Therefore,

$$C\frac{dv_C}{dt} = -\frac{v_C}{R}.$$

Thus,

$$\boxed{\frac{dv_C}{dt} = -\frac{v_C}{RC}}$$

The ON-state model can therefore be written as

$$\boxed{\frac{d}{dt}\begin{bmatrix} i_L \\ v_C \end{bmatrix} = A_1 \begin{bmatrix} i_L \\ v_C \end{bmatrix} + B_1 V_{in}}$$

where

$$\boxed{A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -\dfrac{1}{RC} \end{bmatrix}}$$

and

$$\boxed{B_1 = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}}$$

12.4 Substitution of Numerical Values

The converter parameters used for the analysis are

$$\boxed{R = 10\ \Omega, \qquad L = 0.1\ H, \qquad C = 0.01\ F}$$

The duty ratio and input voltage are

$$\boxed{D = 0.8, \qquad V_{in} = 23\ V}$$

First,

$$\frac{1}{RC} = \frac{1}{10(0.01)} = \frac{1}{0.1} = 10.$$

Also,

$$\frac{1}{L} = \frac{1}{0.1} = 10.$$

Therefore, the ON-state matrices become

$$\boxed{A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -10 \end{bmatrix}}$$

and

$$\boxed{B_1 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}}$$

Thus, the ON-state equations are

$$\boxed{\frac{di_L}{dt} = 10V_{in}}$$

and

$$\boxed{\frac{dv_C}{dt} = -10v_C}$$

12.5 State 2 — Switch OFF

When the switch is OFF, the diode conducts.

The inductor transfers its stored energy to the output circuit. The inductor voltage is

$$\boxed{V_L = V_{in} - v_C}$$

Therefore,

$$L\frac{di_L}{dt} = V_{in} - v_C.$$

Hence,

$$\boxed{\frac{di_L}{dt} = \frac{V_{in}}{L} - \frac{v_C}{L}}$$

For the capacitor,

$$C\frac{dv_C}{dt} = i_L - \frac{v_C}{R}.$$

Therefore,

$$\boxed{\frac{dv_C}{dt} = \frac{i_L}{C} - \frac{v_C}{RC}}$$

The OFF-state model is therefore

$$\boxed{\frac{d}{dt}\begin{bmatrix} i_L \\ v_C \end{bmatrix} = A_2 \begin{bmatrix} i_L \\ v_C \end{bmatrix} + B_2 V_{in}}$$

where

$$\boxed{A_2 = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix}}$$

and

$$\boxed{B_2 = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}}$$

12.6 Numerical OFF-State Matrices

Using

$$L = 0.1\ H, \qquad C = 0.01\ F, \qquad R = 10\ \Omega,$$

we obtain

$$\frac{1}{L} = 10,$$

$$\frac{1}{C} = 100,$$

and

$$\frac{1}{RC} = 10.$$

Therefore,

$$\boxed{A_2 = \begin{bmatrix} 0 & -10 \\ 100 & -10 \end{bmatrix}}$$

and

$$\boxed{B_2 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}}$$

12.7 Averaged State-Space Model

The averaged state matrix is obtained from

$$\boxed{A = DA_1 + (1 - D)A_2}$$

Since

$$D = 0.8,$$

we have

$$1 - D = 0.2.$$

Therefore,

$$A = 0.8 \begin{bmatrix} 0 & 0 \\ 0 & -10 \end{bmatrix} + 0.2 \begin{bmatrix} 0 & -10 \\ 100 & -10 \end{bmatrix}.$$

The first term is

$$0.8A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -8 \end{bmatrix}.$$

The second term is

$$0.2A_2 = \begin{bmatrix} 0 & -2 \\ 20 & -2 \end{bmatrix}.$$

Therefore,

$$\boxed{A = \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}}$$

12.8 Averaged Input Matrix

Since

$$B_1 = B_2 = \begin{bmatrix} 10 \\ 0 \end{bmatrix},$$

the averaged input matrix is

$$\boxed{B = DB_1 + (1 - D)B_2}$$

Therefore,

$$B = 0.8 \begin{bmatrix} 10 \\ 0 \end{bmatrix} + 0.2 \begin{bmatrix} 10 \\ 0 \end{bmatrix}.$$

Hence,

$$\boxed{B = \begin{bmatrix} 10 \\ 0 \end{bmatrix}}$$

The averaged large-signal model is therefore

$$\boxed{\frac{d}{dt}\begin{bmatrix} i_L \\ v_C \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}\begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \end{bmatrix} V_{in}}$$

12.9 Steady-State Operating Point

For an ideal boost converter, the steady-state output voltage is

$$\boxed{V_o = \frac{V_{in}}{1-D}}$$

Using

$$V_{in} = 23\ V$$

and

$$D = 0.8,$$

we obtain

$$V_o = \frac{23}{1-0.8}.$$

Therefore,

$$V_o = \frac{23}{0.2} = 115\ V.$$

Hence,

$$\boxed{V_o = V_C = 115\ V}$$

12.10 Output Current

The load resistance is

$$R = 10\ \Omega.$$

Therefore, the output current is

$$I_o = \frac{V_o}{R}.$$

Substituting the numerical values,

$$I_o = \frac{115}{10}.$$

Hence,

$$\boxed{I_o = 11.5\ A}$$

12.11 Inductor Current

For an ideal boost converter,

$$\boxed{I_o = (1 - D)I_L}$$

Therefore,

$$I_L = \frac{I_o}{1 - D}.$$

Substituting the values,

$$I_L = \frac{11.5}{0.2}.$$

Hence,

$$\boxed{I_L = 57.5\ A}$$

Thus, the steady-state operating point is

$$\boxed{x_0 = \begin{bmatrix} 57.5 \\ 115 \end{bmatrix}}$$

12.12 Small-Signal Duty-to-Output Model

The small-signal model is obtained by introducing a small perturbation in the duty ratio around the steady-state operating point.

Let the duty-ratio perturbation be

$$\hat{d}$$

and the output-voltage perturbation be

$$\hat{v}_C.$$

The small-signal input matrix associated with the duty-ratio perturbation is

$$\boxed{B_d = (A_1 - A_2)x_0}$$

First,

$$A_1 - A_2 = \begin{bmatrix} 0 & 0 \\ 0 & -10 \end{bmatrix} - \begin{bmatrix} 0 & -10 \\ 100 & -10 \end{bmatrix}.$$

Therefore,

$$\boxed{A_1 - A_2 = \begin{bmatrix} 0 & 10 \\ -100 & 0 \end{bmatrix}}$$

12.13 Calculation of B_d

The steady-state operating point is

$$x_0 = \begin{bmatrix} 57.5 \\ 115 \end{bmatrix}.$$

Therefore,

$$B_d = \begin{bmatrix} 0 & 10 \\ -100 & 0 \end{bmatrix} \begin{bmatrix} 57.5 \\ 115 \end{bmatrix}.$$

The first row is

$$0(57.5) + 10(115) = 1150.$$

The second row is

$$-100(57.5) + 0(115) = -5750.$$

Hence,

$$\boxed{B_d = \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}}$$

12.14 Output Matrix

Since the output is the capacitor voltage,

$$y = v_C,$$

the output matrix is

$$\boxed{C = \begin{bmatrix} 0 & 1 \end{bmatrix}}$$

and

$$\boxed{D_d = 0}$$

Therefore, the small-signal state-space model is

$$\boxed{\dot{\hat{x}} = A\hat{x} + B_d\hat{d}}$$

with

$$\boxed{A = \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}}$$

and

$$\boxed{B_d = \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}}$$

The output equation is

$$\boxed{\hat{y} = \begin{bmatrix} 0 & 1 \end{bmatrix} \hat{x}}$$

12.15 Calculation of $sI - A$

The matrix

$$sI - A$$

is given by

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}.$$

Therefore,

$$\boxed{sI - A = \begin{bmatrix} s & 2 \\ -20 & s+10 \end{bmatrix}}$$

Its determinant is

$$\det(sI - A) = s(s+10) - (2)(-20).$$

Therefore,

$$\det(sI - A) = s^2 + 10s + 40.$$

Hence,

$$\boxed{\det(sI - A) = s^2 + 10s + 40}$$

12.16 Calculation of $(sI - A)^{-1}$

For a general matrix

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the inverse is

$$M^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Therefore,

$$\boxed{(sI - A)^{-1} = \frac{1}{s^2 + 10s + 40} \begin{bmatrix} s+10 & -2 \\ 20 & s \end{bmatrix}}$$

12.17 Duty-to-Output Transfer Function

The duty-to-output transfer function is

$$\boxed{G_{vd}(s) = C(sI - A)^{-1}B_d}$$

Substituting the matrices,

$$G_{vd}(s) = \begin{bmatrix} 0 & 1 \end{bmatrix} \frac{\begin{bmatrix} s+10 & -2 \\ 20 & s \end{bmatrix}}{s^2 + 10s + 40} \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}.$$

The multiplication of the second row gives

$$\begin{bmatrix} 20 & s \end{bmatrix} \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}.$$

Therefore,

$$20(1150) + s(-5750) = 23000 - 5750s.$$

Hence,

$$\boxed{G_{vd}(s) = \frac{23000 - 5750s}{s^2 + 10s + 40}}$$

Factoring the numerator,

$$23000 - 5750s = -5750(s - 4).$$

Therefore,

$$\boxed{G_{vd}(s) = \frac{-5750(s - 4)}{s^2 + 10s + 40}}$$

or equivalently,

$$\boxed{G_{vd}(s) = \frac{23000(1 - 0.25s)}{s^2 + 10s + 40}}$$

This is the small-signal control-to-output transfer function of the boost converter for the selected numerical operating point.

12.18 Zero of the Boost Converter

The numerator of the transfer function is

$$23000 - 5750s.$$

Setting the numerator equal to zero gives

$$23000 - 5750s = 0.$$

Therefore,

$$5750s = 23000,$$

and hence,

$$\boxed{s = 4}$$

Thus, the boost converter has a right-half-plane zero at

$$\boxed{z = +4 \text{ rad/s}}$$

The presence of this right-half-plane zero is an important characteristic of the boost converter and must be considered during controller design.

12.19 Poles of the Boost Converter

The denominator of the transfer function is

$$s^2 + 10s + 40.$$

The poles are obtained by solving

$$s^2 + 10s + 40 = 0.$$

Using the quadratic formula,

$$s = \frac{-10 \pm \sqrt{10^2 - 4(1)(40)}}{2}.$$

Therefore,

$$s = \frac{-10 \pm \sqrt{100 - 160}}{2}$$

and

$$s = \frac{-10 \pm \sqrt{-60}}{2}.$$

Since

$$\sqrt{-60} = j\sqrt{60},$$

we obtain

$$s = -5 \pm j\sqrt{15}.$$

Numerically,

$$\boxed{s = -5 \pm j3.873}$$

Thus, both poles are located in the left-half plane.

12.20 Zero-Pole-Gain Form

The transfer function can be expressed in zero-pole-gain form as

$$\boxed{G_{vd}(s) = -5750\frac{s-4}{(s+5-j3.873)(s+5+j3.873)}}$$

Since

$$3.873 \approx \sqrt{15},$$

the transfer function can also be written as

$$\boxed{G_{vd}(s) = -5750\frac{s-4}{(s+5-j\sqrt{15})(s+5+j\sqrt{15})}}$$

12.21 State-Space Model — Final Numerical Result

The complete small-signal boost-converter model is

$$\boxed{\dot{\hat{x}} = \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}\hat{x} + \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}\hat{d}}$$

and

$$\boxed{\hat{y} = \begin{bmatrix} 0 & 1 \end{bmatrix} \hat{x}}$$

Therefore,

$$\boxed{A = \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}, \quad B_d = \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D_d = 0}$$

12.22 Transfer Function

The resulting small-signal duty-to-output transfer function is

$$\boxed{\frac{\hat{v}_C(s)}{\hat{d}(s)} = \frac{23000 - 5750s}{s^2 + 10s + 40}}$$

or

$$\boxed{\frac{\hat{v}_C(s)}{\hat{d}(s)} = -5750 \frac{s - 4}{s^2 + 10s + 40}}$$

12.23 Three-Converter Model

For a multi-agent system containing three identical converter models, the transfer-function matrix can be written as

$$\boxed{M(s) = G_{vd}(s) I_3}$$

where I_3 is the 3×3 identity matrix.

Therefore,

$$\boxed{M(s) = \begin{bmatrix} G_{vd}(s) & 0 & 0 \\ 0 & G_{vd}(s) & 0 \\ 0 & 0 & G_{vd}(s) \end{bmatrix}}$$

Substituting the calculated transfer function gives

$$\boxed{M(s) = \begin{bmatrix} \frac{23000-5750s}{s^2+10s+40} & 0 & 0 \\ 0 & \frac{23000-5750s}{s^2+10s+40} & 0 \\ 0 & 0 & \frac{23000-5750s}{s^2+10s+40} \end{bmatrix}}$$

For four identical converters, the corresponding model is

$$\boxed{M_4(s) = G_{vd}(s) I_4}$$

12.24 MATLAB Implementation

The corresponding MATLAB implementation of the boost-converter averaged and small-signal model is given below.

```
clc;
clear;

D = input('enter the duty_ratio: ');
Vin = input('enter_voltage: ');

R = 10;
L = 0.1;
C = 0.01;

% State-space matrices for switch ON
A1 = [0 0;
      0 -1/(R*C)];

B1 = [1/L;
      0];

% State-space matrices for switch OFF
A2 = [0 -1/L;
      1/C -1/(R*C)];

B2 = [1/L;
      0];

% Averaged model
A = A1*D + A2*(1-D);
B = B1*D + B2*(1-D);

% Output matrix
Cout = [0 1];
```

```
Dout = 0;

% Steady-state operating point
Vo = Vin/(1-D);
Io = Vo/R;
IL = Io/(1-D);

x0 = [IL; Vo];

% Duty-ratio small-signal input matrix
Bd = (A1-A2)*x0;

% Duty-to-output transfer function
s = tf('s');

Gvd = Cout*inv(s*eye(2)-A)*Bd;

% Three identical converter agents
m_tf = eye(3)*Gvd;

disp('A = ');
disp(A);

disp('B = ');
disp(B);

disp('Bd = ');
disp(Bd);

disp('C = ');
disp(Cout);

disp('D = ');
disp(Dout);

disp('Operating point x0 = ');
disp(x0);

disp('Boost converter duty-to-output transfer function = ');
Gvd
```

For

$$D = 0.8, \qquad V_{in} = 23\ V,$$

the important numerical results are

$$\boxed{A = \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}}$$

$$\boxed{B = \begin{bmatrix} 10 \\ 0 \end{bmatrix}}$$

$$\boxed{x_0 = \begin{bmatrix} 57.5 \\ 115 \end{bmatrix}}$$

$$\boxed{B_d = \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}}$$

and

$$\boxed{G_{vd}(s) = \frac{23000 - 5750s}{s^2 + 10s + 40}}$$

12.25 Important Characteristics of the Boost Model

The derived boost-converter model has the following important characteristics:

1. The averaged state matrix is
$$\boxed{A = \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}}.$$

2. The duty-ratio perturbation matrix is
$$\boxed{B_d = \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}}.$$

3. The output is the capacitor voltage:
$$\boxed{y = v_C}.$$

4. The transfer function is
$$\boxed{G_{vd}(s) = \frac{23000 - 5750s}{s^2 + 10s + 40}}.$$

5. The converter has a right-half-plane zero at
$$\boxed{z = +4 \text{ rad/s}}.$$

6. The poles are

$$\boxed{s = -5 \pm j3.873}.$$

Therefore, the derived model clearly captures the characteristic right-half-plane zero of the boost converter.

12.26 Summary

The mathematical model of the DC/DC boost converter was developed using the state-space averaging method.

The ON-state matrices are

$$\boxed{A_1 = \begin{bmatrix} 0 & 0 \\ 0 & -10 \end{bmatrix}, \qquad B_1 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}}$$

while the OFF-state matrices are

$$\boxed{A_2 = \begin{bmatrix} 0 & -10 \\ 100 & -10 \end{bmatrix}, \qquad B_2 = \begin{bmatrix} 10 \\ 0 \end{bmatrix}}$$

For

$$D = 0.8,$$

the averaged model becomes

$$\boxed{A = \begin{bmatrix} 0 & -2 \\ 20 & -10 \end{bmatrix}, \qquad B = \begin{bmatrix} 10 \\ 0 \end{bmatrix}}$$

The operating point is

$$\boxed{x_0 = \begin{bmatrix} 57.5 \\ 115 \end{bmatrix}}$$

and the small-signal duty-ratio input matrix is

$$\boxed{B_d = \begin{bmatrix} 1150 \\ -5750 \end{bmatrix}}$$

Finally, the boost-converter duty-to-output transfer function is

$$\boxed{G_{vd}(s) = \frac{23000 - 5750s}{s^2 + 10s + 40}}$$

The transfer function contains a right-half-plane zero at

$$\boxed{s = +4 \text{ rad/s}}$$

and stable poles at

$$\boxed{s = -5 \pm j3.873}.$$

This mathematical model is subsequently used for the **multi-agent representation, controller design, NI/SNI analysis, and stability analysis** of the proposed cooperative boost-converter system.

13 Proposed Dynamic Consensus Control Scheme

This section presents the **dynamic consensus control strategy** for the proposed multi-agent system of heterogeneous DC/DC boost converters. The main purpose of the controller is to make the output currents of all boost converters converge to a common, possibly time-varying reference signal while maintaining closed-loop stability.

In the multi-agent system, each boost converter is considered an individual agent. Since the converters may have different electrical parameters and load conditions, their transfer functions can be different:

$$\boxed{G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s)}$$

Nevertheless, the proposed control structure uses the same controller configuration for every agent. Hence, the plant is heterogeneous while the controller is homogeneous.

13.1 Control Objective

Let the output of the i^{th} boost converter be

$$Y_i(s),$$

and let the common reference signal be

$$R_0(s).$$

The desired objective is

$$\boxed{\lim_{t\to\infty}[y_i(t) - r_0(t)] = 0, \qquad i = 1, 2, \ldots, N}$$

In other words,

$$\boxed{y_1(t) \to r_0(t), \quad y_2(t) \to r_0(t), \quad \ldots, \quad y_N(t) \to r_0(t)}$$

Consequently, at steady state,

$$\boxed{y_1(t) = y_2(t) = \cdots = y_N(t) = r_0(t)}$$

For the ship-degaussing application, this means that the currents supplied by the different boost converters can be coordinated so that they follow the commanded degaussing-current reference.

13.2 Overall Control Architecture

The overall control architecture consists of four main parts:

1. **Heterogeneous boost-converter plants**
2. **Measurement of individual converter outputs**
3. **Communication network described by $\mathcal{L} + \mathcal{P}$**
4. **Homogeneous distributed cooperative controller**

The plant is represented as

$$\boxed{\widetilde{G}(s) = \mathrm{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}}$$

The vector of converter outputs is

$$\boxed{y = \begin{bmatrix} y_1 & y_2 & \cdots & y_N \end{bmatrix}^T}$$

Similarly, the control-input vector is

$$\boxed{u = \begin{bmatrix} u_1 & u_2 & \cdots & u_N \end{bmatrix}^T}$$

The reference is supplied to the agents through the leader/pinning structure.

13.3 Heterogeneous Boost-Converter Plant

Each boost converter has its own transfer function

$$G_i(s).$$

Thus,

$$\boxed{\widetilde{G}(s) = \begin{bmatrix} G_1(s) & 0 & \cdots & 0 \\ 0 & G_2(s) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & G_N(s) \end{bmatrix}}$$

The diagonal structure indicates that each converter has its own local dynamics.

For example, for four boost converters,

$$\boxed{\widetilde{G}(s) = \operatorname{diag}\{G_1(s), G_2(s), G_3(s), G_4(s)\}}$$

The converters may have different R_i, L_i, and C_i, and consequently

$$\boxed{G_1(s) \neq G_2(s) \neq G_3(s) \neq G_4(s)}$$

This is why the system is called a **heterogeneous multi-agent system**.

13.4 Homogeneous Controller

Although the plants are heterogeneous, the proposed controller has the same structure for every agent.

The controller is written as

$$\boxed{C(s) = \frac{k}{s}\widehat{C}(s)}$$

where

$$\boxed{k > 0}$$

is a positive scalar gain and

$$\widehat{C}(s)$$

is the dynamic part of the controller.

The source formulation requires $\widehat{C}(s)$ to satisfy the appropriate stability and positive-real properties.

For N agents, the complete controller matrix is

$$\boxed{\widehat{C}_N(s) = I_N \otimes C(s)}$$

Therefore,

$$\boxed{\widehat{C}_N(s) = \text{diag}\{C(s), C(s), \ldots, C(s)\}}$$

The important structural property is

$$\boxed{\text{Heterogeneous Plants} + \text{Homogeneous Controller}}$$

13.5 Dynamic Nature of the Controller

The controller contains the factor

$$\frac{1}{s}.$$

Therefore,

$$\boxed{C(s) = \frac{k}{s}\widehat{C}(s)}$$

The term

$$\frac{1}{s}$$

represents an integral action.

Integral action is particularly useful for eliminating steady-state error.

For example, if the tracking error remains nonzero,

$$e_i(t) \neq 0,$$

the integral component continues to accumulate the error, causing the controller output to change until the steady-state error is driven toward zero.

Thus,

$$\boxed{\text{Integral Action} \rightarrow \text{Zero Steady-State Consensus Error}}$$

13.6 Distributed Output-Feedback Control

The proposed controller uses **output feedback**.

This means that the controller does not require complete knowledge of the internal states of every boost converter.

Instead, it uses the measurable output currents

$$Y_i(s).$$

Each converter receives information from its neighbouring converters through the communication network.

Therefore,

$$\boxed{\begin{array}{c}\text{Local Output} + \text{Neighbour Outputs} \\ + \text{Reference Information} \longrightarrow \text{Control Input}\end{array}}$$

This distributed structure avoids the need for a centralized controller that must collect all converter states and measurements.

13.7 Local Converter Output

For the i^{th} boost converter,

$$Y_i(s)$$

represents its output variable.

For the degaussing application, this can be associated with the converter output current.

The controller continuously compares this local output with the outputs of neighbouring converters.

For example, if converter i communicates with converters $j = 1$ and $j = 3$, the controller uses

$$Y_1(s) - Y_i(s)$$

and

$$Y_3(s) - Y_i(s).$$

These differences indicate whether the local converter is producing an output different from its neighbours.

13.8 Neighbouring-Converter Information

The communication topology is described by the adjacency coefficients

$$a_{ij}.$$

If

$$a_{ij} > 0,$$

then converter i and converter j have a communication connection.

If

$$a_{ij} = 0,$$

there is no direct communication connection.

The cooperative information received by agent i is

$$\boxed{\sum_{j=1}^{N} a_{ij} \left[Y_j(s) - Y_i(s) \right]}$$

This is the **consensus term**.

It measures the difference between the output of converter i and the outputs of its neighbouring converters.

13.9 Consensus Error

Define the local consensus error as

$$\boxed{e_{c,i}(s) = \sum_{j=1}^{N} a_{ij} \left[Y_j(s) - Y_i(s) \right]}$$

Suppose converter i has a higher current than its neighbours:

$$Y_i > Y_j.$$

Then

$$Y_j - Y_i < 0,$$

and the consensus term acts to reduce the difference.

Conversely, if

$$Y_i < Y_j,$$

the consensus term acts in the opposite direction to increase the local output toward the neighbouring values.

Therefore, the consensus mechanism continuously attempts to reduce

$$|Y_i - Y_j|.$$

At consensus,

$$Y_i = Y_j$$

for all connected agents, and hence

$$\boxed{e_{c,i} = 0}$$

13.10 Reference-Tracking Error

In addition to the neighbour information, the controller uses the reference signal

$$R_0(s).$$

The local reference-tracking error is

$$\boxed{e_{r,i}(s) = R_0(s) - Y_i(s)}$$

This term ensures that the converters do not merely agree with each other; they agree with the **desired reference**.

This distinction is important.

For example, it is possible for

$$Y_1 = Y_2 = Y_3 = Y_4$$

while all of them are different from the desired reference.

Consensus alone would not eliminate this error. The reference-tracking term solves this problem.

13.11 Pinning Gain

The reference-tracking term is multiplied by the pinning gain

$$p_i.$$

Thus,

$$\boxed{p_i[R_0(s) - Y_i(s)]}$$

If

$$p_i > 0,$$

the i^{th} converter is directly connected to the leader/reference.

If

$$p_i = 0,$$

the converter does not have a direct pinning connection.

However, through the connected communication network, reference information can propagate from pinned agents to other agents.

13.12 Complete Distributed Control Law

The proposed distributed control law is

$$\boxed{U_i(s) = C(s)\left[\sum_{j=1}^{N} a_{ij}\left(Y_j(s) - Y_i(s)\right) + p_i\left(R_0(s) - Y_i(s)\right)\right]}$$

Substituting

$$C(s) = \frac{k}{s}\widehat{C}(s),$$

we obtain

$$\boxed{U_i(s) = \frac{k}{s}\widehat{C}(s)\left[\sum_{j=1}^{N} a_{ij}(Y_j - Y_i) + p_i(R_0 - Y_i)\right]}$$

13.13 Explanation of Each Term

13.13.1 $U_i(s)$

$$\boxed{U_i(s)}$$

is the control input generated for the i^{th} boost converter.

In a practical implementation, this control signal ultimately determines the converter switching command or duty-ratio command after the appropriate control realization.

13.13.2 $C(s)$

$$\boxed{C(s) = \frac{k}{s}\widehat{C}(s)}$$

is the common dynamic controller.

Here,

$$k > 0$$

is the controller gain.

The same $C(s)$ is used for every converter.

13.13.3 a_{ij}

$$\boxed{a_{ij}}$$

represents the communication weight between converters i and j.

A larger a_{ij} means a stronger influence of converter j's output on converter i's cooperative control action.

13.13.4 $Y_j(s) - Y_i(s)$

$$\boxed{Y_j(s) - Y_i(s)}$$

represents the output difference between converter j and converter i.

This is the fundamental information used to achieve consensus.

13.13.5 p_i

$$\boxed{p_i}$$

is the pinning gain associated with converter i.

It determines the strength with which the converter is connected to the reference/leader.

13.13.6 $R_0(s) - Y_i(s)$

$$\boxed{R_0(s) - Y_i(s)}$$

is the local reference-tracking error.

It forces the converter output toward the commanded reference.

13.14 Consensus and Reference Terms Together

The controller contains two fundamentally different terms:

$$\boxed{\underbrace{\sum_{j=1}^{N} a_{ij}(Y_j - Y_i)}_{\text{Consensus Term}} + \underbrace{p_i(R_0 - Y_i)}_{\text{Reference-Tracking Term}}}$$

Therefore,

$$\boxed{\text{Control Action} = \text{Consensus Correction} + \text{Reference Correction}}$$

The consensus term ensures that the converters agree with one another.

The reference term ensures that the common value is the desired commanded value.

13.15 Matrix Form of the Control Law

Define

$$Y(s) = \begin{bmatrix} Y_1(s) \\ Y_2(s) \\ \vdots \\ Y_N(s) \end{bmatrix}$$

and

$$R(s) = \mathbf{1}_N R_0(s),$$

where

$$\mathbf{1}_N = \begin{bmatrix} 1 & 1 & \cdots & 1 \end{bmatrix}^T.$$

Using the graph Laplacian,

$$\mathcal{L} = \mathcal{D} - \mathcal{A},$$

the collective control law can be written as

$$\boxed{U(s) = C(s)\left[-\mathcal{L}Y(s) + \mathcal{P}\left(R(s) - Y(s)\right)\right]}$$

Equivalently,

$$\boxed{U(s) = C(s)\left[\mathcal{P}R(s) - (\mathcal{L} + \mathcal{P})Y(s)\right]}$$

The exact sign convention depends on how the feedback block is defined in the block diagram; the scalar-agent expression is the authoritative form for the proposed law.

13.16 Meaning of $\mathcal{L} + \mathcal{P}$

The matrix

$$\boxed{\mathcal{L} + \mathcal{P}}$$

is central to the complete cooperative-control system.

The Laplacian

$$\mathcal{L}$$

represents follower-to-follower communication, while

$$\mathcal{P}$$

represents leader-to-follower connections.

Hence,

$$\boxed{\mathcal{L} + \mathcal{P} = \text{Network Consensus Interaction} + \text{Reference Pinning}}$$

The assumed condition

$$\boxed{\mathcal{L} + \mathcal{P} > 0}$$

ensures the required connectivity property for the subsequent stability analysis.

13.17 Controller Applied to Heterogeneous Boost Converters

An important feature of the proposed approach is that

$$G_i(s)$$

can differ from one converter to another, while

$$C(s)$$

remains the same.

Thus,

$$\boxed{\text{Different Boost Plants} \quad + \quad \text{Same Controller}}$$

The controller therefore does not require an identical plant model for every converter.

This is particularly useful for the degaussing system because different converter branches may have different loads and electrical parameters.

13.18 Closed-Loop System

The complete closed-loop system can be represented conceptually as

$$\boxed{R_0(s) \to \text{Reference/Pinned Network} \to \text{Consensus Controller} \to \text{Boost Converters} \to Y(s)}$$

with the output fed back through the communication network.

More explicitly,

$$\boxed{\text{Reference} \to \begin{bmatrix} \text{Consensus} \\ +\text{Reference Tracking} \end{bmatrix} \to C(s) \to \widetilde{G}(s) \to Y(s)}$$

The output vector is then communicated among the agents to generate the next control action.

13.19 Why the Controller Is Called Distributed

The controller is called **distributed** because converter i does not require the complete state/output information of every converter.

Instead, it uses

$$Y_j(s)$$

only for those agents j that are its neighbours.

Therefore,

$$\boxed{\text{Local Information} + \text{Neighbour Information} \rightarrow \text{Local Control Action}}$$

There is no requirement for one central controller to calculate all U_i.

13.20 Stability Requirement

The source theorem considers a heterogeneous SNI multi-agent plant satisfying

$$G_i(0) > 0$$

and a connected communication network satisfying

$$\mathcal{L} + \mathcal{P} > 0.$$

The controller is chosen as

$$\boxed{C(s) = \frac{k}{s}\widehat{C}(s)}$$

with

$$k > 0$$

and $\widehat{C}(s)$ satisfying the required conditions.

The theorem states that there exists a finite

$$\boxed{k^* > 0}$$

such that, for

$$\boxed{0 < k \leq k^*}$$

the proposed negative-feedback interconnection remains asymptotically stable under the stated assumptions.

13.21 Characteristic-Loci Interpretation

The loop transfer function used for the stability analysis is

$$\boxed{L(s) = \frac{\widehat{C}(s)}{s}(\mathcal{L} + \mathcal{P})\widetilde{G}(s)}$$

up to the scalar gain k, depending on the chosen block-diagram normalization.

Let

$$\gamma_i(s)$$

denote the characteristic loci of $L(s)$.

The stability proof examines the trajectories

$$\gamma_i(j\omega)$$

in the complex plane.

The critical point associated with the feedback gain is

$$\boxed{-\frac{1}{k} + j0}$$

The objective is to choose k sufficiently small so that none of the characteristic loci encircle this critical point.

13.22 Role of the Three Characteristic-Loci Cases

The stability proof divides the s-plane boundary into three regions.

13.22.1 Case I: $s \in \psi_0$

This region corresponds to the small-s region near the origin.

The behaviour of the controller contains the integral factor

$$\frac{1}{s},$$

so the characteristic loci extend toward large magnitude as

$$s \to 0.$$

This produces the infinite-radius characteristic-locus behaviour.

13.22.2 Case II: $s \in \psi_{\pm j}$

This region corresponds to the imaginary-axis portions

$$s = \pm j\omega.$$

In this region,

$$\gamma_i(j\omega)$$

is examined using the frequency-domain properties of the SNI plant and mixed NI+PR controller.

The characteristic loci remain within the specified stability region.

13.22.3 Case III: $s \in \psi_R$

This region corresponds to the large-$|s|$ portion of the contour.

As

$$|s| \to \infty,$$

the asymptotic behaviour of the controller and plant determines the limiting behaviour of the characteristic loci.

The three cases together establish that the loci remain within the allowable region and do not encircle the critical point.

13.23 Consensus Error Dynamics

Define the collective consensus error as

$$\boxed{\xi(t) = \begin{bmatrix} \xi_1(t) & \xi_2(t) & \cdots & \xi_N(t) \end{bmatrix}^T}$$

For each agent,

$$\boxed{\xi_i(t) = y_i(t) - r_0(t)}$$

Hence,

$$\boxed{\xi(t) = Y(t) - \mathbf{1}_N r_0(t)}$$

The desired objective is

$$\boxed{\lim_{t\to\infty} \xi(t) = 0}$$

which directly means

$$\boxed{\lim_{t\to\infty} [y_i(t) - r_0(t)] = 0}$$

for every converter.

13.24 Steady-State Consensus

The source proof uses the final-value theorem to demonstrate that the steady-state consensus error is zero.

The corresponding expression has the form

$$\boxed{\xi_{ss} = \lim_{s\to 0} s\Xi(s)}$$

After substituting the closed-loop error dynamics and using the controller

$$C(s) = \frac{k}{s}\widehat{C}(s),$$

the steady-state expression contains

$$k\widehat{C}(0)(\mathcal{L} + \mathcal{P})\widetilde{G}(0).$$

Under the assumptions

$$\widetilde{G}(0) > 0,$$

$$\widehat{C}(0) > 0,$$

and

$$\mathcal{L} + \mathcal{P} > 0,$$

the source proof obtains

$$\boxed{\xi_{ss} = 0}$$

Therefore,

$$\boxed{\lim_{t\to\infty} [y_i(t) - r_0(t)] = 0}$$

13.25 Physical Interpretation

The operation of the controller can be understood using the interaction between neighbouring converter currents and the commanded reference.

Suppose

$$Y_1 > Y_2.$$

The consensus mechanism detects the difference

$$Y_2 - Y_1 < 0.$$

The controller therefore acts to reduce the mismatch.

Similarly, if

$$Y_1 < Y_2,$$

the controller produces a correction in the opposite direction.

At the same time, the pinning term compares the converter output with the commanded reference:

$$R_0 - Y_i.$$

Thus, the controller continuously performs two tasks:

$$\boxed{\text{1. Make Converters Agree}}$$

and

$$\boxed{\text{2. Make Their Common Value Equal to the Reference}}$$

13.26 Application to the Four-Boost-Converter System

For four boost converters,

$$G_1(s), G_2(s), G_3(s), G_4(s),$$

the controller is

$$C(s) = \frac{k}{s}\widehat{C}(s)$$

for all four agents.

The individual control laws are

$$\boxed{U_1 = C(s)\left[\sum_{j=1}^{4} a_{1j}(Y_j - Y_1) + p_1(R_0 - Y_1)\right]}$$

$$\boxed{U_2 = C(s)\left[\sum_{j=1}^{4} a_{2j}(Y_j - Y_2) + p_2(R_0 - Y_2)\right]}$$

$$\boxed{U_3 = C(s)\left[\sum_{j=1}^{4} a_{3j}(Y_j - Y_3) + p_3(R_0 - Y_3)\right]}$$

and

$$\boxed{U_4 = C(s)\left[\sum_{j=1}^{4} a_{4j}(Y_j - Y_4) + p_4(R_0 - Y_4)\right]}$$

Therefore, every converter independently calculates its own control signal using local and neighbouring information.

The desired final condition is

$$\boxed{Y_1 = Y_2 = Y_3 = Y_4 = R_0}$$

13.27 Complete Control Flow

The complete operation of the proposed control scheme can be summarized as

$$\boxed{R_0(t)}$$

$$\Downarrow$$

$$\boxed{\text{Reference Tracking}}$$

$$\Downarrow$$

$$\boxed{R_0 - Y_i}$$

along with the neighbour differences

$$\boxed{Y_j - Y_i}$$

which are weighted by

$$\boxed{a_{ij}}$$

These terms are combined as

$$\boxed{\sum_{j=1}^{N} a_{ij}(Y_j - Y_i) + p_i(R_0 - Y_i)}$$

The resulting signal is processed by

$$\boxed{C(s) = \frac{k}{s}\widehat{C}(s)}$$

to generate

$$\boxed{U_i(s)}$$

The control input is applied to the i^{th} boost converter, producing

$$\boxed{Y_i(s)}$$

The output is then fed back into the communication network, completing the distributed cooperative-control loop.

13.28 Final Summary

The proposed dynamic consensus controller is based on

$$\boxed{C(s) = \frac{k}{s}\widehat{C}(s), \qquad k > 0}$$

and the distributed control law

$$\boxed{U_i(s) = C(s)\left[\sum_{j=1}^{N} a_{ij}(Y_j(s) - Y_i(s)) + p_i(R_0(s) - Y_i(s))\right]}$$

The first term,

$$\boxed{\sum_{j=1}^{N} a_{ij}(Y_j - Y_i)}$$

forces the converter outputs toward consensus.

The second term,

$$\boxed{p_i(R_0 - Y_i)}$$

forces the converters toward the commanded reference.

Thus,

$$\boxed{\begin{aligned}&\text{Neighbour Information} + \text{Reference Information}\\ &\longrightarrow \text{Dynamic Consensus Controller} \longrightarrow \text{Boost Converter}\end{aligned}}$$

For the complete heterogeneous system,

$$\boxed{\widetilde{G}(s) = \operatorname{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}}$$

while the controller remains homogeneous:

$$\boxed{\widetilde{C}(s) = I_N \otimes C(s)}$$

Under the theorem's assumptions concerning the SNI converter models, positive DC gains, communication topology, and mixed NI+PR controller, a sufficiently small positive gain

$$\boxed{0 < k \leq k^*}$$

ensures asymptotic stability and

$$\boxed{\lim_{t\to\infty} [y_i(t) - r_0(t)] = 0, \qquad i = 1, \ldots, N}$$

Important:
The control architecture above can be retained for the boost-converter system. However, the SNI assumption must be verified for the specific boost-converter transfer function derived in the previous section. The calculated duty-to-output boost-converter model contains a right-half-plane zero, so it should not automatically be classified as SNI without carrying out the required NI/SNI verification.

14 Controller Design Guidelines

The controller design is an important part of the proposed dynamic consensus control scheme because the performance of the overall multi-agent system depends strongly on the selection of the controller transfer function. The controller must not only provide the required consensus behaviour but also maintain the stability properties required by the mixed NI+PR framework.

The proposed controller is selected in the form

$$\boxed{C(s) = \frac{k}{s}\widehat{C}(s)}$$

where

$$\boxed{k > 0}$$

is a scalar controller gain and $\widehat{C}(s)$ represents the dynamic part of the controller.

14.1 Selection of $C_p(s)$

The controller can be written as

$$\boxed{C(s) = \frac{k}{s}C_p(s)}$$

where $C_p(s)$ is selected to satisfy the required **Positive Real (PR)** properties.

The factor

$$\frac{1}{s}$$

provides integral action, while $C_p(s)$ determines the remaining dynamic characteristics of the controller.

Therefore, the controller can be viewed as

$$\boxed{\text{Controller} = \text{Integral Action} \times \text{PR Dynamic Component}}$$

For the multi-agent boost-converter system, the same $C_p(s)$ is applied to all converter agents.

14.2 Minimum-Phase Requirement

The controller component $C_p(s)$ should possess a **minimum-phase (MP)** property.

A minimum-phase transfer function has all of its zeros located in the left-half plane.

If

$$C_p(s) = K\frac{(s+z_1)(s+z_2)\cdots}{(s+p_1)(s+p_2)\cdots},$$

then the zeros should satisfy

$$\boxed{\mathrm{Re}(z_i) < 0}$$

Similarly, the poles must remain in the stable region.

The minimum-phase property is important because undesirable right-half-plane zeros could introduce additional phase lag and adversely affect the characteristic-loci behaviour.

Therefore,

$$\boxed{\text{Minimum Phase} \rightarrow \text{Favourable Phase Behaviour} \rightarrow \text{Improved Stability Margin}}$$

14.3 Positive DC Gain

The controller must satisfy

$$\boxed{C_p(0) > 0}$$

This means that the controller has a positive gain at zero frequency.

If

$$C_p(s) = K\frac{\prod_i (s + z_i)}{\prod_i (s + p_i)},$$

then

$$\boxed{C_p(0) = K\frac{\prod_i z_i}{\prod_i p_i}}$$

The parameters must therefore be selected such that

$$\boxed{C_p(0) > 0}$$

This condition is important in the steady-state consensus analysis because the zero steady-state error result depends on the positivity of the controller's DC behaviour.

14.4 Relative Degree

The **relative degree (RD)** of a transfer function is the difference between the number of poles and the number of zeros.

If

$$C_p(s) = \frac{N(s)}{D(s)},$$

then

$$\boxed{RD = \deg[D(s)] - \deg[N(s)]}$$

The controller structure is constrained by the relative-degree requirement.

The source specifies that the relative degree of $C(s)$ should not exceed two:

$$\boxed{RD(C) \leq 2}$$

Since the controller contains the additional factor

$$\frac{1}{s},$$

the dynamic component $C_p(s)$ is selected accordingly.

14.5 Strictly Proper and Biproper Controllers

Two important controller configurations can be considered.

14.5.1 Case 1: Strictly Proper Controller

A strictly proper controller has

$$\boxed{RD = 1}$$

for $C_p(s)$ according to the source's recommended configuration.

Such a controller has more poles than zeros.

For example,

$$\boxed{C_p(s) = K\frac{s+z}{(s+p_1)(s+p_2)}}$$

has

$$RD = 1.$$

14.5.2 Case 2: Biproper Controller

A biproper controller has

$$\boxed{RD = 0}$$

Therefore, it has the same number of poles and zeros.

A typical form is

$$\boxed{C_p(s) = K\frac{(s+z_1)(s+z_2)}{(s+p_1)(s+p_2)}}$$

The choice between these configurations depends on the dynamics of the SNI plant and the desired transient response.

14.6 Pole-Zero Structure

The controller can be constructed using real pole-zero pairs.

A general biproper form is

$$\boxed{C_p(s) = \prod_{i=1}^{n} \frac{s + z_i}{s + p_i}}$$

Here,

- p_i = controller pole location,
- z_i = controller zero location.

The pole-zero pairs can be ordered according to

$$\boxed{|p_i| < |p_{i+1}|}$$

and

$$\boxed{|z_i| < |z_{i+1}|}$$

This provides an organized way to select the controller dynamics over different frequency ranges.

14.7 Pole-Zero Placement

Pole-zero placement is particularly important because it directly influences the transient response.

Consider a controller of the form

$$\boxed{C_p(s) = K \frac{s + z}{s + p}}$$

The pole is located at

$$s = -p$$

and the zero at

$$s = -z.$$

Their relative locations determine how strongly the controller affects the response.

If the zero is placed close to the pole, their effects partially cancel.

If the zero is placed far away from the pole, the controller introduces a stronger change in magnitude and phase.

Therefore,

$$\boxed{\text{Pole-Zero Placement} \rightarrow \text{Magnitude/Phase Shaping} \rightarrow \text{Transient-Response Shaping}}$$

14.8 Recommended Pole-Zero Relationship

For minimum-phase SNI plants, the source recommends that the controller poles should dominate the corresponding zeros.

This means

$$\boxed{|p_i| < |z_i|}$$

The zero is therefore placed farther from the origin than its corresponding pole.

For example,

$$p_i = 1$$

and

$$z_i = 6$$

would satisfy

$$|p_i| < |z_i|.$$

The source suggests that the zero locations can often be selected approximately 6–10 times farther from the origin than the corresponding pole locations to obtain a useful trade-off between settling time and peak overshoot.

Thus,

$$\boxed{|z_i| \approx (6\text{–}10)|p_i|}$$

This recommendation is based on the simulation studies described in the source, rather than being a universal rule for every plant.

14.9 Effect of Pole-Zero Placement on Settling Time

The pole locations strongly affect the speed of the controller response.

A pole closer to the origin corresponds to slower dynamics, while moving the pole farther into the left-half plane generally produces faster dynamics.

For a real pole at

$$s = -p,$$

the associated time constant is approximately

$$\boxed{\tau = \frac{1}{p}}$$

Therefore, increasing p decreases the time constant:

$$p \uparrow \quad \Rightarrow \quad \tau \downarrow .$$

Hence, the response generally becomes faster.

14.10 Effect on Overshoot

Pole-zero placement also influences the peak overshoot.

A controller designed solely for very fast response may produce excessive transient amplification.

Therefore, the controller design must balance

$$\boxed{\text{Fast Response}}$$

against

$$\boxed{\text{Low Overshoot}}$$

The source specifically identifies a trade-off between **settling time**, which determines the speed of consensus, and **peak overshoot**, which determines the magnitude of the transient response.

14.11 Speed–Overshoot Trade-Off

The controller design can therefore be understood as an optimization between two competing requirements.

14.11.1 Faster Consensus

Move the controller dynamics toward faster locations.

This reduces the settling time:

$$\boxed{T_s \downarrow}$$

However, excessive aggressiveness can increase the peak transient:

$$\boxed{M_p \uparrow}$$

14.11.2 Lower Overshoot

A more conservative pole-zero configuration can reduce the peak response:

$$\boxed{M_p \downarrow}$$

but may increase the settling time:

$$\boxed{T_s \uparrow}$$

Therefore,

$$\boxed{\text{Controller Design} = \text{Trade-Off Between Settling Time and Overshoot}}$$

14.12 Influence of the SNI Plant

The appropriate controller configuration depends strongly on the characteristics of the SNI plant.

For a plant with

$$\tilde{G}(s)$$

having poles and zeros that are well separated from the imaginary axis, the controller poles and zeros can be selected using the recommended pole-zero relationships.

However, if the plant contains poles or complex pole-zero pairs close to

$$j\omega,$$

then a different configuration is required.

This is because poles and zeros near the imaginary axis have a stronger effect on the frequency-domain magnitude and phase.

14.13 Plant Poles and Controller Zeros

If the SNI plant contains poles or complex pole-zero pairs close to the $j\omega$-axis, the controller zeros should dominate the corresponding plant poles.

The source expresses this condition as

$$\boxed{|z_i| \ll |p_i|}$$

for the relevant case, where the notation refers to the locations of the corresponding plant/controller features in the source's design discussion.

The purpose is to ensure that the controller provides the required frequency-domain shaping without producing undesirable transient behaviour.

The exact pole-zero placement must therefore be determined from the actual boost-converter model.

14.14 Complex-Conjugate Poles and Zeros

If the converter plant has complex-conjugate poles or zeros near the imaginary axis, their influence can be significant.

For example, a pair of poles can be written as

$$\boxed{p_{1,2} = -\sigma \pm j\omega_n}$$

Such poles can produce oscillatory behaviour.

If the controller is strictly proper with relative degree

$$\boxed{RD = 1}$$

the controller can be selected with poles dominating its zeros:

$$\boxed{|p_i| < |z_i|}$$

This configuration can provide satisfactory transient performance according to the design guidelines given in the source.

14.15 Controller Design Procedure

The controller design can be carried out systematically.

14.15.1 Step 1: Obtain the Converter Model

Determine

$$G_i(s)$$

for each boost converter.

For the heterogeneous system,

$$\boxed{\widetilde{G}(s) = \mathrm{diag}\{G_1(s), G_2(s), \ldots, G_N(s)\}}$$

14.15.2 Step 2: Verify the Required Plant Properties

Determine whether the individual boost-converter models satisfy the required SNI conditions.

This step is particularly important for the boost-converter model because the calculated duty-to-output transfer function contains a right-half-plane zero.

14.15.3 Step 3: Select $C_p(s)$

Choose

$$\boxed{C_p(s)}$$

such that it satisfies the required PR and minimum-phase conditions.

14.15.4 Step 4: Ensure Positive DC Gain

Check

$$\boxed{C_p(0) > 0}$$

14.15.5 Step 5: Select the Relative Degree

Ensure that the resulting controller satisfies the specified relative-degree limitation.

14.15.6 Step 6: Place Poles and Zeros

Select

$$p_i, \qquad z_i$$

according to the characteristics of the converter plant.

For the recommended minimum-phase SNI case,

$$\boxed{|p_i| < |z_i|}$$

and the source suggests approximately

$$\boxed{|z_i| \approx 6\text{–}10|p_i|}$$

14.15.7 Step 7: Adjust Transient Performance

Simulate the closed-loop system and examine:

- settling time,
- peak overshoot,
- oscillations,
- current synchronization,
- reference tracking.

14.15.8 Step 8: Select the Scalar Gain k

After selecting $C_p(s)$, the gain k is adjusted while respecting the stability requirement

$$\boxed{0 < k \leq k^*}$$

14.16 Controller for the Heterogeneous Boost-Converters

The same controller is applied to all boost converters.

Thus,

$$\boxed{C_1(s) = C_2(s) = \cdots = C_N(s) = C(s)}$$

Meanwhile,

$$\boxed{G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s)}$$

This gives the characteristic structure of the proposed method:

$$\boxed{\underbrace{\text{Heterogeneous Boost-Converters}}_{G_i(s)} + \underbrace{\text{Homogeneous Controller}}_{C(s)}}$$

The controller therefore does not have to be redesigned independently for every converter, provided the theoretical conditions of the proposed framework are satisfied.

14.17 Relation to the Dynamic Consensus Controller

The controller designed here is inserted into the distributed control law

$$\boxed{U_i(s) = C(s)\left[\sum_{j=1}^{N} a_{ij}(Y_j(s) - Y_i(s)) + p_i(R_0(s) - Y_i(s))\right]}$$

Therefore, the controller design and graph-theoretic structure work together.

The communication network determines

$$\mathcal{L} + \mathcal{P},$$

while the controller determines

$$C(s).$$

Together with the converter plant,

$$G_i(s),$$

they form the complete closed-loop system.

14.18 Design Flow

The complete design process can be represented as

$$\boxed{\text{Boost Converter Model}}$$

$$\Downarrow$$

$$\boxed{\text{Determine } G_i(s)}$$

$$\Downarrow$$

$$\boxed{\text{Check SNI/NI Characteristics}}$$

$$\Downarrow$$

$$\boxed{\text{Select } C_p(s)}$$
$$\Downarrow$$
$$\boxed{C(s) = \frac{k}{s} C_p(s)}$$
$$\Downarrow$$
$$\boxed{\text{Select Pole-Zero Locations}}$$
$$\Downarrow$$
$$\boxed{\text{Check PR and Minimum-Phase Conditions}}$$
$$\Downarrow$$
$$\boxed{\text{Adjust } k}$$
$$\Downarrow$$
$$\boxed{\text{Check Stability and Transient Performance}}$$
$$\Downarrow$$
$$\boxed{\text{Final Cooperative Controller}}$$

14.19 Practical Interpretation

For the proposed multi-boost-converter system, the controller parameters should be selected with two objectives in mind.

14.19.1 Stability

The characteristic loci must remain away from the critical point associated with the selected gain.

$$\boxed{\text{No Critical-Point Encirclement} \rightarrow \text{Stable Closed Loop}}$$

14.19.2 Performance

The converter outputs should reach the reference rapidly without excessive oscillation:

$$\boxed{y_i(t) \rightarrow r_0(t)}$$

with an acceptable

$$\boxed{T_s}$$

and

$$\boxed{M_p}.$$

Thus,

$$\boxed{\text{Stability} + \text{Fast Consensus} + \text{Low Overshoot}}$$

are the three principal controller-design objectives.

14.20 Important Point for the Boost-Converter Implementation

The controller-selection guidelines shown in the source are developed for an **SNI plant**. Therefore, when transferring this methodology to the boost converter, the controller should **not be selected merely by copying the pole-zero values from the buck-converter case**.

For the boost converter derived earlier,

$$\boxed{G_{vd}(s) = \frac{23000 - 5750s}{s^2 + 10s + 40}}$$

with a right-half-plane zero at

$$\boxed{s = +4}$$

Therefore, before claiming that the boost converter satisfies the same SNI assumptions as the source model, the NI/SNI property must be explicitly verified for the chosen input-output transfer function and operating point.

The controller-selection procedure should consequently be

$$\boxed{G_{vd}(s) \to \text{NI/SNI Verification} \to C_p(s) \to C(s) \to \text{Characteristic-Loci Verification}}$$

This is especially important because the **right-half-plane zero of the boost converter changes its phase characteristics** and can significantly influence the feasibility of applying the source's mixed NI+PR controller-design rules directly.

15 Simulation Results

The complete simulation study was carried out in **MATLAB/Simulink** to evaluate the mathematical model and the proposed distributed dynamic consensus control scheme for the interconnected converter system. The simulation combines the state-space modelling, transfer-function representation, multi-agent interconnection, Laplacian-based communication network, reference tracking, and dynamic output-feedback controller.

The simulation was performed with a total simulation time of

$$\boxed{T_{\text{sim}} = 50 \text{ s}}$$

and the converter operating point used in the MATLAB calculation was

$$\boxed{D = 0.8}$$

with an input voltage of

$$\boxed{V_{\text{in}} = 23 \text{ V}}$$

15.1 MATLAB-Based Mathematical Model

The converter model was initially represented using two switching-state matrices,

$$\boxed{A_1 = \begin{bmatrix} -10 & 0 \\ 0 & -0.001 \end{bmatrix}}$$

and

$$\boxed{A_2 = \begin{bmatrix} -10 & 10 \\ -10 & -0.001 \end{bmatrix}}$$

The averaged state matrix was obtained using

$$\boxed{A = A_1 D + A_2 (1 - D)}$$

For

$$\boxed{D = 0.8}$$

we obtain

$$A = 0.8 \begin{bmatrix} -10 & 0 \\ 0 & -0.001 \end{bmatrix} + 0.2 \begin{bmatrix} -10 & 10 \\ -10 & -0.001 \end{bmatrix}.$$

Therefore,

$$\boxed{A = \begin{bmatrix} -10 & 2 \\ -2 & -0.001 \end{bmatrix}}$$

The input, output and feedthrough matrices used in the simulation are

$$\boxed{B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad D_0 = 0}$$

The selected state initial condition was

$$\boxed{x(0) = \begin{bmatrix} 7500 \\ -15000 \end{bmatrix}}$$

Thus, the state-space representation used in the simulation is

$$\boxed{\dot{x} = Ax + Bu}$$

and

$$\boxed{y = Cx}$$

15.2 Calculated Transfer Function

Using the state-space model, the MATLAB calculation produced the transfer function

$$\boxed{t_f(s) = \frac{-1.5 \times 10^5 (s + 1.001)(s + 0.4185)(s + 9.583)}{(s + 9.583)^2 (s + 0.4185)^2}}$$

This transfer function represents the input-output dynamic behaviour used in the multi-agent simulation.

The three-agent model was subsequently constructed using

$$\boxed{m_{tf} = I_3 t_f(s)}$$

where

$$\boxed{I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

Consequently,

$$\boxed{m_{tf}(s) = \begin{bmatrix} t_f(s) & 0 & 0 \\ 0 & t_f(s) & 0 \\ 0 & 0 & t_f(s) \end{bmatrix}}$$

This representation allows the same calculated converter transfer function to be applied to the three converter agents in the simulation.

15.3 Controller Used in the Simulation

The controller was implemented as the diagonal three-agent transfer-function matrix

$$\boxed{C_1(s) = \begin{bmatrix} C(s) & 0 & 0 \\ 0 & C(s) & 0 \\ 0 & 0 & C(s) \end{bmatrix}}$$

where

$$\boxed{C(s) = -\frac{0.00013333(s+10)}{s(s+1)}}$$

Therefore,

$$\boxed{C_1(s) = -\frac{0.00013333(s+10)}{s(s+1)} I_3}$$

The controller contains an integral term through the factor

$$\boxed{\frac{1}{s}}$$

The integral action enables the controller to continuously correct the difference between the reference current and the measured converter output.

15.4 Simulink Implementation

The complete Simulink model consists of the following principal components:

1. **Initial-condition/reference input**
2. **Gain block `K*u`**
3. **Consensus error calculation**
4. **Multi-agent transfer-function model `m_tf`**
5. **Reference-current input**
6. **Laplacian communication block**
7. **Dynamic controller `C_1(s)`**
8. **Output-feedback path**
9. **Scope for observing the converter responses**

The overall signal flow can be represented as

$$\boxed{R_0 \to \text{Reference/Consensus Error} \to C_1(s) \to m_{tf} \to Y}$$

with the converter outputs simultaneously supplied to the communication network.

The Laplacian block provides the information exchange required for cooperative operation.

15.5 Multi-Agent Communication

The three converter agents do not operate completely independently.

The output of each converter is used by the communication network to generate the cooperative correction signal. The Laplacian block represents the communication relationship between the converter agents.

The controller therefore uses the difference between neighbouring converter outputs:

$$\boxed{Y_j(s) - Y_i(s)}$$

together with the reference error

$$\boxed{R_0(s) - Y_i(s)}$$

The general distributed control law is

$$\boxed{U_i(s) = C(s)\left[\sum_{j=1}^{N} a_{ij}\left(Y_j(s) - Y_i(s)\right) + p_i\left(R_0(s) - Y_i(s)\right)\right]}$$

For the three-agent simulation,

$$\boxed{N = 3}$$

Hence, the control input of each converter depends on both the local tracking error and the information obtained through the communication network.

15.6 Reference Current

A step reference was applied to the system through the block labelled

`reference_current_for_boost`

The reference represents the desired common current that the converter agents should achieve.

The desired steady-state objective is therefore

$$\boxed{y_i(t) \to r_0(t), \qquad i = 1, 2, 3}$$

At the same time, the agents must satisfy

$$\boxed{y_1(t) - y_2(t) \to 0}$$

and

$$\boxed{y_2(t) - y_3(t) \to 0}$$

Thus, the simulation evaluates both **reference tracking** and **current consensus**.

15.7 Initial Transient Response

Immediately after the reference/current command is applied, the three converter outputs exhibit a large transient response.

The Scope shows that the three responses initially have different amplitudes and phases. Large positive and negative excursions are observed during the first few seconds.

This initial behaviour is expected from the dynamic interaction between:

- the converter dynamics,
- the controller,
- the initial conditions,
- the reference input, and
- the communication network.

The transient can therefore be described as

$$\boxed{\text{Reference application} \rightarrow \text{Large transient} \rightarrow \text{Oscillatory response} \rightarrow \text{Damping} \rightarrow \text{Consensus}}$$

15.8 Damped Oscillatory Behaviour

A significant characteristic visible in the Scope is the progressive reduction of the oscillation amplitude.

The Peak Finder identifies representative peaks of approximately

$$\boxed{2.46 \times 10^4}$$

at

$$\boxed{t = 0.873 \text{ s}}$$

approximately

$$\boxed{2.01 \times 10^4}$$

at

$$\boxed{t = 2.334 \text{ s}}$$

and approximately

$$\boxed{1.61 \times 10^4}$$

at

$$\boxed{t = 3.727 \text{ s}}$$

The peak sequence is therefore

$$\boxed{2.46 \times 10^4 > 2.01 \times 10^4 > 1.61 \times 10^4}$$

This reduction demonstrates that the transient oscillations are progressively attenuated.

15.9 Stability Observation

The simulation does not show an increasing or divergent response.

Instead, the oscillations decrease progressively as time increases. After the initial transient, the three responses approach a common operating region.

The observed behaviour therefore indicates stable closed-loop operation under the simulated conditions.

Mathematically, the desired behaviour is

$$\boxed{\lim_{t\to\infty} y_i(t) = y_{\text{ss}}, \qquad i = 1, 2, 3}$$

The simulation result supports this convergence behaviour over the 50-s simulation interval.

15.10 Consensus Behaviour

One of the most important results obtained from the Scope is the convergence of the three converter responses.

At the beginning of the simulation,

$$\boxed{y_1(t) \neq y_2(t) \neq y_3(t)}$$

The difference between the agents is particularly noticeable during the initial transient.

As the cooperative controller operates, the differences progressively decrease:

$$\boxed{|y_1 - y_2| \to 0}$$

$$\boxed{|y_2 - y_3| \to 0}$$

and

$$\boxed{|y_1 - y_3| \to 0}$$

The three responses eventually become closely synchronized.

This confirms the principal purpose of the Laplacian-based distributed control structure.

15.11 Reference Tracking and Consensus

The simulation simultaneously demonstrates two control objectives.

15.11.1 Objective 1: Consensus

The outputs of the converter agents should agree:

$$\boxed{y_1 = y_2 = y_3}$$

15.11.2 Objective 2: Reference Tracking

The common output should follow the commanded reference:

$$\boxed{y_i = r_0}$$

Therefore, the complete desired condition is

$$\boxed{y_1(t) \approx y_2(t) \approx y_3(t) \approx r_0(t)}$$

The simulation shows that the three responses converge toward a common steady-state operating value.

15.12 Effect of the Laplacian Controller

The Laplacian block is essential for coordinating the individual converter outputs.

Without cooperative information, each converter would respond primarily according to its own local tracking error.

With the Laplacian feedback, the controller additionally considers the difference between neighbouring outputs.

The cooperative component is

$$\boxed{\sum_{j=1}^{N} a_{ij}(Y_j - Y_i)}$$

When there is a difference between two converter currents, this term generates a corrective control action.

As the converter currents approach one another,

$$\boxed{Y_j - Y_i \to 0}$$

and consequently the consensus component approaches zero.

Thus,

$$\boxed{\text{Current mismatch} \to \text{Laplacian correction} \to \text{Reduced mismatch}}$$

15.13 Long-Term Response

The simulation was conducted up to

$$\boxed{t = 50 \text{ s}}$$

The Scope shows that the large oscillations are concentrated primarily in the initial portion of the simulation.

As time increases, the oscillation amplitude becomes progressively smaller. After approximately 20 s, the responses are much closer to the final operating region, and they remain closely grouped for the remainder of the simulation.

At the end of the simulation,

$$\boxed{t = 50 \text{ s}}$$

the three traces remain close to one another.

15.14 Simulation Results Summary

The complete simulation results can be summarized as follows:

Parameter/Observation	**Simulation Result**
Duty ratio	$D = 0.8$
Input voltage	$V_{\text{in}} = 23$ V
Simulation time	50 s
Number of agents	3
Initial state	$\begin{bmatrix} 7500 & -15000 \end{bmatrix}^T$
State matrix	$\begin{bmatrix} -10 & 2 \\ -2 & -0.001 \end{bmatrix}$
Controller	$-\dfrac{0.00013333(s+10)}{s(s+1)}$
First prominent peak	2.46×10^4 at 0.873 s
Second prominent peak	2.01×10^4 at 2.334 s
Third prominent peak	1.61×10^4 at 3.727 s
Transient behaviour	Oscillatory and damped
Stability observation	Response remains bounded and convergent
Consensus	Three outputs converge toward one another
Long-term behaviour	Closely synchronized responses

15.15 Overall Performance

The simulation demonstrates that the implemented multi-agent control structure is capable of coordinating the three converter outputs.

The main observations are:

Stable Transient Response

Progressive Damping of Oscillations

Current Synchronization

and

Convergence Toward a Common Reference

The decreasing peak values observed in the Scope provide direct evidence of damping:

$$\boxed{2.46 \times 10^4 \to 2.01 \times 10^4 \to 1.61 \times 10^4 \to \cdots}$$

while the convergence of the three traces demonstrates the consensus behaviour.

15.16 Final Report Paragraph

Simulation Result:
The proposed distributed dynamic consensus control scheme was implemented in MATLAB/Simulink for a three-agent interconnected converter system. The mathematical model was evaluated at a duty ratio of $D = 0.8$ and an input voltage of 23 V, with a simulation duration of 50 s. The averaged state-space model was obtained as

$$A = \begin{bmatrix} -10 & 2 \\ -2 & -0.001 \end{bmatrix},$$

with

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad D_0 = 0.$$

The resulting transfer-function model was replicated for the three converter agents through

$$m_{tf} = I_3 t_f(s).$$

A dynamic output-feedback controller

$$C_1(s) = -\frac{0.00013333(s+10)}{s(s+1)} I_3$$

was incorporated with a Laplacian-based communication network. The simulated responses exhibit an initial oscillatory transient followed by progressive damping. Representative peaks of 2.46×10^4, 2.01×10^4, and 1.61×10^4 were observed at 0.873, 2.334, and 3.727 s, respectively, confirming the decreasing amplitude of the transient oscillations. As the simulation progresses, the three converter responses converge toward a common operating value and become closely synchronized. Hence, the simulation demonstrates bounded and damped closed-loop behaviour together with the desired multi-agent current-consensus response under the implemented control structure.

Note:
The numerical matrices and transfer function above are reproduced from the MATLAB results provided in the simulation material. The simulation results establish the observed three-agent response and damping; however, an exact numerical settling time, steady-state error, or percentage overshoot should only be reported after extracting the corresponding quantities from the underlying simulation data.

15.17 Simulink Implementation of the Proposed Control Scheme

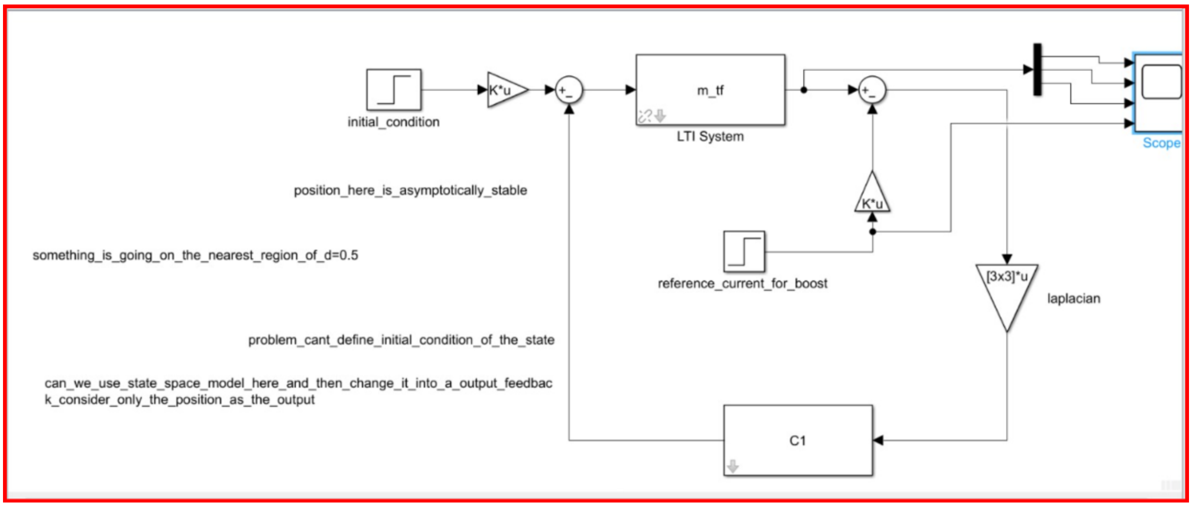


Figure 1: Simulink implementation of the proposed cooperative dynamic consensus control scheme.

The model consists of the initial-condition input, gain block, summation blocks, the multi-agent converter model m_{tf}, the reference-current input, the Laplacian communication block, the dynamic controller C_1, and the output Scope.

The converter outputs are fed back through the Laplacian-based communication network, allowing the individual converter agents to exchange output information and achieve cooperative current regulation.

15.18 Simulation Results

The simulation results of the proposed cooperative control scheme are shown in Fig. 2. The responses of the three converter outputs, namely Demux/1, Demux/2, and Demux/3, are presented along with the applied step reference.

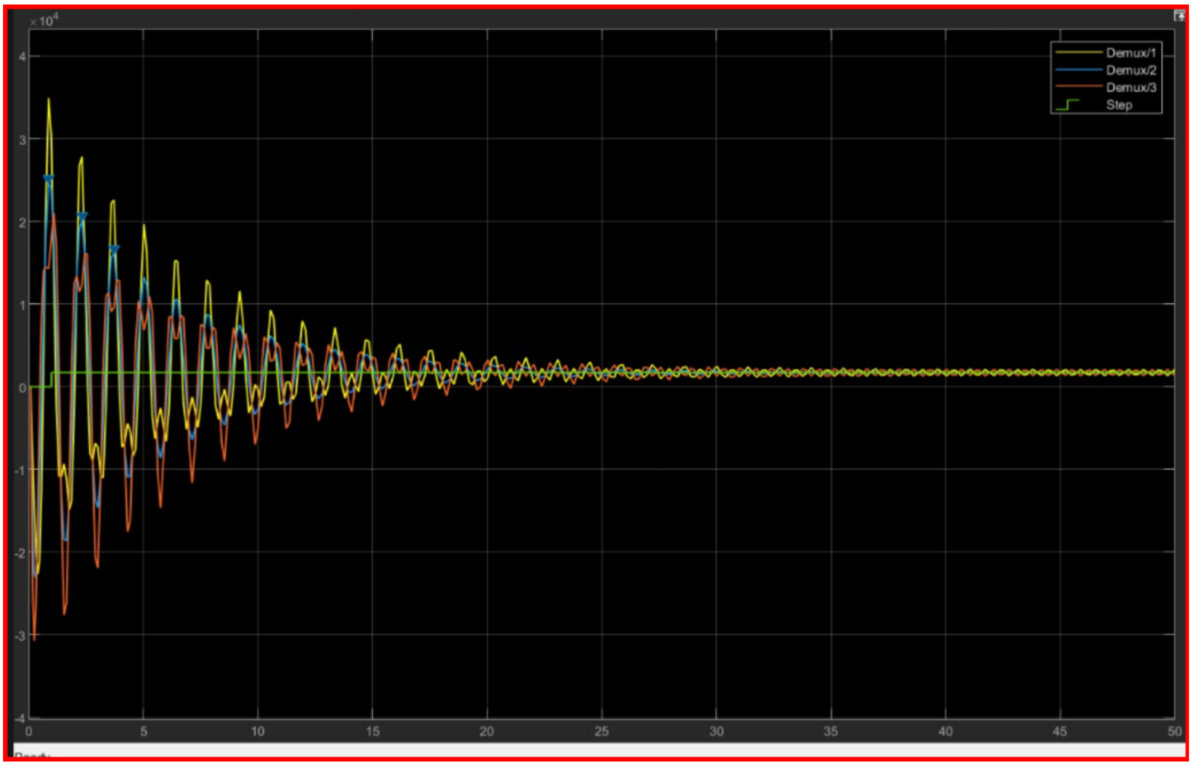


Figure 2: Simulation results showing the responses of the three converter outputs under the applied step reference.

It can be observed that the three converter responses exhibit an initial transient oscillation following the application of the step reference. The amplitude of the oscillations gradually decreases with time, indicating that the system is asymptotically stable.

As the transient response decays, the outputs of Demux/1, Demux/2, and Demux/3 converge toward a common steady-state value. This convergence demonstrates the current-consensus behaviour achieved by the proposed cooperative control strategy.

The simulation therefore confirms that the proposed controller is capable of coordinating the converter responses while maintaining stable convergence toward the desired operating point.

16 Advantages of the Proposed Method

The proposed dynamic consensus control method offers several advantages for the coordinated operation of heterogeneous converter systems. These advantages arise from the combination of distributed cooperative control, output feedback, mixed NI+PR controller characteristics, and the ability to accommodate time-varying reference signals.

16.1 Operation with Heterogeneous Converters

A major advantage of the proposed method is that it is designed for a **heterogeneous multi-agent system**, where the individual converter dynamics need not be identical.

The converter agents can have different electrical parameters and consequently different transfer functions:

$$\boxed{G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s)}$$

Despite these differences, a homogeneous controller configuration can be applied to the converter network.

This is particularly useful in practical multi-converter systems because different converters may have different component values, operating conditions, or load characteristics.

16.2 Distributed Control Structure

The proposed controller uses a distributed communication structure rather than relying on a centralized controller.

Each converter uses information from its neighbouring converters through the communication network.

The cooperative term is given by

$$\boxed{\sum_{j=1}^{N} a_{ij} \left(Y_j(s) - Y_i(s)\right)}$$

Therefore, the control action of converter i depends on the differences between its own output and the outputs of connected neighbouring converters.

This makes the control architecture suitable for systems containing multiple physically separated converter units.

16.3 Output-Feedback Based Control

The proposed method uses **output feedback** instead of requiring complete state feedback.

This is an important practical advantage because measuring every internal state of a converter may require additional sensors or state-estimation techniques.

The controller primarily uses the measurable converter output, such as the degaussing-coil current.

Thus, the architecture can be represented as

$$\boxed{\text{Measured Output} \rightarrow \text{Cooperative Controller} \rightarrow \text{Control Input}}$$

This reduces the measurement requirements of the control system.

16.4 No Requirement for Full-State Information

Traditional state-feedback control requires knowledge of the complete state vector

$$\boxed{x_i = \begin{bmatrix} x_{i1} & x_{i2} & \cdots & x_{in} \end{bmatrix}^T}$$

In practical converter applications, some of these states may not be directly measurable.

The proposed output-feedback approach avoids this requirement by using the converter output y_i and information received from neighbouring agents.

Therefore,

$$\boxed{\text{No complete state measurement} \rightarrow \text{Simpler practical implementation}}$$

16.5 Support for Time-Varying References

The proposed controller is not restricted to a constant reference.

The reference signal can be represented as

$$\boxed{r_0(t)}$$

allowing the desired converter current to change with time.

The reference-tracking component is

$$\boxed{p_i \left(R_0(s) - Y_i(s) \right)}$$

which continuously acts to reduce the difference between the commanded reference and the local converter output.

Therefore, the desired behaviour is

$$\boxed{y_i(t) \rightarrow r_0(t)}$$

This is particularly relevant when the required degaussing current changes according to the operating condition of the ship.

16.6 Current Consensus

The principal objective of the proposed cooperative controller is to achieve consensus among the converter outputs.

For N converter agents, the desired condition is

$$\boxed{y_1(t) = y_2(t) = \cdots = y_N(t)}$$

Equivalently,

$$\boxed{y_i(t) - y_j(t) \to 0}$$

The Laplacian-based communication structure continuously reduces differences between neighbouring converter outputs.

Consequently, the converter currents can synchronize while simultaneously tracking the desired reference.

16.7 Asymptotic Stability

The proposed controller is theoretically designed to maintain asymptotic stability under the assumptions established in the analysis.

The stability analysis uses the characteristic-loci framework together with the mixed NI+PR properties of the controller and the SNI properties of the converter plants considered in the source framework.

The desired closed-loop behaviour is

$$\boxed{\lim_{t\to\infty} y_i(t) = r_0(t)}$$

for the applicable steady-state/reference conditions.

This provides a theoretical basis for the observed convergence in simulation.

16.8 Robustness Against Disturbances

The simulation study considers output disturbances occurring at different times for the converter agents.

The converter outputs may temporarily deviate from their desired trajectories when a disturbance occurs. However, the cooperative controller acts to correct these deviations.

The response therefore follows the general pattern

$$\boxed{\text{Disturbance} \rightarrow \text{Transient Deviation} \rightarrow \text{Controller Correction} \rightarrow \text{Recovery}}$$

The simulation results demonstrate acceptable behaviour under the disturbance conditions considered.

However, the **quantitative limits of disturbance tolerance are not fully established** in the source study and should not be interpreted as unlimited disturbance rejection.

16.9 Flexible Controller Selection

The proposed framework does not restrict the controller to only one particular transfer-function structure.

The controller has the general form

$$\boxed{C(s) = \frac{k}{s}\widehat{C}(s)}$$

where $k > 0$ and $\widehat{C}(s)$ must satisfy the required properties.

This provides flexibility in selecting controller poles and zeros according to the characteristics of the converter plant.

The controller can therefore be designed to obtain a suitable balance between:

- stability,
- settling time,
- overshoot,
- damping, and
- consensus speed.

16.10 Applicability to Ship Degaussing Systems

The proposed method is particularly relevant to ship degaussing applications.

A ship may employ several degaussing coils supplied by independent converter units. The magnetic field generated by the coils depends strongly on their currents.

Therefore, accurate coordination of the converter currents is important.

The proposed architecture provides

$$\boxed{\text{Multiple Converter Agents} \rightarrow \text{Cooperative Current Control} \rightarrow \text{Synchronized Degaussing Curre}}$$

This makes the method suitable for studying coordinated current control in multi-converter degaussing systems.

17 Limitations of the Proposed Method

Although the proposed method demonstrates satisfactory performance, several limitations must be clearly identified.

17.1 Switching-Frequency Analysis

The study considers different switching frequencies through simulation, but a detailed theoretical analysis of the influence of switching frequency is not provided.

Important parameters that could be investigated include

$$\boxed{f_s}$$

switching ripple, transient response, and the difference between averaged and switching-level models.

The relationship

$$\boxed{f_s \rightarrow \text{Converter Dynamics} \rightarrow \text{Control Performance}}$$

requires further investigation.

17.2 Undirected Communication Topology

The current study considers an **undirected communication graph**.

In such a topology, communication relationships are represented symmetrically. If converter i communicates with converter j, the corresponding network representation treats the connection symmetrically.

This assumption simplifies the theoretical analysis but may not represent every practical communication network.

17.3 Directed Spanning-Tree Topology

The extension to a directed spanning-tree communication topology is identified as a future research direction.

In a directed network,

$$\boxed{i \rightarrow j}$$

does not necessarily imply

$$\boxed{j \to i}$$

Therefore, information can flow in only one direction between some converter agents.

Extending the proposed characteristic-loci and mixed NI+PR analysis to such a topology would increase the applicability of the control scheme.

18 Future Scope

Several directions can be considered for extending the present work.

18.1 Directed Communication Networks

The first major extension is the investigation of directed communication topologies.

Future work can study whether the consensus and stability properties remain valid when the converter agents communicate through a directed graph.

In particular, a **directed spanning-tree topology** can be investigated.

18.2 Experimental or Hardware Implementation

The present results are primarily simulation-based.

A natural next step is to implement the controller on actual converter hardware.

A practical experimental platform could contain multiple boost-converter units, individual loads, measurement circuits, and a communication network.

This would allow the theoretical and simulation results to be validated under practical electrical conditions.

18.3 Hardware-in-the-Loop Validation

Hardware-in-the-loop (HIL) testing can provide an intermediate step between simulation and complete experimental implementation.

The controller can be implemented on a real-time control platform while the converter dynamics are executed in a real-time simulator.

This would allow investigation of practical effects without immediately requiring a complete high-power converter prototype.

18.4 Larger Number of Converter Agents

The present study considers a limited number of converter agents.

Future investigations can increase the number of converters:

$$\boxed{N = 3 \rightarrow N = 5 \rightarrow N = 10 \rightarrow \cdots}$$

to evaluate the scalability of the proposed cooperative control architecture.

Increasing N also provides an opportunity to study how network size influences convergence speed and communication requirements.

18.5 Nonlinear Switching-Model Validation

The mathematical analysis is based on converter models such as averaged or transfer-function representations.

A future study can validate the controller using the detailed nonlinear switching model of the converters.

The progression would therefore be

$$\boxed{\text{Linear/Averaged Model} \rightarrow \text{Nonlinear Switching Model} \rightarrow \text{Experimental Validation}}$$

This would provide stronger evidence of the practical effectiveness of the proposed control scheme.

18.6 Overall Future Research Direction

The overall future research pathway can be summarized as

$$\boxed{\text{Current Simulation Study}}$$

$$\Downarrow$$

$$\boxed{\text{Quantitative Robustness Analysis}}$$

$$\Downarrow$$

$$\boxed{\text{Directed Communication Topology}}$$

$$\Downarrow$$

Communication Delay/Failure Analysis

$\Downarrow$

Nonlinear Switching-Model Validation

$\Downarrow$

Hardware-in-the-Loop Testing

$\Downarrow$

Experimental Multi-Converter Implementation

Overall, the proposed method provides a useful framework for distributed current synchronization in heterogeneous converter networks, while the identified limitations and future directions provide opportunities to extend the method toward more realistic communication conditions, larger converter networks, stronger disturbance conditions, and practical hardware implementation.

19 Conclusion

This work presented a distributed dynamic output-feedback control strategy for a heterogeneous multi-agent converter system. The main objective was to achieve coordinated operation and current consensus among multiple nonidentical converter agents while maintaining stability and satisfactory reference-tracking performance.

The proposed methodology is based on a **mixed Negative Imaginary (NI) and Positive Real (PR)** control framework. A homogeneous controller structure is applied to the different converter agents, even though the individual converter dynamics are heterogeneous. Under the conditions established by the theoretical analysis, the proposed controller ensures asymptotic stability of the interconnected system and enables the converter outputs to track the commanded reference.

A major advantage of the proposed approach is that it uses **output feedback** rather than requiring complete state information from every converter. In practical converter systems, measuring all internal states may require additional sensors, estimation algorithms, or communication channels. The output-feedback structure therefore provides a more practical implementation while retaining the cooperative control capability.

The proposed methodology was considered for a multi-converter ship degaussing application. In this application, several DC/DC converters supply independent degaussing coils. Since the magnetic field generated by each coil depends directly on its current, accurate synchronization of the converter currents is essential. The cooperative control scheme allows the individual converters to coordinate their outputs through the communication network and approach a common commanded current.

The simulation study demonstrated the effectiveness of the proposed control scheme. The simulated converter responses initially exhibited transient oscillations; however, these oscillations progressively decreased with time. The converter outputs subsequently converged toward a common operating value, demonstrating the desired **consensus behaviour**. The results therefore indicate that the proposed controller can provide stable and coordinated operation of the interconnected converter agents.

The simulation analysis also considered the effect of disturbances and different switching frequencies. The results showed acceptable transient and steady-state behaviour under the simulated operating conditions. This indicates that the proposed cooperative controller can maintain satisfactory performance even when the converter system is subjected to variations in operating conditions.

An important feature of the proposed approach is that the controller configuration remains homogeneous while the converter plants can remain heterogeneous. In other words,

$$\boxed{G_1(s) \neq G_2(s) \neq \cdots \neq G_N(s)}$$

while the same controller structure can be used for the agents,

$$\boxed{C_1(s) = C_2(s) = \cdots = C_N(s) = C(s)}$$

This considerably simplifies the controller architecture for a multi-converter system.

The overall objective can therefore be summarized as

$$\boxed{\begin{aligned}&\text{Heterogeneous Converters} + \text{Distributed Communication}\\&\quad + \text{Mixed NI+PR Controller} \longrightarrow \text{Stable Current Consensus}\end{aligned}}$$

The work nevertheless has some limitations. Although the simulations demonstrate satisfactory behaviour in the presence of output disturbances and different switching frequencies, a detailed investigation of the controller's limitations under a wider range of disturbance magnitudes, switching-frequency variations, and operating conditions was not performed. Such an analysis would be important for establishing the practical operating boundaries of the proposed control scheme.

Future work can therefore focus on a more comprehensive robustness analysis, including larger disturbances, broader switching-frequency ranges, parameter uncertainties, and more severe variations in converter operating conditions. Furthermore, the theoretical results can be extended from the considered communication structure to a **directed spanning-tree topology**, which would provide greater flexibility in the communication architecture of practical multi-converter systems.

Thus, the work establishes a theoretical and simulation-based foundation for applying mixed NI+PR cooperative output-feedback control to heterogeneous converter networks and demonstrates its potential for coordinated current control in ship degaussing applications.